

Lupus Nephritis and Other Kidney Manifestations of Systemic Inflammatory and Autoimmune Diseases

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CHAPTER OUTLINE

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KEY POINTS

- Kidney biopsy remains the gold standard for diagnosis of lupus nephritis (LN). Resolution of histologic changes on kidney biopsy may take years.
- Class III and IV lupus may present both clinically and histologically with a wide spectrum of involvement, and it is unclear whether current recommendations for the treatment of these classes of LN applies to those who present with milder disease or pathologic findings on biopsy.
- Novel treatments including calcineurin inhibitors and those targeting B cells are gaining prominence in the therapy of LN.
- Nonadherence to therapy is a major problem in the management of lupus and mandates clear and compassionate communication with patients.
- Holding anticoagulation before and after a kidney biopsy in antiphospholipid syndrome, even if only for a few days, may be enough to activate systemic thrombosis in susceptible patients, and there is a risk of late bleeding with resumption of anticoagulation. Therefore a risk benefit analysis should be undertaken before considering kidney biopsy.
- The incidence of renal involvement in mixed connective tissue disease (MCTD) varies from 10% to 26% of adults and from 33% to 50% of children with MCTD. Many patients have mild clinical manifestations with only microhematuria and <500 mg proteinuria daily; however, heavier proteinuria, severe hypertension, and acute kidney injury reminiscent of "scleroderma renal crisis" may occur.
- Although scleroderma renal crisis (SRC) typically presents with hypertension and microangiopathic hemolytic anemia, normotensive SRC may also occur in a minority of cases. There is a higher rate of renal recovery from dialysis in SRC, but among those who develop end-stage kidney disease, there is a higher rate of mortality compared with the general population.

SYSTEMIC LUPUS ERYTHEMATOSUS

Lupus nephritis (LN) is a frequent and potentially serious complication of systemic lupus erythematosus (SLE, lupus).^{1,2} Kidney disease influences morbidity and

mortality, both directly related to the consequences of the disease itself and indirectly, through complications of therapy. The key challenge therefore is timely diagnosis and tailoring appropriate therapy to achieve remission of kidney disease and preventing renal flares and the

progression of kidney disease while minimizing adverse complications of therapy.

EPIDEMIOLOGY

The incidence and prevalence of SLE depend on the age, sex, geographic locale, and ethnicity of the population studied, as well as the diagnostic criteria for defining SLE.³ SLE is newly diagnosed in approximately 400,000 people each year, with a prevalence of approximately 3.4 million.^{4,5} One review found that Poland, the United States, Barbados, and China have the highest incidences of SLE.⁴ The incidence of SLE in Poland is the highest, at around 80 per 100,000 person-years.⁴ In the United States the SLE incidence is around 5 to 12 per 100,000 person-years.^{4,6} Globally, SLE is more common in women compared with men being reported to be around 8.8 versus 1.5 per 100,000 person-years. Among women the prevalence is around 79 per 100,000 compared with 9.3 per 100,000 among men.⁴ These data must be interpreted with caution, however, as data included in the global analysis were only available for 20% of countries. The onset of lupus peaks between 15 and 45 years of age with >85% of patients being younger than 55 years. SLE is more often associated with severe nephritis in children and males. SLE and LN are more common and are associated with more severe kidney involvement, in populations of Asian, Hispanic, and African descent. Nearly half of all patients who develop end-stage kidney failure (ESKF) from SLE in the United States are African-American.⁷⁻¹³ The precise roles of biologic-genetic versus socioeconomic factors underlying this heterogeneity have not been clearly defined. The incidence of kidney involvement in SLE varies depending on the populations studied, the diagnostic criteria for kidney disease, and whether involvement is defined by biopsy or clinical findings. Approximately 25% to 50% of unselected lupus patients will have kidney disease at presentation, whereas up to 60% of adults with SLE will develop kidney involvement during their course.¹⁴⁻¹⁶

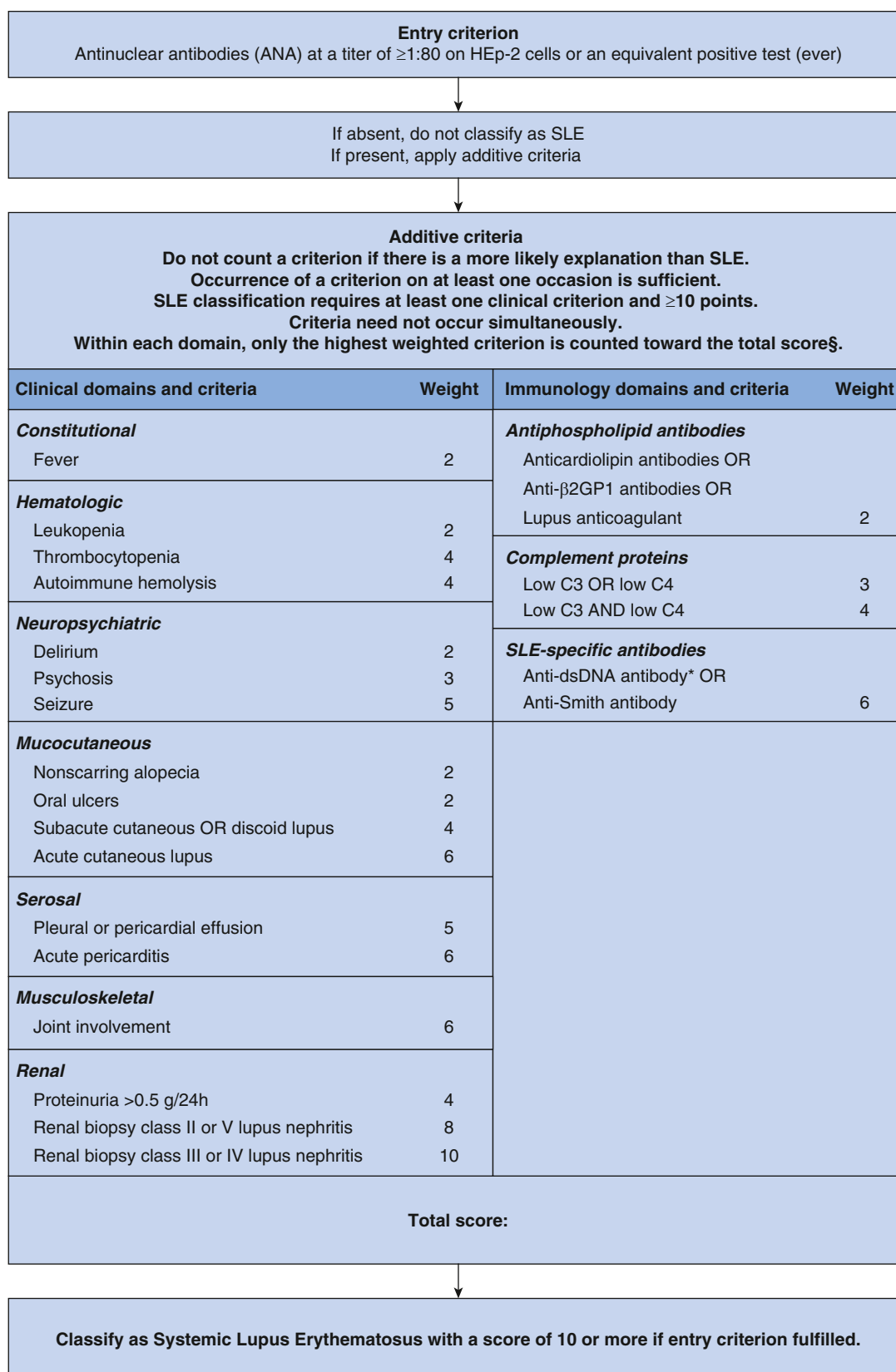
A number of genetic, hormonal, and environmental factors influence the course and severity of SLE.¹⁷⁻¹⁹ A multiplicity of genes are involved in both SLE and LN.²⁰ A genetic predisposition is supported by a higher concordance rate in monozygotic twins (25%) than fraternal twins (<5%), the greater risk of relatives of SLE patients developing SLE or other autoimmune disease, the association with distinct major histocompatibility complex class I and II (MHC) or human leukocyte antigen (HLA) genotypes (e.g., HLA-B8, HLA-DR2, and HLA-DR3), inherited deficiencies in complement components (e.g., homozygous C1q, C2, and C4 deficiencies), and Fc receptor polymorphisms. Some HLA alleles in SLE patients appear to be protective against LN (HLA DR4 and DR11), whereas others increase the risk of developing kidney involvement (HLA DR3 and DR13).²⁰⁻²³ Genome-wide association studies (GWASs) have identified more than 90 different genetic loci associated with an increased risk of SLE.²⁴ These candidate susceptibility genes regulate diverse immune functions such as T-cell activation, B-cell signaling, Toll-like receptors (TLRs), signal transduction, neutrophil function, and interferon (IFN) production.^{21,25} A meta-analysis of GWAS studies in SLE patients looking for risk alleles for LN has mapped these alleles to individual genes such as PDGF receptor A

gene.²⁶ Likewise, high-risk groups for LN such as those of African descent have a high frequency of genetic markers that may explain the increased risk, including certain Fc gamma RIIA –R131 alleles and apolipoprotein L1 (APOL1) risk alleles, which can almost triple the risk of ESKD in this population.^{27,28} Inbred spontaneous genetic murine models of SLE and LN are commonly used to model the disease and include the NZB B/W F1 hybrid, BXSB, and MRL/lpr mouse. SLE is inducible in other murine strains through injection of autoantibodies against DNA or by injection of Smith antigen peptides. Evidence for the role of hormonal factors includes the strong predominance of SLE in females of childbearing age and the increased incidence of lupus flares during or shortly after pregnancy. In the F1 NZB/NZW mice, females have more severe disease than males and disease severity is ameliorated by oophorectomy or androgen therapy. Environmental factors other than estrogens also modulate disease expression; these factors include viral or bacterial antigens, exposure to sunlight and ultraviolet radiation, and certain medications.^{17,18,21,29}

The updated definition of SLE in 2019 by the European League Against Rheumatism (EULAR) and the American College of Rheumatology (ACR) involves a minimum score assigned based on signs and symptoms.³⁰ For the kidney criterion of the scoring system, histology demonstrating class III or IV LN is assigned a higher score compared with class II LN, class V LN, or proteinuria >0.5 g/day (Fig. 31.1).

PATHOGENESIS OF SYSTEMIC LUPUS ERYTHEMATOSUS AND LUPUS NEPHRITIS

In patients with SLE, abnormalities of immune regulation lead to a loss of self-tolerance and production of a variety of autoantibodies and immune complexes.^{21,29,31-35} Genetic susceptibility may impact loss of tolerance to SLE antigens.³⁶ SLE is associated with defective regulation of T cells with decreased numbers of cytotoxic and suppressor T cells; increased helper (CD4⁺) T cells; dysfunctional T-cell signaling; and abnormal T_H1, T_H2, and T_H17 cytokine production.^{17,21,31-35,37} There is also polyclonal activation of B cells and defective B-cell tolerance. Autoreactive B-cell clones differentiate into plasma cells in the bone marrow and memory B cells located in lymphoid organs.³⁸ B cells also drive the expansion of antigen-specific T-cell clones.³⁹ The failure of apoptotic mechanisms to delete autoreactive B-cell and T-cell clones can promote their expansion and may trigger immune responses through interactions with TLRs leading to autoantibody production. The role of apoptosis, mediated via the Fas receptor or its ligand (FasL) in the generation of lupus, is unclear. Mutations in the FasL have been observed to lead to lupus-like systemic autoimmune diseases in mice and in some humans, but specific defects in FasL function are not evident in the majority of patients with SLE.⁴⁰ In comparison with single antigen glomerular diseases, such as antineutrophil cytoplasmic antibody (ANCA) vasculitis and PLA2R-positive membranous nephropathy, SLE has a greater number of autoreactive T-, B-, and plasma cell clones (Fig. 31.2).⁴¹ The result of these autoreactive clones in SLE is the production of a wide range of autoantibodies including those directed against nucleic acids, nucleosomes (double-stranded DNA [dsDNA]) in association with a core of positively charged

Fig. 31.1 Classification criteria for systemic lupus erythematosus.³⁰

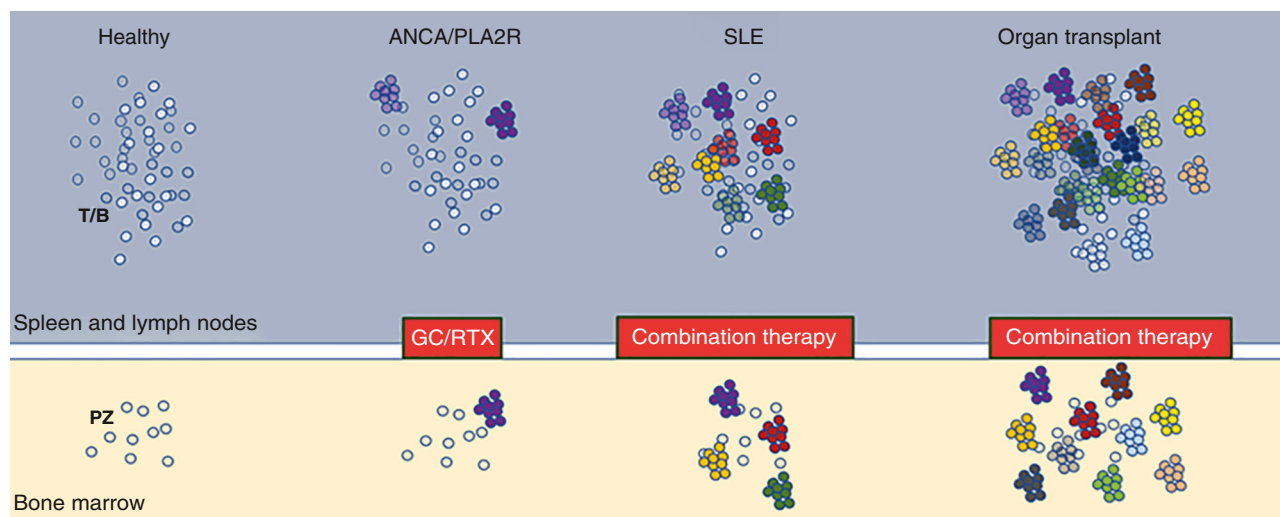


Fig. 31.2 Number of autoreactive T, B, and plasma cell clones in glomerular diseases with single antigen targets, systemic lupus erythematosus, and organ transplant. Immunological activity of all chronic autoimmune diseases is controlled by the size and antigen affinity of autoreactive clones of memory T cells and memory B cells (T/B) in the lymphoid organs as well as by long-lived plasma cells (PZ) in the bone marrow. In “simple” autoimmune diseases with single antigens such as, e.g. proteinase-3 in granulomatous polyangiitis or phospholipase A2 receptor in primary membranous glomerulonephritis, few immunotherapy is needed to control the few clones. In contrast, SLE involves autoimmunity against many, maybe 10-100 antigens, which goes together with a high number of T-, B-, and plasma cell clones to control, which requires combination therapy. Recipients of an organ transplant are in an even worst condition as each transplanted alloantigen will prime such a set of allo-reactive clones, which can go into the 10,000s of clones, difficult to control even with combination therapy. Different auto-reactive T, B, and plasma cell clones are depicted with different colours. ANCA, Antineutrophil cytoplasmic antibody; PLA2R, phospholipase A2 receptor; SLE, systemic lupus erythematosus; GC, glucocorticoids; RTX, rituximab. (From Parodis I, Depascale R, Doria A, Anders HJ. When should targeted therapies be used in the treatment of lupus nephritis: early in the disease course or in refractory patients? *Autoimmun Rev.* 2024;23(1):103418.)

histones), chromatin antigens, and nuclear and cytoplasmic ribonuclear proteins.^{17,21,29,35,42} Viral or bacterial peptides containing sequences similar to native antigens may lead to “antigen mimicry” and stimulate autoantibody production.

In SLE, autoantibodies combine with self-antigens to produce circulating immune complexes that deposit in the glomeruli, activate complement, and incite an inflammatory response. Immune complexes are also detectable in the skin at the dermal-epidermal junction, in the choroid plexus, pericardium, and pleural spaces. Renal involvement in SLE has been considered a human prototype of classic experimental chronic immune complex-induced glomerulonephritis.⁴³ The chronic deposition of circulating immune complexes plays a major role in the mesangial and endocapillary proliferative patterns of LN. The size, charge, and tissue avidity of the immune complexes, local hemodynamic factors, and ability of the mesangium to clear the complexes all influence the localization of circulating immune complexes within the glomerulus.^{17,32,44} In diffuse proliferative LN, the deposited complexes consist of nuclear antigens (e.g., DNA) and high-affinity complement-fixing immunoglobulin G (IgG₁ and IgG₃) antibodies.⁴⁴ In some SLE patients, the initiating event may be the local binding of cationic nuclear antigens such as histones to the subepithelial region of the glomerular capillary wall, followed by in situ immune complex formation. Once glomerular immune deposits form, the complement cascade is activated, leading to complement-mediated damage, activation of procoagulant factors, leukocyte Fc receptor activation with leukocyte infiltration, release of proteolytic enzymes, and production of various cytokines regulating glomerular cellular proliferation and matrix

synthesis. Additionally, studies suggest TGFβ1 may serve as a potential mediator of complement-related fibrosis.^{45,46} Transcriptome analysis of glomeruli from LN biopsies shows variable expression of B-cell genes, myelomonocytic genes, IFN-inducible genes, and fibrosis genes in both human and murine models.⁴⁷ Characterizing the immune profile of the kidney biopsy at the time of LN flare differentiates early treatment responders from nonresponders.⁴⁸ Autoantibodies to complement components (C1q and C3b), which are found in some patients, may enhance autoantigen exposure and facilitate immune complex deposition, while autoantibodies to C-reactive protein (CRP) may also lead to further immune activation and worsening of LN.^{49,50} Neutrophils undergoing cell death may release chromatin meshworks (called neutrophil extracellular traps [NETs] composed of chromatin, histones, and neutrophil proteins). These NETs, which are detectable in LN biopsies, are not degraded properly in patients with lupus and are a source of autoantigen presentation and induction of IFN-α by plasmacytoid dendritic cells.^{51,52} There is also evidence for intrarenal autoantibody production in patients with LN.⁵³ Glomerular damage may be potentiated by mechanisms distinct from immune complex deposition, such as hypertension and coagulation or complement regulatory abnormalities. For example, genetic polymorphism in complement factor H (CFH) influences susceptibility to SLE, and its deficiency accelerates the progression of LN in mice.⁵⁴ Some lupus patients with associated ANCA have documented focal segmental necrotizing glomerular lesions without significant immune complex deposition, resembling a “pauci-immune” glomerulonephritis.^{55,56} The presence of APL antibodies, with their attendant

Table 31.1 International Society of Nephrology/Renal Pathology Society (2003) Classification of Lupus Nephritis

Class I Minimal mesangial lupus nephritis (LN)
Class II Mesangial proliferative LN
Class III Focal LN ^a (<50% of glomeruli)
III (A) Active lesions
III (A/C) Active and chronic lesions
III (C) Chronic lesions
Class IV Diffuse LN ^b (≥50% of glomeruli)
Diffuse segmental (IV-S) or global (IV-G) LN
IV (A) Active lesions
IV (A/C) Active and chronic lesions
IV (C) Chronic lesions
Class V ^c Membranous LN
Class VI Advanced sclerosing LN
(≥90% globally sclerosed glomeruli without residual activity)

^aIndicates the proportion of glomeruli with active and with sclerotic lesions.

^bIndicates the proportion of glomeruli with fibrinoid necrosis and cellular crescents.

^cClass V may occur in combination with III or IV, in which case both will be diagnosed.

Indicate and grade (mild, moderate, or severe) tubular atrophy, interstitial inflammation and fibrosis, severity of arteriosclerosis or other vascular lesions.

alterations in endothelial and platelet function (including reduced production of prostacyclin and other endothelial anticoagulant factors, activation of plasminogen, inhibition of protein C or S, and enhanced platelet aggregation), can also potentiate glomerular and vascular lesions. Infrequently, SLE patients can develop podocytopathy with no evidence of immune complex deposition.^{56,57}

HISTOPATHOLOGY OF LUPUS NEPHRITIS

The histopathology of LN is pleomorphic.^{21,43,58,59} This diversity is evident when comparing biopsy findings from different patients or even adjacent glomeruli in a single biopsy. Moreover, the lesions have the capacity to transform from one pattern to another spontaneously or following treatment.^{43,60} The World Health Organization (WHO)^{61,62} classified LN by combining glomerular light microscopic (LM), immunofluorescence (IF), and electron microscopic (EM) findings. The 2003 International Society of Nephrology/Renal Pathology Society (ISN/RPS) classification of LN (Table 31.1) has proven more reproducible and provides more standardized definitions for precise clinical pathologic correlations.^{63,64} In 2018, a minor revision of this classification was published.⁶⁵ This revision provided a new cutoff for mesangial hypercellularity (>3 mesangial cells per mesangial area) and more precise definitions of cellular and fibrocellular crescents. It replaced the term “endocapillary proliferation” with “endocapillary hypercellularity” to acknowledge the major contribution of infiltrating leukocytes rather than proliferation of resident cells. It also eliminated the subdivisions IV-S and IV-G of class IV LN and applied modified National Institutes of Health (NIH) activity and chronicity indices to all LN classes.

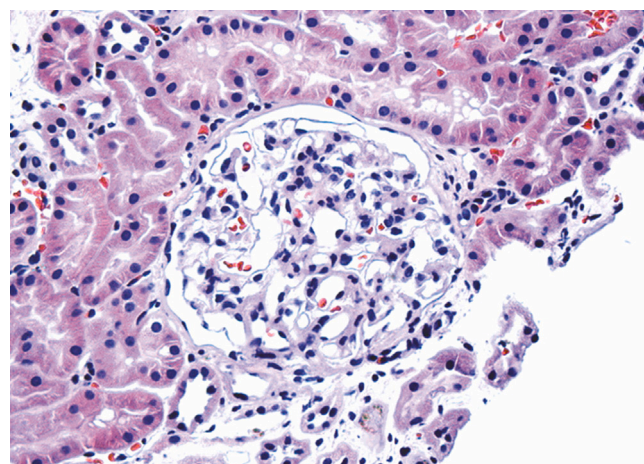


Fig. 31.3 Class II mesangial proliferative lupus nephritis. Increase in mesangial matrix and cellularity. Peripheral capillary loops are normal. The adjacent tubules in the interstitium are uninvolved (hematoxylin-eosin, ×400). (From Fogo A, Kashgarian M. *Diagnostic atlas of renal pathology*. 3rd ed. Elsevier, 2017.)

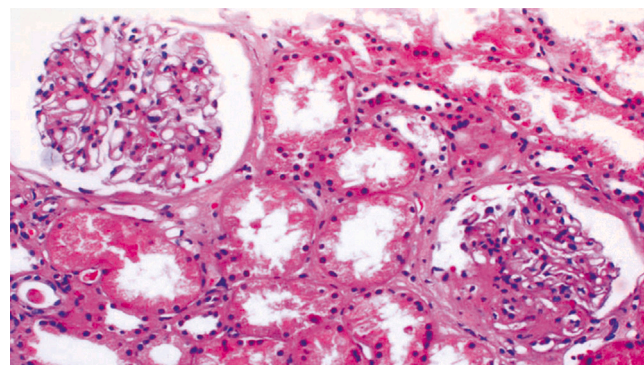


Fig. 31.4 Class III lupus nephritis. Focal and segmental glomerulonephritis. The glomerulus in the upper left pole is normal appearing, whereas the one on the right demonstrates a segmental area of adhesion and necrosis (hematoxylin-eosin, ×400). (From Fogo A, Kashgarian M. *Diagnostic atlas of renal pathology*. 3rd ed. Elsevier, 2017.)

ISN/RPS class I has normal-appearing glomeruli by LM but with mesangial immune deposits detected by IF and EM. Even patients without clinical renal disease often have mesangial immune deposits when studied carefully by the more sensitive techniques of IF and EM.⁶³

ISN/RPS class II is defined as mesangial hypercellularity, which is defined as more than three cells in mesangial regions distant from the vascular pole in 3-μm-thick sections, with mesangial immune deposits on IF and EM (Fig. 31.3).

ISN/RPS class III, focal LN, is defined as focal segmental and/or global endocapillary and/or extracapillary glomerulonephritis affecting less than 50% of the total glomeruli sampled (Figs. 31.4 and 31.5). Both active and chronic lesions are taken into account when determining the percentage of total glomeruli involved. There is typically focal segmental endocapillary hypercellularity, which variably includes mesangial cells, endothelial cells, and infiltrating mononuclear or polymorphonuclear leukocytes. Class III biopsies can have any combination of active and

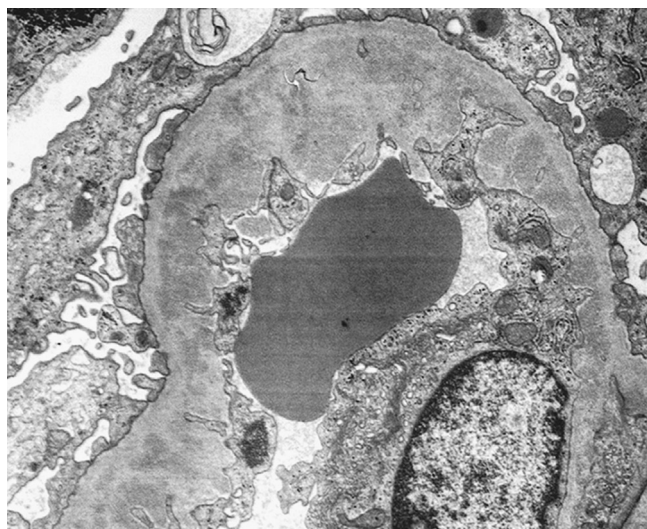


Fig 31.5 Class III lupus nephritis. Electron microscopy demonstrating subendothelial and mesangial deposits (transmission electron microscopy, $\times 4000$). (From Fogo A, Kashgarian M. *Diagnostic atlas of renal pathology*. 3rd ed. Elsevier, 2017.)

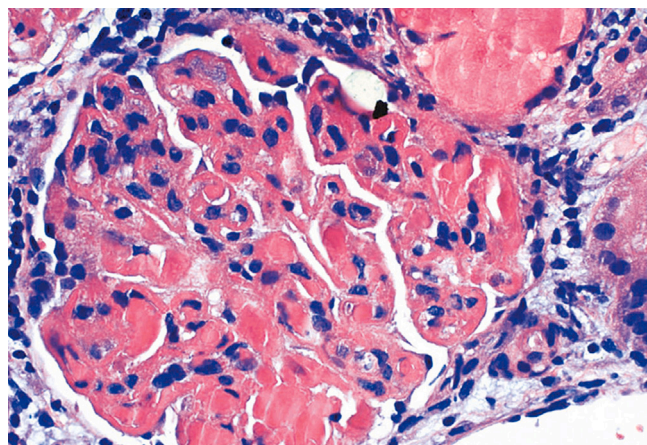


Fig 31.6 Class IV LN. Diffuse proliferative LN involves most or all of the glomeruli. The capillaries here contain “hyaline thrombi,” a lesion caused by massive subendothelial deposits bulging into the capillary lumen. Hyaline thrombi may be thickened to the point to be considered “wire-loop” lesions, due to the extensive subendothelial deposits (hematoxylin-eosin, $\times 400$). (From Fogo A, Kashgarian M. *Diagnostic atlas of renal pathology*. 3rd ed. Elsevier, 2017.)

chronic features. Active lesions may display cellular crescents, fibrinoid necrosis, nuclear pyknosis or karyorrhexis, and rupture of the glomerular basement membrane (GBM). Hematoxylin bodies, swollen basophilic nuclear material resulting from binding to ambient ANAs, are occasionally found within the necrotizing lesions. Subendothelial immune deposits may be visible by LM as “wire-loop” thickenings of the glomerular capillary walls or large intraluminal masses known as “hyaline thrombi.” Chronic glomerular lesions consist of segmental and/or global glomerular sclerosis owing to scarred glomerulonephritis with or without fibrous crescents.⁶³ In class III biopsies, glomeruli adjacent to those with severe histologic changes may show only mesangial abnormalities by LM. In class III,

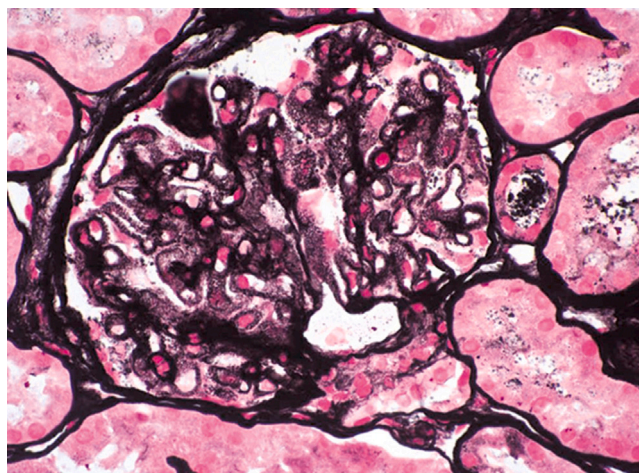


Fig. 31.7 Class V lupus membranous nephritis. Silver methenamine (Jones) stains reveal a spike and dome pattern to be present along the peripheral capillary loops. Where the wall of the capillaries is cut tangentially, there is a moth-eaten appearance of the capillary wall due to the deposits not staining with silver (Jones, $\times 400$). (From Fogo A, Kashgarian M. *Diagnostic atlas of renal pathology*. 3rd ed. Elsevier, 2017.)

diffuse mesangial and focal and segmental subendothelial immune deposits are typically identified by IF and EM. The segmental subendothelial deposits are usually present in the distribution of the segmental endocapillary hypercellularity.

ISN/RPS class IV, diffuse LN, has qualitatively similar glomerular endocapillary and/or extracapillary lesions as class III but involves more than 50% of the total glomeruli sampled (Fig. 31.6).^{61,63,66,67} Again, both active (hypercellular) and chronic (sclerosing) lesions are included when determining the percentage of glomeruli affected. All the active features described earlier for class III (including fibrinoid necrosis, leukocyte infiltration, wire-loop deposits, hyaline thrombi, hematoxylin bodies, and crescents) may be encountered in class IV LN. In general, there is more extensive peripheral capillary wall subendothelial immune deposition, and extracapillary hypercellularity in the form of crescents is not uncommon. Class IV lesions may have features similar to those of primary membranoproliferative glomerulonephritis (MPGN; also known as mesangiocapillary glomerulonephritis) with mesangial interposition along the peripheral capillary walls and double contours of the GBMs. Some class III and IV biopsies will have focal necrotizing and crescentic lesions akin to those seen in small vessel vasculitides. Some of these patients have circulating ANCA, especially of the peripheral IF (P-ANCA) or antimyeloperoxidase ELISA variety.^{55,68}

The $>50\%$ or $<50\%$ glomerular involvement is somewhat arbitrary, and there is no prognostic significance in this finding alone.^{69,70} For example, if a kidney biopsy contains 11 glomeruli and 6 ($>50\%$) of the glomeruli show proliferation and the findings described earlier, the diagnosis would be class IV LN. If 5/11 glomeruli showed these changes, it would be reported as class III LN. Therefore the distinction between class III and IV may be somewhat random and should not necessarily have implications for therapy.

ISN/RPS class V is defined by regular subepithelial immune deposits producing a membranous pattern (Fig. 31.7).^{63,71-73} The coexistence of mesangial immune deposits

and mesangial hypercellularity in most cases helps to distinguish membranous LN from primary membranous glomerulopathy (and may also be responsible for the microhematuria sometimes noted in membranous LN). Early membranous LN class V may have no identifiable abnormalities by LM, but subepithelial deposits are detectable by IF and EM.⁷⁴ In well-developed membranous lesions, there is typically thickening of the glomerular capillary walls and “spike” formation between the subepithelial deposits. Because sparse subepithelial deposits may also be encountered in other classes (III or IV) of LN, a diagnosis of “pure” lupus membranous LN should be reserved only for those cases in which the membranous pattern predominates. When the membranous alterations involve more than 50% of the total glomerular capillaries and are accompanied by focal or diffuse endocapillary hypercellular lesions and subendothelial immune complex deposition, they are classified as class III + V or class IV + V, respectively.

ISN/RPS class VI, advanced sclerosing LN is reserved for biopsies with more than 90% of the glomeruli sclerotic and no residual activity.⁶³ In such cases, it may be difficult even to establish the diagnosis of LN without the identification of residual glomerular immune deposits by IF and EM or a biopsy history of prior active LN.

IMMUNOFLUORESCENCE

In LN, immune deposits can be found in the glomeruli, tubules, interstitium, and blood vessels.^{43,58-60,74} IgG is almost universal, with codeposits of IgM, IgA, C3, and C1q commonly present. IgG dominant or codominant staining on immunofluorescence has a reported sensitivity of 0.98 (95% confidence interval [CI] 0.96–0.99) to distinguish it from non-LN; however, it has a poor specificity of 0.53 (95% CI 0.49–0.57).⁷⁵ The presence of all three immunoglobulins (IgG, IgA, and IgM) along with the two complement components (C1q and C3) is known as “full house” staining and is highly suggestive of LN. The reported sensitivity and specificity of “full house” staining are 0.71 (95% CI 0.66–0.76) and 0.90 (95% CI 0.87–0.92), respectively, to distinguish it from non-LN.⁷⁵ The resolution of immunoglobulin staining after successful treatment of proliferative LN may take many years.⁷⁶ Staining for fibrin–fibrinogen is common in crescents and segmental necrotizing lesions. The “tissue ANA”⁷⁴ (i.e., nuclear staining of renal epithelial cells in sections stained with fluorescent antisera to human IgG) is an infrequent finding in LN but is highly specific.⁷⁵ It results from the binding of patient’s own ANA to nuclei exposed in the course of cryostat sectioning.

ELECTRON MICROSCOPY

The distribution of glomerular, tubulointerstitial, and vascular deposits seen by EM correlates closely with that observed by IF.^{17,43,58-60} Deposits are typically electron dense and granular. Some exhibit focal organization with a “fingerprint” substructure composed of curvilinear parallel arrays measuring 10 to 15 nm in diameter.⁴³ Tubuloreticular inclusions (TRIs), which are intracellular branching tubular structures measuring 24 nm in diameter located within dilated cisternae of the endoplasmic reticulum of glomerular and vascular endothelial cells, are commonly observed in SLE biopsies.^{43,59,60} TRIs are inducible upon

exposure to α -IFN (so-called interferon footprints) and are also present in biopsies of human immunodeficiency virus (HIV)-infected patients and those with some other viral infections.⁷⁷

MASS SPECTROMETRY

The use of mass spectrometry has been emerging as an invaluable tool in the pathology of glomerular diseases. It may play a key role in proteome profiling and novel antigen/autoantibody discovery in LN. Exostosin 1 and 2 (EXT1 and 2) are two antigens that have been identified in kidney tissue samples of patients with membranous nephropathy associated with systemic autoimmune diseases like SLE. Approximately 30% of patients with class V LN have positive IHC staining for EXT1 or 2,⁷⁸ although patients typically do not have circulating antiexostosin antibodies. One retrospective study showed that, compared with EXT 1/2 negative membranous LN, EXT1/2 positive LN had fewer features of chronicity and was less likely to progress to ESKD.⁷⁹ Additionally, there are numerous additional antigens being reported in membranous LN, which are less frequent than EXT1 and EXT2 including neural cell adhesion molecule-1 (NCAM1), transforming growth receptor β receptor 3 (TGFB3), and early endosome antigen 1 (EEA1).⁸⁰ Although these novel antigens have been reported in class V LN, they may play a role in the discovery of antigens and subsequent antibodies in other classes of LN.

ACTIVITY AND CHRONICITY

Current guidelines advocate that renal biopsies should be accorded an activity and chronicity score, as modified from the NIH system.⁶⁵ The purpose is to identify and quantify active (potentially reversible) lesions and chronic (irreversible) lesions. In the modified NIH system, activity index is calculated by grading the biopsy on a scale of 0 to 3+ for each of six histologic features; these features are endocapillary hypercellularity, glomerular neutrophil infiltration and/or karyorrhexis, wire-loop deposits/hyaline thrombi, fibrinoid necrosis, cellular and/or fibrocellular crescents, and interstitial inflammation. The severe lesions of crescents and fibrinoid necrosis are assigned double weight. The sum of the individual components yields a total histologic activity index score from 0 to 24. Likewise, a chronicity index of 0 to 12 is derived from the sum of global and/or segmental glomerulosclerosis, fibrous crescents, tubular atrophy, and interstitial fibrosis, each graded on a scale of 0 to 3+. Studies at the NIH correlated both a high activity index (>12) and especially a high chronicity index (>4) with a poor 10-year renal survival rate. However, in several other large studies, neither the activity index nor the chronicity index correlated well with long-term prognosis. Other NIH studies concluded that a combination of an elevated activity index (>7) and chronicity index (>3) predicts a poor long-term outcome.⁸¹ A major value of calculating the activity and chronicity indices is in the comparison of sequential biopsies in individual patients. This provides an understanding of the efficacy of therapy and the relative degree of reversible versus irreversible lesions.^{43,58-60,82,83} More recently, Malvar and colleagues⁷⁶ investigated repeat biopsies for patients with LN undergoing treatment and found that the activity index may take years to resolve. Some

of the lesions such as neutrophils/karyorrhexis, fibrinoid necrosis, and cellular/fibrocellular crescents may resolve earlier than endocapillary hypercellularity, subendothelial hyaline deposits, and interstitial inflammation.

TUBULOINTERSTITIAL DISEASE, VASCULAR LESIONS, AND LUPUS PODOCYTOPATHY

Some SLE patients have prominent changes in the tubulointerstitial compartment in association with significant glomerular activity, while much more rarely interstitial nephritis may be the major renal involvement in patients with SLE.⁸⁴⁻⁸⁸ Active tubulointerstitial lesions include edema and inflammatory infiltrates of T lymphocytes (both CD4⁺ and CD8⁺ cells), B lymphocytes, monocytes, and plasma cells.⁸⁷ Tubulointerstitial immune deposits of immunoglobulin and/or complement may be present along the basement membranes of tubules and interstitial capillaries. Severe acute interstitial changes and tubulointerstitial immune deposits are most commonly found in patients with active proliferative class III and IV LN. The degree of interstitial inflammation does not correlate well with the presence or quantity of tubulointerstitial immune deposits.^{84,85} Interstitial fibrosis, tubular atrophy, or both are commonly encountered in the more chronic phases of LN. One study documented a strong inverse correlation between the degree of tubular damage and renal survival.⁸⁵ In addition, the renal survival rate was higher in patients who had lower expression levels of the intercellular adhesion molecule-1 (ICAM-1) in kidney biopsy tissue.⁸⁶ Increased urinary levels of C3, CFI, and C9-to-CD59 ratio have been associated with increased tubulointerstitial fibrosis, suggesting that urinary complement studies could serve as biomarkers for chronic histologic changes in LN.⁴⁵

Vascular lesions are not included in either the ISN/RPS classification or in the NIH activity and chronicity indices despite their frequent occurrence and clinical significance.^{56,89-91} The most frequent vascular lesion is simple vascular immune deposition, most common in patients with active class III and IV biopsies. Vessels may be normal by LM, but by IF and EM, there are granular immune deposits in the media and intima of small arteries and arterioles. Noninflammatory necrotizing vasculopathy, most common in arterioles in active class IV LN, is a fibrinoid necrotizing lesion without leukocyte infiltration that severely narrows or occludes the arteriolar lumen. True inflammatory vasculitis resembling polyangiitis is extremely rare in SLE patients. It may be renal limited or part of a more generalized systemic vasculitis.^{56,90,91} Thrombotic microangiopathy involving vessels and glomeruli may be associated with anticardiolipin/APL antibody or hemolytic-uremic/thrombotic thrombocytopenic purpura (HUS/TTP)-like syndrome.^{56,90,91}

A number of other renal diseases have been documented on biopsy in SLE patients including podocytopathies with features of minimal change disease (MCD), focal segmental glomerulosclerosis (FSGS), or collapsing glomerulopathy.^{57,92-94} In some, the relationship between SLE and podocytopathy suggests that this is not a coincidence but related to SLE-induced cytokine-mediated effects on podocytes. A collapsing pattern of focal sclerosis in SLE patients of African descent has been associated with APOL1 risk alleles.⁹⁵

CLINICAL MANIFESTATIONS

Although SLE predominantly affects young females, the clinical manifestations are similar in both sexes and in adults and children. Organ systems commonly affected include the kidneys, joints, serosal surfaces (including pleura and pericardium), central nervous system, and skin. In addition, involvement of the heart, liver, lungs, hematopoietic, and gastrointestinal organ systems is not uncommon.

Renal involvement often develops concurrently or shortly after the onset of SLE and may follow a protracted course with periods of remissions and exacerbations. Clinical renal involvement usually, but not always, correlates with the extent of glomerular involvement. However, some patients may have disproportionately severe vascular or tubulointerstitial lesions that dominate the clinical course.^{56,84,91}

Patients with ISN/RPS class I biopsies often have little evidence of clinical renal disease and are often diagnosed following a protocol biopsy. Likewise, most patients with mesangial lesions (ISN/RPS class II) have mild or minimal clinical renal findings.^{21,37,58-61} They may have active lupus serology (a high anti-DNA antibody titer and low serum complement), but the urinary sediment may be inactive, hypertension is infrequent, proteinuria is usually <1 g/day, and the serum creatinine concentration and glomerular filtration rate (GFR) are usually normal. Nephrotic-range proteinuria is extremely rare with class II unless there is a superimposed podocytopathy.^{92,93}

Class III, focal proliferative LN, is often associated with active lupus serology, although the degree of serologic activity does not necessarily correlate with the histologic severity.^{61,66} Hypertension and active urinary sediment are common. Proteinuria is often more than 1 g/day, and one-quarter to one-third of patients with focal LN have nephrotic syndrome at presentation. Many patients, in addition, have an elevated serum creatinine concentration at presentation. Patients with less extensive glomerular proliferation, fewer necrotizing features, and no crescents are more likely to be normotensive and have preserved renal function.

Patients with ISN/RPS class IV, similar to class III, often have high anti-DNA antibody titers, low serum complement levels, and very active urinary sediment, with erythrocytes, and cellular casts on urinalysis.^{21,58-61,63,67,96} Virtually all have proteinuria and up to 50% of the patients will have nephrotic syndrome. Hypertension and renal dysfunction are typical. Even when the serum creatinine level is in the “normal range,” the GFR is usually depressed. As mentioned earlier, the distinction between “focal” and “diffuse” proliferative LN may depend on tissue biopsy sampling. Class III and IV lupus may present both clinically and histologically with a wide spectrum of involvement, and it is unclear whether current recommendations for the treatment of these classes of LN apply to those who present with milder disease or pathologic findings on biopsy.

Patients with membranous LN, ISN/RPS class V, typically present with proteinuria, edema, and other manifestations of nephrotic syndrome.^{21,58-61,63,71-73} However, up to 40% will have proteinuria of <3 g/day, and 16% to 20% <1 g/day. In contrast to the proliferative variants, only about 60% of membranous LN patients have low serum complement concentration and an elevated anti-DNA antibody titer at presentation.⁶¹ The

ANA is usually positive. However, hypertension and renal dysfunction may occur without superimposed proliferative lesions. Patients with membranous LN may present with nephrotic syndrome before developing other clinical and laboratory manifestations of SLE.^{61,71-73} In addition, they are predisposed to thrombotic complications such as renal vein thrombosis and pulmonary emboli.^{56,90} Patients with mixed membranous and proliferative biopsies have clinical features that reflect both disease components.

Advanced sclerosing LN, ISN/RPS class VI, is usually the result of “burned-out” LN of long duration.⁶³ Some renal histologic damage may represent nonimmunologic progression of sclerosis mediated by hyperfiltration in remnant nephrons. Although the lesions are sclerosing and inactive, class VI patients may still have microhematuria and proteinuria. Virtually all have hypertension and a decreased GFR. Levels of anti-DNA antibodies and serum complement levels often normalize at this late stage of disease.

“Silent LN”^{21,42,97} has been described in patients without clinical evidence of renal involvement despite biopsy evidence of active proliferative LN. Some define silent LN as active biopsy lesions without active urinary sediment, proteinuria, or a depressed GFR, whereas others require negative lupus serologies as well. Although silent LN is well described in some studies, others have been unable to find even isolated examples.^{21,37} It appears to be uncommon, and it is highly likely that even patients with true “silent disease” will manifest clinical renal involvement over time.

Lupus cystitis is an underrecognized manifestation of SLE.⁹⁸ Permanent bladder dysfunction has been reported in rare cases. It affects an estimated 0.01% to 2% of SLE patients, mainly women.⁹⁹ Most of the reported cases are from East Asia, particularly Japan.¹⁰⁰ The pathophysiology is proposed to be due to immune complex vasculitis in the arterioles of the bladder. This can result in progressive bladder fibrosis and thinning and rarely obstruction. Both a dilated urinary tract and stenosis have been reported.¹⁰¹ Symptoms may include urgency, dysuria, frequency, nocturia, and lower abdominal pain; reports of copresentation with lupus enteritis have been described.^{102,103} Imaging and cystoscopy are required for diagnosis. Glucocorticoids and cyclophosphamide have been reported as treatment, with other immunosuppression regimens for refractory cases.¹⁰⁴⁻¹⁰⁶

BIOMARKERS

SEROLOGIC TESTS

The presence of antibodies directed against nuclear antigens (ANAs) and especially against DNA (anti-DNA) antibodies is included in the ACR criteria for SLE and commonly used to monitor the disease course.^{17,21,29} ANAs are a highly sensitive screen for SLE, being found in more than 90% of untreated patients, but they are not specific for SLE and occur in many other rheumatologic and nonrheumatologic conditions.^{17,21,29,43} Neither the particular pattern of ANA fluorescence (homogeneous, speckled, nucleolar, or rim) nor the titer correlates well with the presence or the severity of renal involvement in SLE.

Autoantibodies directed against dsDNA (anti-dsDNA) are a more specific but less sensitive marker of SLE, but even so are found in almost three-fourths of untreated active

SLE patients.^{17,21,29} Anti-dsDNA IgG antibodies of high avidity that fix complement (i.e., IgG₁ and IgG₃) correlate best with the presence of renal disease,^{17,21,29} and these anti-dsDNA antibodies have been found in the glomerular immune deposits of murine and human LN.^{21,29,107,108} High anti-dsDNA antibody titers correlate well with clinical activity.^{17,21,29} Anti-single-stranded DNA antibodies (anti-ssDNA), commonly found in SLE and other collagen vascular diseases, do not correlate with clinical lupus activity. It has been argued that anti-ssDNA may be helpful diagnostically when the anti-dsDNA antibody screen is negative.

Autoantibodies directed against ribonuclear antigens are commonly present in lupus patients and include anti-Sm and anti-nRNP against extractable nuclear antigen (ENA).^{17,29,35,109} Anti-Sm antibodies, although specific for SLE, are found in only about 25% of lupus patients and are of unclear prognostic value. Anti-nRNP antibodies, found in more than one-third of SLE patients, are also present in many other rheumatologic diseases, particularly mixed connective tissue disease (MCTD).^{29,35,110} Anti-Ro/SSA antibodies are directed against the protein complex of a cytoplasmic RNA and are present in 25% to 30% of SLE patients. Anti-La/SSB autoantibodies, directed against a nuclear RNP antigen, are present in 5% to 15% of lupus patients. Neither of the latter two antibodies is specific for SLE, and both are found in other collagen vascular diseases, especially Sjögren syndrome. Maternal anti-Ro antibodies are important in the pathogenesis of neonatal lupus including the development of cardiac conduction abnormalities in the newborn.¹¹¹

Levels of total hemolytic complement (CH50) and complement components are usually decreased during active SLE, especially with active proliferative LN.^{17,58-60,109} Levels of C4 and C3 often decline before a clinical flare of SLE. Serial monitoring of complement levels, with a decline in levels predicting a flare, is considered more useful clinically than an isolated depressed C3 or C4 value.⁵⁹ Likewise, normalization of depressed serum complement levels is often associated with improved renal outcomes.¹¹² Some lupus patients are serologically *concordant*, where the complement levels and dsDNA antibodies correlate with active disease, whereas other patients are *discordant* and continue to show active lupus serology even in the face of quiescent disease. Levels of total complement and C3 may be decreased in the absence of active systemic or renal disease in patients with extensive dermatologic involvement. Several heritable complement deficiency states (including C1r, C1s, C2, C4, C5, and C8) have been associated with SLE, and these patients may have depressed total complement levels despite inactive disease.¹¹³

Other immunologic test results commonly found in lupus patients include elevated levels of circulating immune complexes, a positive lupus band test, and the presence of cryoglobulins. None correlates well with SLE or LN activity.¹¹⁴⁻¹¹⁶ Patients with SLE commonly have a false-positive Venereal Disease Research Laboratory (VDRL) test result due to the presence of APL antibodies.²¹ The surprising finding that patients with SLE often have low CRP levels was initially used to argue against the importance of using CRP levels as a biomarker for inflammation in other diseases. It is now appreciated that this reflects high titer antibodies to CRP in some SLE patients, although the clinical correlates of these unique antibodies are not

clear. In patients who are homozygous C2 deficient, anti-Ro antibodies are associated with a unique dermal psoriasiform type of lupus with a vasculitic disease associated with central nervous system involvement and cutaneous ulcers.¹¹⁷ In addition, lupus patients can develop antibodies directed against histones, endothelial cells, phospholipids, the *N*-methyl-D-aspartate receptor (associated with central nervous system disease in SLE), and neutrophil cytoplasmic antigens (ANCA).¹¹⁸⁻¹²⁰

URINARY BIOMARKERS

The presence of active urinary sediment may serve as an indicator of glomerular inflammation in many different glomerular diseases including LN.¹²¹ Active urinary sediment can generally be defined as ≥ 10 red blood cells per high-powered field or hemoglobinuria 1+. Similar to IgA nephropathy, the mechanism of microscopic hematuria may be in the context of immune complexes subsequently releasing inflammatory cytokines and complement cascade, which result in glomerular injury and passage of red blood cells into the urinary space.¹²² Appropriate handling of urine samples is required, and it is important to be aware of urologic and gynecologic causes of hematuria including menstruation in women. Unlike in IgA nephropathy,¹²¹ the prognostic relevance of hematuria in LN has not been demonstrated. Clinical trials in LN have variably included resolution of microscopic hematuria as part of the response definition. The role of home urinalysis testing for detecting early LN and flares is also currently under investigation.¹²³ Other indicators of glomerular inflammation on urine microscopy include dysmorphic red blood cells and casts, and the significance of their resolution after treatment is even less defined than microscopic hematuria.

Several novel urinary biomarkers in LN have been independently validated across multiple cohorts.¹²⁴ Updates in technologies such as multiplexed immune assays, transcriptomics, proteomics, and mass cytometry have led to a plethora of biomarkers being investigated for LN.¹²⁴ Urinary CD163 is a transmembrane receptor predominantly expressed by activated macrophages, which has been associated with active kidney inflammation. It has demonstrated associations with histologic activity and treatment response.¹²⁵⁻¹²⁷ Additionally, urinary biomarkers of monocyte/neutrophil degranulation, wound healing/matrix degradation, and IL-16 correlate with histologic activity, and a decline of these biomarkers after 3 months of treatment predicted 1-year remission better than proteinuria alone.¹²⁸ However, the diagnostic performance of these biomarkers needs to be validated further in longitudinal studies in diverse populations. They may serve as valuable investigations in the future.

MONITORING CLINICAL DISEASE

It is important to be able to predict systemic and renal relapses and prevent their occurrence through the judicious use of immunosuppressive agents. There may be clinical differences between patients with relapsed disease versus new presentations of LN. Particularly, hypertension may be less frequent in relapsed patients compared with newly diagnosed LN.¹²⁹ Serial measurements of many serologic tests (including complement components, autoantibodies,

erythrocyte sedimentation rate [ESR], CRP, circulating immune complexes, and, recently, levels of cytokines and interleukins [ILs]) have been used to predict lupus flares. Although there is controversy regarding the value of serum C3 and C4 levels and anti-DNA antibody titers in predicting clinical flares of SLE or LN, these have yet to be replaced by new biomarkers.^{17,58,116} Additional factors impacting the use of anti-dsDNA to predict LN activity include the use of immunosuppression, which may affect levels, as well as overlap with class 5 disease, the majority of which are known to be anti-dsDNA negative.¹³⁰ In most patients with lupus activity, however, serum levels of anti-dsDNA typically rise and serum complement levels typically fall as the clinical activity of SLE increases, often preceding clinical renal deterioration. A particularly challenging group of patients may be those that remain serologically active but clinically quiescent for long periods of time.¹³¹ However, these patients tend to have more disease activity in concert with a lupus flare.

In patients with active renal involvement, the urinalysis frequently reveals dysmorphic erythrocytes, red blood cell casts, and other formed elements. Given the vagaries of centrifuging urine and preparing a slide, the number of red cells or red cell casts per high-power field should not be used as a measure of the degree of kidney involvement. An increase in proteinuria from levels of <1 g/day to nephrotic levels is a clear indication of either increased activity or a change in renal histologic class. When there is concern about the degree of activity of the SLE and LN, a repeat renal biopsy will often clarify whether to change therapy.^{58,61}

Some data suggest that specific serologic tests, in conjunction with clinical parameters and urine investigations, may help predict the likelihood of histologic activity with LN.¹³² Gao and colleagues¹³² developed a prediction model composed of mean arterial pressure, hematocrit, serum albumin, eGFR, urinary erythrocytes, serum C3, and anti-dsDNA to predict an activity index >2 on kidney biopsy with good correlation. External validation is still required in diverse patient populations before this prediction tool is recommended for clinical use, but this study suggests that the constellation of the clinical assessment and laboratory parameters may prove useful. The predictability of kidney lesions with alternative biomarkers would be of value where kidney biopsy may be high risk or unavailable. Until this is achieved, in some contexts it may be reasonable to proceed with therapy without subjecting the patient to the biopsy.

DRUG-INDUCED LUPUS

A variety of medications may induce a lupus-like syndrome or exacerbate an underlying predisposition to SLE. Those medications metabolized by acetylation, such as procainamide and hydralazine, have been common causes.^{133,134} This occurs more commonly in patients who are slow acetylators due to genetic variants associated with a decrease in hepatic *N*-acetyltransferase activity. Diltiazem, minocycline, penicillamine, isoniazid, methyl dopa, chlorpromazine, and prazosin, a selective β_1 -blocker, are other potential causes of drug-induced lupus.¹³³⁻¹³⁶ Other drugs that have been associated less frequently with this syndrome include phenytoin, quinidine, propylthiouracil, sulfonamides, lithium, β -blockers, nitrofurantoin, para-aminosalicylic

acid, captopril, glyburide, hydrochlorothiazide, IFN- α , carbamazepine, sulfasalazine, rifampin, and tumor necrosis factor- α (TNF- α) blockers.^{133,137,138}

Clinical manifestations of drug-induced lupus include fever, rash, myalgias, arthralgias and arthritis, and serositis. Central nervous system and renal involvement are relatively uncommon.^{133,139,140} While elevated anti-DNA antibodies and depressed serum complement levels are less common in drug-induced lupus, antihistone autoantibodies are present in more than 95% of patients.¹³³ These are usually formed against a complex of the core histone dimer H2A-H2B and DNA and other histone components.^{133,141} Antibodies are also present in the vast majority of idiopathic, nondrug-related SLE patients, but they are directed primarily against different histone antigens (linker H1 and core H2B).¹³³ The presence of antihistone antibodies in the absence of anti-DNA antibodies and other serologic markers for SLE is also indicative of drug-induced disease. The diagnosis of drug-induced lupus depends on documenting the offending agent and achieving a remission following withdrawal of the drug. The primary treatment consists of discontinuing the offending drug. For patients with an established diagnosis of SLE and LN, it may be reasonable to allow them to take medications associated with drug-induced lupus as long as they are in remission. However, if they are in a flare of SLE, it may be safest to withhold these drugs given their potential to confound lupus symptomatology.

PREGNANCY AND SYSTEMIC LUPUS ERYTHEMATOSUS

Because SLE occurs so commonly in women of childbearing age, the issue of pregnancy arises often in the care of this population. Independent but related issues are the health of the mother (in terms of both flares of lupus activity and progression of renal disease) and the fate of the fetus. For more discussion on pregnancy and kidney disease, see [Chapter 58](#). These maternal and fetal complications in SLE patients include, but are not limited to, higher rates of cesarean section, preeclampsia and hypertension, spontaneous abortion, thromboembolic disease, postpartum infections, premature births and babies small for gestational age, neonatal intensive care unit requirements, congenital defects, and lower live birth rates.¹⁴² Approximately 1 in 5 pregnancies in women with SLE experience an adverse pregnancy outcome.¹⁴³ One systematic review and meta-analysis has described the adverse pregnancy risks associated with SLE but recognizes attrition bias and confounding.¹⁴⁴ SLE disease activity at conception or preconception has been associated with increased preterm birth (OR 2.91 [95% CI 1.96–433]) and preeclampsia (OR 2.32 [1.40–3.83]).¹⁴⁴ Other risk factors for adverse pregnancy outcomes include primigravida mothers, use of antihypertensives, and thrombocytopenia.^{143,145} Frequency of adverse pregnancy outcomes also varies by race and ethnicity, with higher rates of complications among patients of African descent.¹⁴⁶

It is unclear whether flares of SLE occur more commonly during pregnancy or shortly after delivery, but reported flare rates in pregnancy are approximately 30% to 60%.^{147,148} Patients with quiescent lupus for at least 6 months before conception are less likely to experience an exacerbation of SLE during pregnancy.¹⁴⁹ Decreased serum complement

levels, particularly C4, during the first trimester have been associated with risk of flare and adverse pregnancy outcomes.^{149,150} Therefore both immunologic and clinical SLE quiescence is optimal before pregnancy to reduce the risk of adverse maternal and fetal outcomes.

Pregnancy in patients with preexisting LN has also been associated with worsening of renal function.^{151,152} This is less likely to occur in patients who have been in remission for at least 6 months. However, even previous LN has also been associated with decreased live birth rate (OR 0.62 [95% CI 0.47–0.81]), increased risk of preterm birth (OR 2.00 [1.55–2.57]), and increased risk of preeclampsia (OR 3.11 [2.35–4.12]).¹⁴⁴ Patients with hypertension are likely to develop higher blood pressure, and those with proteinuria are likely to have increased proteinuria during pregnancy, at least in part as a result of hyperfiltration. Chronic hypertension has been associated with increased risk of disease flare in pregnancy (OR 2.50 [95% CI 1.74–3.58]), preterm birth (OR 2.65 [1.87–3.77]), and preeclampsia (OR 5.86 [3.41–10.06]).¹⁴⁴ Patients with elevated serum creatinine levels are most likely to suffer worsening of renal function and to be at highest risk for fetal loss.

It is generally recommended to continue with hydroxychloroquine during pregnancy, with some studies reporting lower rates of SLE flare and a decrease in congenital heart block.^{153,154} It is worth mentioning one study that demonstrated hydroxychloroquine was associated with a small increased risk of congenital anomalies (OR 1.26) with a daily dose >400 mg in the first trimester.¹⁵⁵ Although it is difficult to dismiss these findings, hydroxychloroquine has long been considered safe in pregnancy and notable confounders were not accounted for in this study.¹⁵⁶ Although high-dose corticosteroids, cyclosporine, tacrolimus, and azathioprine have all been used in pregnant lupus patients, their safety is unclear. Cyclophosphamide is contraindicated due to its teratogenicity. Mycophenolate is also highly teratogenic and associated with first-trimester pregnancy loss and facial deformities such as cleft lip/palate, cartilage abnormalities of the outer ear, absent auditory canals, and limb abnormalities.¹⁵⁷ It should be discontinued at least 6 weeks before planned pregnancies. Newer agents such as rituximab are not recommended due to a lack of data about their safety. Although calcineurin inhibitors have been used in pregnancy, voclosporin is contraindicated in pregnancy due to the high alcohol content. Belimumab, a human monoclonal antibody that inhibits B-cell activating factor (BAFF), is not currently recommended for use in pregnancy. Postmarketing reports in pregnant women with SLE exposed to belimumab are small in number and heterogeneous and provide neither clear risk of harm nor sufficient evidence of safety.^{158,159} The judicious use of rituximab and belimumab in pregnancy should only be considered in those at highest risk of maternal and fetal complications.

The rate of fetal loss in most series is 20% to 40% and may even approach 50% in other selected series.^{151,152,160,161} While fetal mortality is increased in SLE patients with renal disease, it may be decreasing in the modern treatment era.¹⁶² Patients with anticardiolipin or APL antibodies, hypertension, or heavy proteinuria are at higher risk for fetal loss. Secondary antiphospholipid syndrome (APS) has been associated with decreased probability of livebirth (OR 0.40 [0.27–0.58]), increased risk of pregnancy loss after 20

weeks of gestation (OR 2.77 [1.44–5.31]), and increased risk of preterm birth (OR 1.65 [1.29–2.11]).¹⁴⁴ One review of 10 studies in more than 550 women with SLE found that fetal death occurred in 38% to 59% of all pregnant SLE patients with APL antibodies compared with 16% to 20% of those women with SLE without these antibodies.¹⁶³ Prospective studies are needed to better define risk and overcome bias in retrospective studies.¹⁶⁴

COURSE AND PROGNOSIS OF LUPUS NEPHRITIS

There is significant morbidity and mortality associated with LN, associated with autoimmune disease activity and progression of kidney disease, as well as treatment-related adverse effects from immunosuppressive therapies. Compared with the standard population, those with LN may have up to a sixfold higher rate of mortality.¹⁶⁵ Even in comparison with other primary glomerular diseases, patients with LN experience a higher risk of death and major adverse cardiovascular events.¹⁶⁶ This increased risk is likely driven by progression of atherosclerotic disease from both LN^{167,168} and CKD,¹⁶⁹ but serious infectious complications are an important contributor to mortality in this population.¹⁷⁰ This mortality risk may be partially mitigated by achieving either complete or partial renal remission.^{171,172}

Among patients with LN, about 10% may progress to ESKF over 10 years.^{165,173,174} This course is defined by the initial pattern and severity of renal involvement as modified by therapy, exacerbations of the disease, adherence to therapy, and complications of treatment. The prognosis has clearly improved in recent decades with wider and more judicious use of new immunosuppressive medications.^{66,175-177}

Patients with lesions limited to the renal mesangium generally have an excellent course and prognosis.^{21,43,58-60} If these patients do not transform into other patterns, they are unlikely to develop progressive renal failure and any excess mortality is due to extrarenal manifestations and complications of therapy. Patients with focal proliferative disease have an extremely varied course. Those with mild proliferation involving a small percentage of glomeruli respond well to therapy, and <5% progress to renal failure over 5 years.^{21,58-61,177,178} Patients with more proliferation, necrotizing features, and/or crescent formation have a prognosis more akin to patients with class IV diffuse LN. Class III patients may transform into class IV over time and, as mentioned, sometimes the distinction between the two may be the result of random sampling. Some patients with active segmental proliferative and necrotizing lesions resembling ANCA-associated small vessel vasculitis have a worse renal prognosis than other patients with focal proliferative lesions.^{55,68,179}

Patients with diffuse proliferative disease have the least favorable prognosis in most older series.^{61,66,67} Nevertheless, the prognosis for this group has markedly improved, with renal survival rates now exceeding 90% in some series of patients treated with modern immunosuppressive agents.^{67,178,180} In trials from the NIH, the 5-year risk of doubling the serum creatinine concentration, a surrogate marker for progressive renal disease, in patients with diffuse proliferative lupus treated with cyclophosphamide-containing regimens ranged from 35% to <5%.¹⁸⁰⁻¹⁸² In an Italian study of diffuse proliferative LN, patient survival was

77% at 10 years and more than 90% of extrarenal deaths were excluded.⁶⁷ In a U.S. study of 89 patients with diffuse proliferative LN, renal survival was 89% at 1 year and 71% at 5 years.¹⁸³ It is unclear whether the improved survival rates in these recent series are largely due to improved immunosuppression or better supportive care and more judicious clinical use of these medications.

In the past, some studies have found age, sex, and race to be as important prognostic variables as clinical features in patient and renal survival in SLE.^{8-10,21,58,81,176,183-186} A consistent finding is that those of African ancestry have a greater frequency of LN and a worse renal and overall prognosis. This worse prognosis appears to relate to both biologic/genetic and socioeconomic factors.^{9,10,21,58,81,183,185-187} In a study from the NIH of 65 patients with severe LN, clinical features at study entry associated with progressive renal failure included age, African ancestry, hematocrit <26%, and serum creatinine concentration >2.4 mg/dL.⁸¹ Patients with a combined activity index (>7) plus chronicity index (>3) on renal biopsy, as well as those with the combination of cellular crescents and interstitial fibrosis also had a worse prognosis. In another U.S. study of 89 patients with diffuse proliferative LN, none of the following features affected renal survival: age, gender, SLE duration, uncontrolled hypertension, or any individual histologic variable.¹⁸³ Entry serum creatinine level higher than 3.0 mg/dL, increased activity and chronicity indices on biopsy, and African ancestry predicted a poor outcome. Five-year renal survival rate was 95% for Caucasian patients but only 58% for Black patients. In a study of more than 125 LN patients with WHO class III or IV from New York, both racial and socioeconomic factors were associated with the worst outcomes in African-Americans and Hispanics.⁹ An evaluation of 203 patients from the Miami area confirmed worse renal outcomes in African-Americans and Hispanics, argued to be related to both biologic and economic factors.¹⁰

More rapid and more complete renal remissions are associated with improved long-term prognosis.^{171,188} Repeated renal flares during the course of SLE, including those as a result of nonadherence to therapy, also may predict a poor renal outcome.¹⁸⁹⁻¹⁹¹ Relapses of severe LN over 5 to 10 years of follow-up occur in up to 50% of patients and usually respond less well and more slowly to repeated courses of therapy.^{58,189,192-195} A retrospective analysis of 70 Italian patients in whom more than half had diffuse proliferative disease found excellent patient survival (100% at 10 years and 86% at 20 years), as well as preserved renal function with the likelihood of not doubling the serum creatinine concentration to be 85% at 10 years and 72% at 20 years. Most patients in this study were Caucasian, which likely influenced the excellent long-term prognosis. Patients with renal flares of any type had seven times the risk of renal failure, and those with rapid rises in creatinine had 27 times the chance of doubling their serum creatinine concentration. Another Italian study of 91 patients with diffuse proliferative LN showed more than 50% having a renal flare, which correlated with a younger age at biopsy (<30 years old), higher activity index, and karyorrhexis on biopsy.¹⁹⁰ The number of flares, nephritic flares, and flares with increased proteinuria correlated with a doubling of the serum creatinine. The role of relapses

in predicting progressive disease has been noted by others as well, although relapse does not invariably predict a bad outcome.¹⁹⁶

Although an elevated anti-dsDNA antibody titer and low serum complement levels may correlate with active renal involvement, they do not correlate with long-term renal prognosis.^{21,61,183,189,197} In several studies, anemia has been a poor prognostic finding regardless of the underlying cause.^{81,189} Severe hypertension has also been correlated with renal prognosis in some, but not all, studies.⁶¹ Renal dysfunction, as noted by elevated serum creatinine or decreased GFR, heavy proteinuria, and especially nephrotic syndrome, is indicative of a poor renal prognosis in the vast majority of series.^{17,21,58-60,66} However, not all studies have found an elevation of the initial serum creatinine to predict a poor long-term outcome, and in some the initial serum creatinine only predicted short-term renal survival.¹⁸³ Other renal features, such as duration of LN and rate of decline of GFR, may also predict prognosis.^{178,197}

Finally, histologic features such as the class, the degree of activity and chronicity, and the severity of tubulointerstitial damage have also predicted outcomes. In a number of studies, the pattern of renal involvement, especially when using the ISN/RPS or older WHO classification, has been a useful guide to prognosis.^{61,64,96} In NIH trials, patients with severe proliferative LN with a higher activity index or chronicity index were more likely to have progressive renal failure.¹⁷⁸ However, studies with different referral populations could not confirm this finding.^{58,60,61,198} Regardless, the contribution of chronic renal scarring to a poor long-term outcome has been confirmed by many studies.^{86,190,196,197} Some studies have found the initial renal biopsy to have little predictive value; rather, certain features on a repeat biopsy at 6 months proved to be a strong predictor of doubling the serum creatinine or progression to renal failure.^{85,196} The lack of predictive value of the initial biopsy also suggests that a high-risk kidney biopsy at presentation may not always be necessary. These include ongoing inflammation with cellular crescents, macrophages in the tubular lumens, persistent immune deposits (especially C3) on IF microscopy, and persistent subendothelial and mesangial deposits or an active NIH activity score.^{199,200} Other studies suggest that reversal of interstitial fibrosis and glomerular segmental scarring along with remission of initial inflammation and immune deposition are important favorable prognostic findings on the 6-month biopsy.¹⁹⁶ Thus this argues that chronic changes on the initial biopsy are not always cumulative or immutable. Surveillance kidney biopsies for prognosis may not be all that useful and should be weighed against the risk of biopsy.

The natural history of membranous LN is less clear. In early studies, the prognosis appeared far better than the prognosis of active proliferative disease.³⁷ Subsequent studies with longer follow-up suggested a worse outcome for membranous LN with persistent nephrotic syndrome.⁶¹ Even nonnephrotic patients with membranous LN may be at risk for sustained loss of kidney function.²⁰¹ Retrospective analyses show that 5-year renal survival rates largely depend on whether patients have pure membranous lesions (class V) or superimposed proliferative lesions in a focal (class III + V) or diffuse (IV + V) distribution.^{72,73} A U.S.-based study found that the 10-year survival rate from death or dialysis

was 72% for patients with pure membranous lesions but only 20% to 48% for those with superimposed proliferative lesions.⁷² A larger Canadian study described a similar good survival from ESKF of 92% over a median of 8 years for patients with only class 5 lesions; however, nearly 20% of patients developed a sustained 40% decline in eGFR.²⁰¹ Black race, younger age, elevated serum creatinine, higher degrees of proteinuria, hypertension, higher NIH chronicity index, and transformation to another WHO pattern all portended a worse outcome.²⁰¹⁻²⁰³ The poor survival in those of African descent with membranous LN may explain the excellent results in retrospective Italian studies, which largely follow Caucasian cohorts. One such Italian study found the 10-year survival rate of membranous LN patients to be 93%.⁷² Even in this Italian population, survival for pure membranous LN was far better than in patients with superimposed proliferative lesions. Thus at least in part, the variability of prognosis can be at least partially explained by the differences in racial background and histology.

DIALYSIS AND TRANSPLANTATION

The percentage of patients with severe LN who progress to dialysis or transplantation varies from 5% to 50% depending on the population studied, the length of follow-up, the response to therapy, and medication adherence.^{17,21,59,61,181,182,197,204} Many with slowly progressive renal failure have resolution of their extrarenal disease manifestations and serologic activity.^{205,206} With more prolonged time on dialysis, the prevalence of clinically active patients declines further, decreasing in one study from 55% at the onset of dialysis to <10% by the fifth year and 0% by the tenth year of dialysis.²⁰⁶ Patients with ESKF due to LN have increased mortality during the early months of dialysis, likely due to infectious complications of immunosuppressive therapy.^{205,206} Long-term survival of SLE patients on chronic hemodialysis or peritoneal dialysis is similar to that of nonlupus patients, with the most common cause of death being cardiovascular.²⁰⁵⁻²⁰⁷ There are no data to suggest one dialysis modality is favorable in patients with SLE and ESKF.

Most renal transplant programs suggest that patients with active SLE undergo a period of dialysis from 3 to 12 months to allow clinical and serologic disease activity to become quiescent before transplantation.²⁰⁶ This arbitrary time on dialysis before kidney transplant in clinically quiescent patients is controversial. The updated KDIGO 2024 LN guidelines suggest transplantation may be carried out as soon as disease is quiescent.²⁰⁸ Allograft survival rates in patients with LN are comparable with the nonlupus transplant population.^{21,59,207,209-212} The rate of recurrent SLE in the allograft has been low, <4% in most series,^{207,209-212} although in some reports a higher recurrence rate had been noted.²¹⁰ The prevalence of recurrent LN was only 2.4% in a 20-year study of nearly 7000 lupus transplant recipients and was more common in younger and female patients and those with African ancestry.²¹¹ When surveillance biopsies were used, however, recurrences could be detected in up to 54% of a small cohort of lupus transplant recipients, although this was mostly subclinical mild mesangial LN.²¹² The low rate of clinically important recurrence may be due, in part, to the immune suppressant action of the renal failure before transplantation and, as expected, the immunosuppressive antirejection regimens used following transplantation.

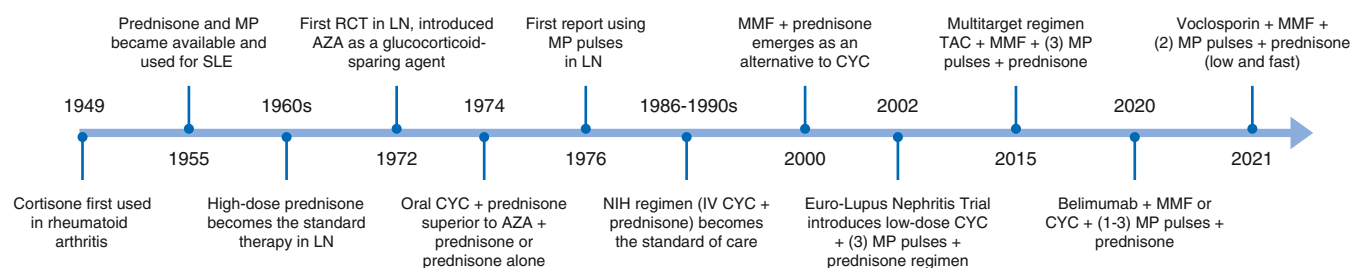


Fig. 31.8 Evolution of immunosuppressive therapies in lupus nephritis.²²³

Lupus patients with an APL antibody may benefit from anticoagulation therapy during the posttransplant period.^{213,214}

TREATMENT OF PROLIFERATIVE LUPUS NEPHRITIS

OVERVIEW

The arsenal of immune-related treatments for LN continues to grow, but the most effective and least toxic regimen for any given patient is not always clear (Fig. 31.8). Although cyclophosphamide is an effective therapy for many patients with LN, newer regimens have been developed in the hope of attaining equal or greater efficacy with less toxicity. The concept of more vigorous initial therapy during an “induction” treatment phase, followed by more prolonged lower-dose therapy during a “maintenance phase,” is the current paradigm of treatment.^{203,215,216} A caveat to this would be LN class VI where the risks of immunosuppressive treatment outweigh benefit. Hydroxychloroquine has been shown to increase rates of complete remission, lower the incidence of LN flares, and reduce progression to ESKD.²¹⁷ Given this, it is recommended that all classes of LN receive treatment with hydroxychloroquine.²¹⁶

Patients with ISN/RPS class I and II biopsies have an excellent renal prognosis and need no therapy directed to the kidney. Transformation to another histologic class is usually heralded by increasing proteinuria and activity of the urinary sediment, although class II itself may be associated with microhematuria. At this point in the patient’s clinical course, a repeat renal biopsy may serve as a guide to therapy.²¹

Patients with diffuse proliferative disease, ISN/RPS class III and IV lesions, require intensive and timely treatment to avoid irreversible renal damage and progression to ESKD.^{17,180,188} The ideal immunosuppressive regimen should be individualized and based on the patient’s prior therapy, risk and concern over potential side effects, compliance, and tolerability.

THERAPEUTIC AGENTS

Glucocorticoids

Glucocorticoids have potent antiinflammatory effects, mediated through both genomic and nongenomic pathways. Prednisone, despite the lack of controlled trials, is included in most treatment regimens for LN. Its use for LN was first described in the 1960s.²¹⁸ In retrospective studies, higher initial doses of corticosteroids appeared more effective than lower-dose therapy (<30 mg prednisone daily).²¹⁹

Traditional regimens use 1 mg/kg/day of prednisone, tapering after 4 to 6 weeks of treatment so that patients are receiving 30 mg/day or less by 3 months. Some clinicians start with daily pulses of IV methylprednisolone for 1 to 3 days followed by oral corticosteroids. More recently, lower initial doses of prednisone have been used after receiving pulse methylprednisolone therapy.²²⁰ Participants in the AURA-LV trial received initial prednisone doses of 0.4 mg/kg/day with a taper over 3 months to 2.5 mg/day. The complete renal response in this study for the placebo group was comparable with other studies in LN.^{8,221} Both the 2024 KDIGO and 2023 EULAR LN guidelines highlight the use of lower initial doses of prednisone when paired with pulse methylprednisolone. Importantly, the use of IV methylprednisolone has not been associated with osteonecrosis.²²²

Pulse therapy exerts rapid effects through nongenomic pathways.²²³ The decision to use pulse methylprednisolone should be considered within the clinical context of the presentation or flare. In a hospitalized patient with a severe flare and acute kidney injury (AKI), pulse steroid is an important initial treatment. However, in the setting of an outpatient with a normal GFR but modest hematuria and low-grade proteinuria where the kidney biopsy shows class III or IV LN, it is reasonable to commence oral corticosteroids rather than proceeding with pulse therapy. The exception would be where there are serious concerns about patient adherence to medications.

Appropriate counseling for patients receiving glucocorticoid is required given the number of adverse events that may be associated with its use, including but not limited to hypertension, diabetes, weight gain, cosmetic and dermatologic changes, increased risk of infections, psychiatric effects, adrenal suppression, and osteoporosis. Osteoporosis prophylaxis with vitamin D and calcium supplementation, along with bone mineral density scans in high-risk patients, is suggested.²²⁴

Cyclophosphamide

A series of landmark trials from the 1980s demonstrated the efficacy of cyclophosphamide for the treatment of LN.^{181,225} Cyclophosphamide is an alkylating agent that prevents cell division by decreasing DNA synthesis. Rapidly dividing cells, such as the immune cells in proliferative LN, are predominantly affected. Reports from the NIH and others have helped establish the role of cyclophosphamide in the treatment of proliferative LN.^{82,180-182,215,226,227} These studies demonstrated superior remission and relapse rates when comparing cyclophosphamide and corticosteroids against azathioprine or corticosteroids alone and formed

the standard of care for induction therapy of LN. Further trials refined dosing of IV cyclophosphamide from every 3 months to monthly.^{192,227} Thereafter, monthly IV cyclophosphamide for 6 months with glucocorticoids was the standard induction therapy of LN until the Euro-Lupus protocol was published.^{82,183,192,227} This landmark trial demonstrated similar efficacy with less toxicity using lower induction doses of cyclophosphamide. The Euro-Lupus Nephritis Trial, a multicenter prospective trial of 90 patients with severe LN, compared low-dose versus “conventional” high-dose intravenous cyclophosphamide.¹⁸⁸ Patients were randomized to either 6 monthly intravenous pulses of 0.5 to 1 g/m² cyclophosphamide, followed by two quarterly pulses, or only 500 mg intravenously every 2 weeks for a total of six doses (“Euro-Lupus regimen”), both followed by oral azathioprine as maintenance therapy. At 40 months’ follow-up, there were no statistically significant differences in treatment failures, renal remissions, or renal flares, but twice as many infections occurred in the high-dose group. This study is often cited as comparing low-dose to high-dose cyclophosphamide, but it should be recognized that the “low-dose cyclophosphamide” group received azathioprine for the bulk of the study. This regimen may also speak to the usefulness of azathioprine in the maintenance phase of LN. Although this trial may have included some patients with milder renal disease (mean creatinine, 1–1.3 mg/dL; mean proteinuria, 2.5–3.5 g/day for both groups)

and a predominantly Caucasian patient population, it supported the use of shorter-duration and lower total-dose cyclophosphamide for induction therapy. Longer follow-up of this population confirmed these data and suggests that early response to therapy is predictive of a good long-term outcome and that the long-term results are excellent.¹⁸⁸ Subsequent studies have investigated reduced-dose IV cyclophosphamide in those of Indian, African, and Latino ancestry.^{228,229} The 2024 KDIGO,²⁰⁸ 2024 ACR, and 2023 EULAR²¹⁶ guidelines all suggest that the Euro-Lupus protocol of cyclophosphamide be used when using cyclophosphamide as an induction agent.

Patients receiving cyclophosphamide require counseling regarding the adverse effects (Table 31.2). In most patients treated with intravenous cyclophosphamide, side effects such as hemorrhagic cystitis, alopecia, and neoplasms have been infrequent.^{17,21} Regarding infertility risk, women treated after the age of 30 are at higher risk.²³⁰ Total cumulative dose of cyclophosphamide is an independent risk factor for infertility. In women with breast cancer, cumulative doses of cyclophosphamide of 20 gm for women in their 20s, 10 gm for women in their 30s, and 5 gm in women in their 40s were associated with greater rates of premature ovarian failure.²³¹ Therefore the total 3 gm cumulative dose of cyclophosphamide in the Euro-Lupus protocol is unlikely to have significant impacts on ovarian reserve,²³² yet cumulative cyclophosphamide exposure may become a bigger concern for women with refractory/relapsing SLE or LN requiring retreatment with IV or oral cyclophosphamide. For males, infertility risks are low in prepubescent males if the cumulative doses of cyclophosphamide are kept under 30 gm.²³³ For sexually mature males, the risks of infertility with doses as low as 7.5 gm were 20%, rising to about 100% risk of infertility when reaching doses exceeding 30 gm.²³³ Men with LN receiving treatment with cyclophosphamide may be considered for long-term sperm preservation. With respect to risk of malignancy, cumulative doses of cyclophosphamide under 10 gm have only been associated with risk of squamous cell cancers but not solid organ malignancy.²³⁴ Cumulative doses of cyclophosphamide >36 gm have been associated with the greatest risk of solid organ malignancies (SIR 3.4).²³⁴ The dose of intravenous cyclophosphamide must be reduced for significant renal impairment and adjusted for some removal by hemodialysis. The leukopenic response to cyclophosphamide can be unpredictable in CKD, and even a reduced dose may result in prolonged and severe leukopenia. The cytoprotective agent mesna has been used successfully by some to reduce bladder complications from cyclophosphamide.

Table 31.2 Risks Associated With Cyclophosphamide Use and Mitigation Strategies

Side Effect	Mitigation/Monitoring Strategies
Bone marrow suppression	Regularly monitor complete blood counts (CBC) and adjust dosage as needed in response to leukopenia.
Hemorrhagic cystitis	Ensure adequate hydration and administer mesna to protect the bladder.
Infertility	Discuss fertility preservation options with patients before starting treatment. Ensure cumulative doses do not exceed 10 gm in men and women over the age of 30.
Secondary malignancies	Long-term monitoring for signs of secondary cancers. Ensure secondary screening up to date. Risks of malignancy low, aside from squamous cell cancers, when cumulative doses below 10 gm. Highest risk of solid organ cancers when cumulative doses exceed 36 gm.
Infections	Screen for chronic infections and provide appropriate immunizations (influenza, pneumococcal, SARS-CoV-2) before starting treatment. Consider if prophylactic antibiotics required and regular monitoring for signs of infection.

Mycophenolate Mofetil

MMF has proven to be an effective immunosuppressive in transplant patients and a variety of other immunologic renal diseases.²³⁵ It is a reversible inhibitor of inosine monophosphate dehydrogenase required for de novo purine synthesis and blocks B- and T-cell proliferation, inhibits antibody formation, and decreases expression of adhesion molecules, among other effects. MMF was shown to have good efficacy and reduced complications when compared with cyclophosphamide treatment regimens in a number of uncontrolled trials in LN.^{235,236} Treatment failures, relapses following therapy, discontinuation of

therapy, mortality, and time to remission have been similar when compared with cyclophosphamide. Longer follow-up at 4 years has shown MMF to have comparable efficacy to cyclophosphamide with no significant difference in complete or partial remissions, doubling of baseline creatinine, or relapses. Additionally, there have been lower rates of severe infections, leukopenia, or amenorrhea, and all deaths and renal failure in the MMF group.²³⁷ There has been a trend toward superior results (higher rates of remission and lower rates of ESKF) and improved tolerability of MMF in African Americans, as well as other ethnic groups, when compared with cyclophosphamide-based regimens.^{238,239} These results have been observed with up to 3 years of follow-up. In patients with reduced eGFR and crescentic LN (>50%), similar findings of equal efficacy and reduced adverse event profile have been observed when comparing MMF with cyclophosphamide-based regimens.^{240,241}

Calcineurin Inhibitors

The calcineurin inhibitors voclosporin and tacrolimus have been proven to increase the remission rate in a number of uncontrolled and controlled trials.²⁴² Calcineurin inhibitors reduce T-cell activation and are involved in podocyte cytoskeleton stabilization, both of which are thought to contribute to proteinuria and activity of LN.²⁴³ Tacrolimus has been successful in increasing remissions as part of a multidrug regimen for severe LN patients with combined ISN/RPS class IV and V lesions.²⁴⁴ Intravenous cyclophosphamide resulted in complete remission in 5% and partial remissions in 40% at 6 months versus a “multitargeted regimen” of tacrolimus, MMF, and corticosteroids, which led to a 50% complete and a 40% partial remission rate in this period. In a large multicenter trial from China containing more than 350 patients, multitargeted therapy with a CNI added to mycophenolate and corticosteroids proved superior to cyclophosphamide and corticosteroid induction therapy.²⁴⁵ An extension trial of the 200 patients of this study population who responded initially to this therapy compared maintenance with the multitarget therapy with maintenance with azathioprine for the cyclophosphamide group. Both groups had similar low renal relapse rates (5% and 7%, respectively). Moreover, serum creatinine and GFR remained stable in both groups. Of note, the azathioprine group had more adverse events (44% vs. 16%) and more withdrawals due to adverse events.²⁴⁶

Voclosporin

More recently, data from the phase 3 clinical trial investigating voclosporin in combination with MMF demonstrate improved renal remission rates compared with MMF alone. The Aurinia Renal Response in Active Lupus with Voclosporin (AURORA 1) study investigated whether voclosporin versus placebo when added to MMF and corticosteroids improved complete renal remission at 52 weeks. Complete renal remission was defined as a composite of UPCr ≤ 0.5 mg/mmol, eGFR >60 mL/min or $<20\%$ decrease from baseline, no rescue medication, and achieving prednisone reduction targets near the end of the trial.²⁴⁷ More patients in the voclosporin group achieved the composite renal outcome versus placebo (41% vs. 23%, OR 2.65, 95% CI 1.64–4.27, $P < 0.0001$). For the phase 2

study Aurinia Urinary Protein Reduction Active Lupus with Voclosporin (AURORA-LV) investigating voclosporin versus placebo in patients with class 3 to 5 LN receiving MMF and corticosteroids, there was a significantly increased number of adverse events in the voclosporin group including more deaths.²²⁰ However, this was designated a center effect. In the phase 3 study the serious adverse events were comparable between the voclosporin and placebo groups, with the event rate being approximately 21% in both groups. A pooled analysis of both phase 2 and phase 3 studies for voclosporin suggests no major safety signals, as well as efficacy across racial and ethnic subgroups.²⁴⁸ Compared with cyclosporine and tacrolimus, voclosporin has a more consistent pharmacokinetic-pharmacodynamic profile and drug monitoring may not be required.²⁴⁹ Voclosporin has not been investigated in those with eGFR <45 mL/min or in combination therapy with cyclophosphamide. Long-term safety and efficacy data remain under investigation. Although cyclosporine and tacrolimus have similar drug profiles, cyclosporine may reduce mycophenolic acid exposure by as much as 40% and therefore voclosporin or tacrolimus may be the preferred CNI when coprescribing with mycophenolic acid treatment of LN.²⁵⁰ It is unclear whether the predominant effects of CNIs are through stabilization of the podocyte cytoskeleton, reduction of T-cell activity in LN, or both. The frequent observation of relapse of proteinuria with discontinuation of CNIs suggests a nonimmunologic mechanism for the reduction of proteinuria. Studies designed to examine reduction in proteinuria as an endpoint may be reported as positive with the use of CNIs, but it is unclear whether they successfully treat LN immunologically. Furthermore, in many jurisdictions, voclosporin is costly and not feasible in the absence of insurance or government coverage.

Belimumab

In 2021, belimumab received U.S. Food and Drug Administration (FDA) approval for an induction regimen add-on to cyclophosphamide or mycophenolate for the treatment of LN. Belimumab is a monoclonal antibody against BAFF, also known as B Lymphocyte Stimulator (BLyS), which impacts B-cell survival and development of autoreactive B cells.²⁵¹ Serum BAFF is elevated in lupus and belimumab was already approved for extrarenal lupus.²⁵² The Belimumab International Study in Lupus Nephritis (BLISS-LN) trial evaluated the efficacy and safety of belimumab plus standard therapy (mycophenolate or cyclophosphamide-azathioprine) in patients with LN.²⁵³ In this phase 3 multicenter study of class III or IV LN (with or without a class V lesion), patients were treated with standard-of-care therapy and randomized to receive intravenous belimumab (10 mg/kg) or placebo for 104 weeks. The primary outcome was a composite of UPCr <0.7 mg/mmol, eGFR within 20% of baseline or >60 mL/min, and no use of rescue therapy (labeled primary efficacy renal response). More patients in the belimumab group than placebo group achieved the primary efficacy renal response (43% vs. 32%, odds ratio 1.6, $P = 0.03$). It is worth highlighting that there was no difference in complete renal response rates in the cyclophosphamide-treated group, and although the rate of primary efficacy renal response trended toward favoring belimumab in this subgroup, this finding was not

statistically significant. Further investigation is required to see if belimumab works better in combination with MMF than cyclophosphamide.

In a post hoc analysis of the BLISS-LN trial, the rate of LN flare at any time point with belimumab relative to placebo was reduced by 59% (HR 0.41, CI 0.23–0.73, $P = 0.002$).²⁵⁴ Importantly, it was noted that patients with proteinuria >3 g/g or a pure membranous LN class 5 did not have a better renal response when compared with placebo. Although renal clearance of belimumab is increased when there is a greater degree of proteinuria, the exposure response analysis in BLISS-LN did not suggest that this was the reason for lack of treatment effect in these patients.²⁵⁵ Additional post hoc subgroup analyses suggest that relapsed patients, versus those that were newly diagnosed, who were treated with belimumab and standard of care therapy had greater odds of having a complete renal response or a primary efficacy renal response.²⁵⁶ Additionally, they concluded that initial glucocorticoid pulses did not impact complete renal response or primary efficacy renal response rates for both newly diagnosed and relapsed patients.

Rituximab

Other therapies remain available for LN and are generally reserved for aggressive cases. Rituximab, a chimeric monoclonal antibody, depletes CD20 B cells through multiple mechanisms including complement-dependent cell lysis; Fc γ -dependent, antibody-dependent, cell-mediated cytotoxicity; and induction of apoptosis. It is approved by the FDA for the treatment of rheumatoid arthritis (RA), granulomatosis with polyangiitis (GPA; formerly designated Wegener granulomatosis), and microscopic polyangiitis (MPA). It has been used with varying success in many other immunologic and autoimmune diseases, including a variety of primary glomerular diseases.²⁶⁶ It has been used in more than 300 LN patients, mostly in case reports and open-label uncontrolled trials.^{202,203} However, two large randomized controlled trials have given disappointing results. In one trial of 257 SLE patients without severe renal disease, patients were randomized to receive rituximab or placebo.²⁶⁷ Although subgroup analyses suggested a beneficial effect in the African-American and Hispanic subgroups, there were no significant overall differences between the placebo and the rituximab arms of therapy. In the Lupus Nephritis Assessment with Rituximab (LUNAR) trial, 140 patients with class III and IV LN were randomized to rituximab or placebo in addition to an induction regimen of MMF (goal 3 g/day) and tapering corticosteroids.²²¹ Although the rituximab group had a greater fall in anti-dsDNA antibody titers and rise in serum complement levels, there was no statistically significant difference in the primary renal response between treatment groups at 1 year.²⁶⁸ At present therefore rituximab is not a first-line agent for induction therapy of most patients with severe LN. It continues to be used in patients resistant to other treatments and in those who do not tolerate conventional treatment.^{269,270} A study of the use of rituximab and MMF in 50 LN patients without the use of oral corticosteroids (the patients received intravenous corticosteroids at the time of rituximab infusion) reported excellent results for complete and partial remission.²⁷¹

Approach to combining therapies

Given the updates with the EULAR and KDIGO 2024 guidelines and the additions of calcineurin inhibitors and belimumab to induction regimens for treatment of proliferative LN combination therapies are becoming an attractive therapeutic option, although who will benefit most may not yet be entirely clear. In addition, the optimal timing of add-on therapies to either MMF or cyclophosphamide with prednisone to counterbalance the adverse effect profiles of these medications is also not yet clear. The ACR has released a summary of its 2024 guidelines on the screening, treatment, and management of LN, in which they advocate all patients receive triple therapy with either CNi or belimumab. Another approach may be to assess clinical risk factors, such as race, eGFR, and relapsing disease before deciding if upfront multitarget therapy should be used. Additionally, as data emerge regarding worse renal outcomes with repeat flares of LN, the focus of treatment may shift from solely obtaining a complete renal remission to avoiding renal flares (Fig. 31.9).²⁵⁷ Drug cost coverage and patient preference are important issues that will factor into the decision regarding multitarget therapy at the outset. The post hoc analysis of the BLISS-LN trial suggests that belimumab in combination with MMF or cyclophosphamide with prednisone may not confer additional benefit for nephrotic patients.²⁵⁴ The EULAR guidelines suggest if a 25% reduction in urine protein is not seen at 3 months with initial therapy with MMF or cyclophosphamide and prednisone, then the additional therapies may be considered. Future studies on optimal timing of initiation of CNi or belimumab with standard of care therapies are needed.

Maintenance therapies

A number of studies have focused on the optimal maintenance therapy for LN with the goal of avoiding relapse and flares while minimizing the long-term immunosuppressive toxicity. Earlier studies, with good representation of patients with African and Hispanic ancestry, demonstrated greater toxicity and higher rates of ESKF and relapses when continuing cyclophosphamide as maintenance treatment when compared with MMF or azathioprine.²⁵⁸ The results of two large, randomized trials further delineate the role of these oral agents in the maintenance of patients with proliferative LN.^{259,260} In the European MAINTAIN trial, 105 patients were randomized to either azathioprine or MMF for at least 3 years of maintenance (mean, 53 months).²⁵⁹ There was no difference between these medications in the time of renal flares or renal remission. In the worldwide Aspreva Lupus Management Study (ALMS) maintenance trial, 227 patients who achieved remission after induction therapy with either intravenous cyclophosphamide or MMF were rerandomized in double-blind fashion to either MMF or azathioprine maintenance for 3 years.²⁶⁰ MMF proved superior to azathioprine with respect to the primary endpoint of time to treatment failure (death, ESKF, doubling of serum creatinine, LN flare, or requirement for rescue therapy).²⁶⁰ Differences between the two studies likely explain the differing results. The MAINTAIN trial was prerandomized from day 1, included patients who were largely Caucasian, and used the endpoint of renal flare because few patients in this population progress to renal failure. Even so, there

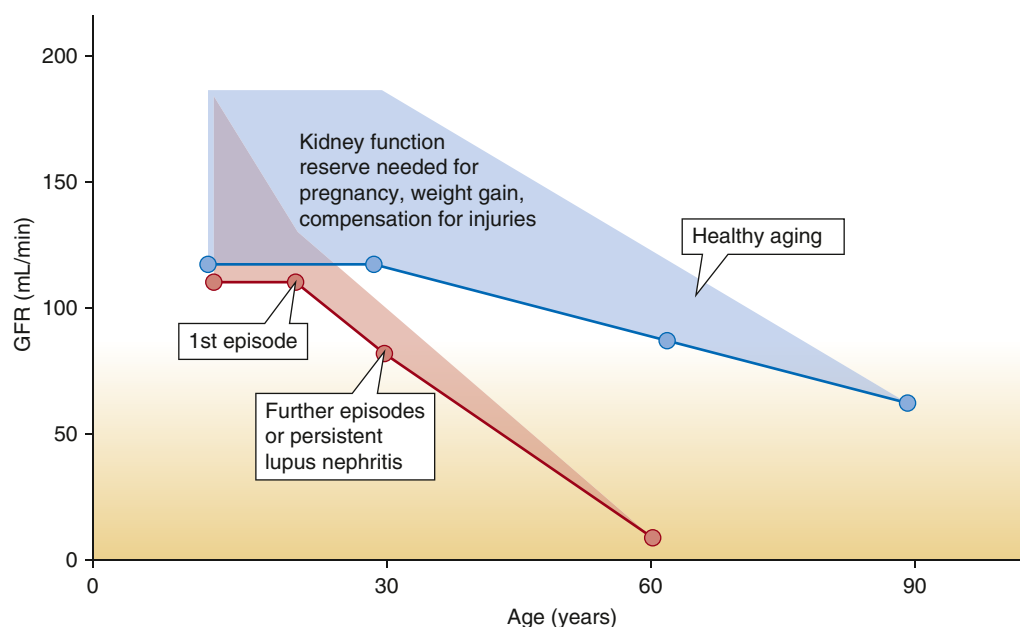


Fig. 31.9 Kidney function decline in healthy adults versus those with lupus nephritis.³⁹

were 26% flares in the azathioprine group compared with 19% in the MMF group, although this difference was not statistically significant and perhaps not clinically meaningful. The ALMS maintenance trial included only those patients who achieved remission after induction; was international, including multiracial and diverse populations; and used harder endpoints for response (doubling creatinine, ESKF, etc.).

At present, the 2024 KDIGO guidelines²⁰⁸ recommend MMF as maintenance therapy, and the 2023 EULAR guidelines²¹⁶ suggest MMF or azathioprine. Azathioprine is a better choice for maintenance therapy in those who are planning pregnancy or likely to become pregnant and those who have gastrointestinal problems with MMF. Total duration of immunosuppression (induction and maintenance) is suggested for a minimum of 36 months, or 3 years.²⁶¹ Additionally, there are data to support histologic features guiding withdrawal of maintenance immunosuppression for patients in remission.²⁶² Malvar and colleagues²⁶² repeated kidney biopsies for patients with LN on treatment for at least 42 months and in remission for at least 12 months. Maintenance immunosuppression was withdrawn if the activity index was zero. The LN flare rate was 1 per 50 patient-years, which is significantly less than previous reported rates.²⁶³⁻²⁶⁵ Importantly, this study highlights that a repeat biopsy may be helpful in the decision to prolong immunosuppression if there are features of histologic activity, but not withdrawing it in any event before the recommended minimum duration of 36 months.

MEMBRANOUS LUPUS NEPHROPATHY

For patients with class V membranous LN, there have been conflicting data regarding the course, prognosis, and response to treatment.²⁰³ It is reasonable to assume that membranous LN should be treated similarly to other causes of nephrotic syndrome with antiproteinuric and lipid-lowering measures. The long-term renal benefit of acute

initiation of RAS inhibitors is unclear, as the priority of initial management is immune therapy. The possible adverse effects of treatments with consideration of the number of different medications being prescribed to a young cohort not used to taking pills should be kept in mind. The use of anticoagulation to prevent thromboembolic events in patients with severe nephrotic syndrome should be individualized.

There is disagreement about which patients with class V membranous LN should be treated with immunosuppressive therapy. The degree of superimposed proliferative lesions greatly influences outcomes in class V patients, and it is unclear if older trials included only pure membranous LN patients. Thus early trials reported low and inconsistent response rates with oral corticosteroids.⁷¹ Excellent long-term results with intensive immunosuppressive regimens from Italian studies and others raise questions of whether the results are related to the therapeutic intervention, the population studied, or better supportive treatments.⁷³ A retrospective Italian trial found better remission rates with a regimen of chlorambucil and methylprednisolone than corticosteroids alone.²⁷² In a small nonrandomized trial of cyclosporine in membranous LN, there was an excellent remission rate of nephrotic syndrome with mean proteinuria decreasing from 6 to 1 or 2 g/day by 6 months.⁸³ At long-term follow-up and rebiopsy, there was no evidence of cyclosporine-induced renal damage, but two patients had developed superimposed proliferative lesions over time. An NIH trial of 42 nephrotic patients with membranous LN compared cyclosporine, prednisone (alternate-day), and intravenous cyclophosphamide and found superior remission rates for the cyclosporine and cyclophosphamide regimens, but more relapses when the cyclosporine was withdrawn.¹³⁰ This tendency to relapse suggests that calcineurin inhibitors may suppress proteinuria principally via nonimmunologic mechanisms. Although cyclosporine has been investigated more extensively than tacrolimus in lupus membranous nephropathy, given the similar mechanism of action, and

that tacrolimus has been studied in multitarget therapy of proliferative LN, it may be reasonable to use tacrolimus for treatment of those with class V LN and heavy nephrotic syndrome to target reduction in proteinuria. This may be particularly relevant if there is concern about cosmetic or metabolic side effects of cyclosporine.

Both azathioprine and mycophenolate have been investigated in lupus membranous nephropathy. A study of 38 patients with pure membranous LN evaluated long-term treatment with prednisone plus azathioprine.²⁷³ At 12 months 67% of the patients had experienced a complete remission and 22% a partial remission. At 3 years only 12% had relapsed, at 5 years only 16%, and at 90 months only 19% relapsed. At the end of follow-up, no patient had doubled serum creatinine. Clearly in this population, a regimen of steroids plus azathioprine was highly effective. The response of patients with membranous LN to MMF has been varied.²⁷⁴⁻²⁷⁶ There were 84 patients with pure ISN class V membranous LN among the 510 patients enrolled in two similarly designed randomized controlled trials comparing MMF and intravenous cyclophosphamide induction therapy.²⁷⁶ Rates of remissions, relapse, and course were similar in both treatment groups (and also may suggest that it was the corticosteroid alone that was responsible for the treatment effect). Thus MMF can also be considered a first-line therapy for certain patients with membranous LN.

Given limited data, the treatment of membranous LN should be individualized.²⁰³ Patients with pure membranous LN and a good renal prognosis (subnephrotic levels of proteinuria and preserved GFR) can be managed expectantly. For those at higher risk of progressive disease (African descent, those fully nephrotic), options include cyclosporine, tacrolimus, monthly intravenous pulses of cyclophosphamide, MMF, or azathioprine plus corticosteroids. There are reports of patients responding to rituximab as well. Patients with mixed membranous and proliferative LN are treated in the same way as those with proliferative disease alone, although we have seen patients respond to immunosuppressive therapy for proliferative LN and in parallel develop de novo membranous lupus nephropathy.²⁷⁷

Importantly, the time needed for resolution of proteinuria can often be months to years with the current therapies.^{278,279} There may be “residual proteinuria” that can take years to resolve, if at all. If all other indices are favorable, it may put the patient at risk to intensify or continue immunosuppression in attempting to eradicate proteinuria. There may be a greater role in optimizing nonimmune therapies, such as RAS inhibitors, at this stage.

NONIMMUNOLOGIC THERAPIES

CARDIOVASCULAR RISK

As effective and safer therapies for LN have evolved, greater attention has been directed to other causes of morbidity and mortality in the SLE population. Lupus patients have accelerated atherogenesis and a disproportionate rate of atypical coronary vascular disease, leading to a high mortality rate.²⁸⁰ The high cardiovascular risk rate has been attributed to concurrent hypertension, hyperlipidemia, nephrotic syndrome, prolonged corticosteroid use, APL syndrome (APS), and, in some, the added vascular risks of systemic

inflammation and chronic kidney disease (CKD).^{281,282} Despite limited data on therapeutic interventions in this population, aggressive management of modifiable cardiovascular risk factors may alter the morbidity and mortality of this population. Recognizing that there are few directed studies in the population with lupus nephritis, but extrapolating from other proteinuric CKD populations, closely monitored blood pressure control (<130/80 mm Hg), the use of angiotensin-converting enzyme (ACE) inhibitors and/or angiotensin receptor blockers, and correction of dyslipidemia with statins are all reasonable in LN patients, taking into account number of medications and the risk of nonadherence. Another source of caution in this approach is the teratogenicity of RAS inhibitors. In addition, calcium, vitamin D supplements, and bisphosphonates to prevent glucocorticoid-induced osteoporosis may be useful.²²⁴ Annual eye examinations should be performed for patients on hydroxychloroquine.

In murine models of LN, the use of SGLT2 inhibitors has been associated with decreases in proteinuria and stable eGFR with evidence of attenuated damage on renal pathology.²⁸³ Studies in humans have not been performed on a large scale.^{284,285} There remain concerns about infectious complications with SGLT2 inhibitors and immunosuppression for LN and potentially masking immune-mediated activity via proteinuria reduction. Other nephroprotective medications such as finerenone, glucagon-like peptide one receptor agonists (GLP-1RA), and endothelin receptor antagonists (ERAs) have not been investigated in SLE or LN.²⁸⁶

INFECTION RISK

Pneumocystis jirovecii pneumonia (PJP) prophylaxis has traditionally been considered when prescribing cyclophosphamide, and coprescribing with glucocorticoids has become more common in patients with glomerular diseases after the results of a study involving patients with IgA nephropathy.²⁸⁷ However, this risk may have been driven by ethnicity and age, relative to patients with LN. The literature has generally described low rates of PJP pneumonia in SLE patients.^{288,289} Additionally, SLE patients have experienced adverse effects of trimethoprim-sulfamethoxazole such as neutropenia, hyperkalemia, nephropathy, and other risks.^{288,290} There is no consensus regarding prophylaxis in SLE patients, and larger studies are needed to investigate the benefit/risk ratio of PJP pneumonia prophylaxis prescribing in patients with SLE.²⁹¹ Until then, it may be reasonable to consider prophylaxis in those with additional underlying respiratory comorbidities, increased local endemic rates of PJP pneumonia, and prolonged courses of intensive immunosuppression.

THROMBOSIS RISK

Some form of APL antibodies is present in 40% to 75% of lupus patients.²⁹²⁻²⁹⁴ Because most do not experience thrombotic complications, they require no special treatment. However, some would recommend low-dose aspirin and hydroxychloroquine for prophylaxis of asymptomatic patients with APL antibodies. In patients with evidence of a clinical thrombotic event, most investigators use chronic anticoagulation with warfarin as long as the antibody persists. While the standard practice has been to not anticoagulate

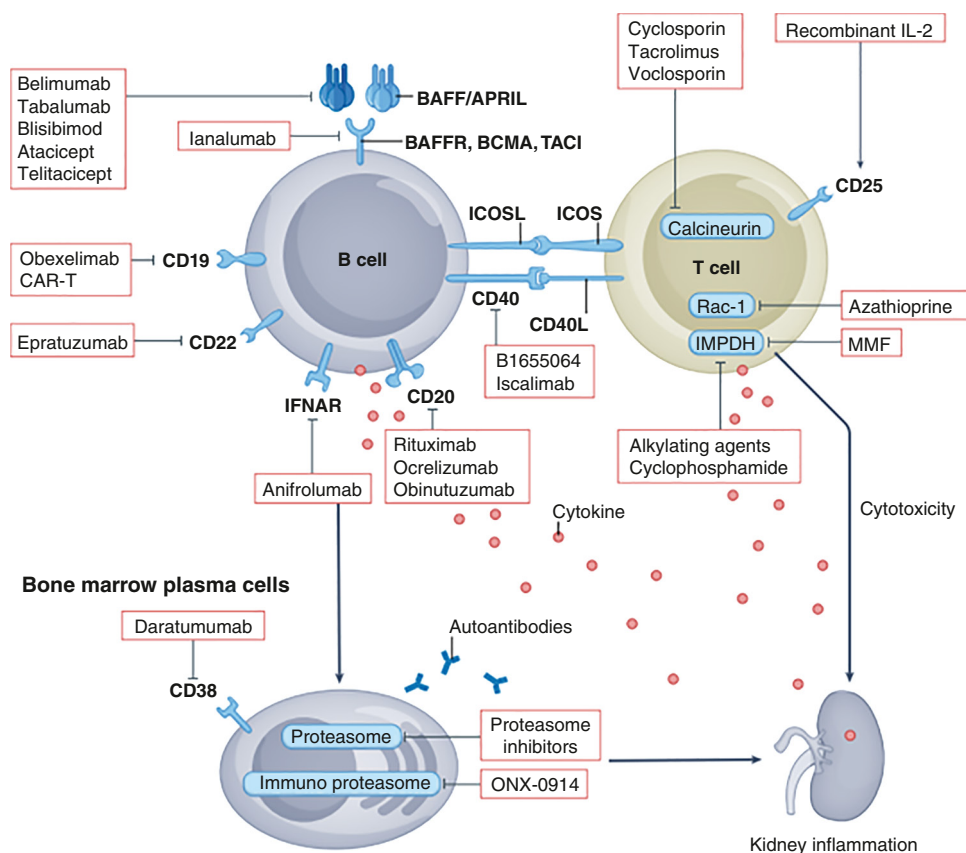


Fig. 31.10 Current targets of investigational therapies in lupus nephritis.³⁹ B cells are precursors for plasma cells. B cells present autoantigens to T cells and hence drive the expansion of autoantigen-specific T cell clones. For this reason, drug developers primarily invest in B-cell targeting therapies to minimize autoimmune activity in lupus nephritis, which is mediated, among others, by autoantibodies, cytokines and T cell cytotoxicity.

other patients, in one recent series of more than 100 SLE patients, more than one-fourth had APL antibodies, of whom almost 80% had a thrombotic event. The antibody-positive patients also had a greater incidence of chronic renal failure than the antibody-negative patients.²⁹⁴ (See discussion in “Antiphospholipid Syndrome” later.)

THERAPIES UNDER INVESTIGATION

The majority of therapies currently under investigation target B cells in LN (Fig. 31.10). Obinutuzumab, an FDA-approved fully humanized anti-CD20 monoclonal antibody, has shown promising results for their phase 2 clinical trial in proliferative LN.²⁹⁵ One-hundred and twenty-five patients with proliferative LN receiving MMF and prednisone were randomized to receive obinutuzumab or placebo on day 1 and weeks 2, 24, and 26 and followed through week 104. The primary endpoint was complete renal remission (CRR) at weeks 52 and 104. The CRR was greater for obinutuzumab than placebo at weeks 52 (35% vs. 23%, $P = 0.115$) and 104 (41% vs. 23%, $P = 0.26$). Although there were infusion-related reactions (nonsevere), obinutuzumab was not associated with severe adverse events. Compared with rituximab in the LUNAR study, obinutuzumab resulted in a faster, more uniform, and greater degree of B-cell depletion, which may account for the impact seen. The number of autoreactive B-, T-, and plasma cell clones in LN compared with other glomerular diseases may explain why obinutuzumab yielded promising results in LN, whereas rituximab did not.⁴¹

The results of the phase 3 study of obinutuzumab for patients with LN have been released.²⁹⁶ Patients were

randomized to receive obinutuzumab in one of two dose schedules (1000 mg on day 1 and at weeks 2, 24, 26, and 52, with or without a dose at week 50). The CRR was again greater for the obinutuzumab group than placebo at week 76 (42.7% vs. 30.9%, $P = 0.05$). There were more serious adverse effects in the obinutuzumab group including COVID-19 pneumonia and neutropenia. The authors highlight that most of the SARS-CoV-2 infections were early in the pandemic before vaccines were available, and no serious infections were observed in the latter half of the study once vaccination programs were commenced. Nonetheless, this increased risk of COVID-19 pneumonia when obinutuzumab was added to mycophenolate mofetil must be considered in the treatment pathway and overall immunosuppression burden.

Ocrelizumab, another fully humanized anti-CD20 monoclonal antibody, FDA approved for severe multiple sclerosis, has the advantage of avoiding first-dose infusion reactions and the development of human antichimeric antibodies that are potential problems with rituximab therapy.²⁹⁷ However, a controlled randomized trial using this agent in patients with LN was terminated early due to adverse events.

Atacicept, a soluble fully humanized recombinant fusion protein that inhibits B-cell stimulating factor (BLISS) and a proliferation-inducing ligand (APRIL), failed in initial trials with patients with LN.²⁹⁸

Epratuzumab, a humanized monoclonal antibody against CD22, a marker of mature B cells but not plasma cells, is currently being studied. IFN- α is an antiviral cytokine induced in SLE patients that may lead to inflammation, autoimmunity, and renal damage.

Anifrolumab, a monoclonal antibody against the IFN- α type I receptor, has been shown in a trial of more than 300 SLE patients to add to the efficacy of standard therapy.²⁹⁹

Chimeric antigen receptor (CAR)-T cell therapy is a promising therapy for SLE. Mackensen and colleagues³⁰⁰ conducted a phase I study of 5 patients with refractory SLE who received CAR-T cell therapy. Autologous T cells were transduced with a lentiviral anti-CD19 CAR vector and reinfused after lymphodepletion with fludarabine and cyclophosphamide. This resulted in a depletion of B-cells, improvement in clinical symptoms, and normalization of laboratory values. All five patients achieved remission by 3 months. This resulted in a decrease in the median SLEDAI (Systemic Lupus Erythematosus Disease Activity Index) score from 16 to 0. Treatment was well tolerated with only a mild cytokine release syndrome. Follow-up did not show any relapses of SLE while still being off any SLE-associated medications. A 24-month update of these 5 patients, with an additional 3 patients, was released.³⁰¹ This update demonstrated a sustained reduction in proteinuria. Five of the 8 patients developed grade 1 cytokine release syndrome, for which 3 patients received tocilizumab. Given these promising results, other CAR-T cell therapies are in clinical trials for patients with SLE.³⁰² Given that cytokine release syndrome can be fatal, larger-scale studies are needed to confirm the safety in addition to efficacy. These results also indicate, as mentioned earlier, that targeting B cells appears to be a promising therapy in LN. Numerous other therapies targeting B cells are under investigation for LN.^{303,304}

T-lymphocyte activation requires two signals.^{266,305} The first occurs when the antigen is presented to the T-cell receptor via major histocompatibility complex (MHC) class II molecules on antigen-presenting cells. The second mechanism of activation is by the interaction of costimulatory molecules on T lymphocytes and antigen-presenting cells. Disruption of costimulatory signals interrupts the (auto)immune response. Two clinical trials using different humanized anti-CD40L monoclonal antibodies in LN patients to block B and T-cell costimulation have not been successful.^{306,307} Another costimulatory pathway is mediated through the interaction of CD28 with CD80/86. CTLA4 Ig, abatacept, a fusion molecule that combines the extracellular domain of human CTLA4 with the constant region (Fc) of the human IgG1 heavy chain, interrupts the interaction between CD28 and CD80/86. CTLA-4 is similar to the T-cell costimulatory protein CD28, and both bind to CD80 and CD86, also known as B7-1 and B7-2, on antigen-presenting cells. It is FDA approved for the treatment of RA. Two major randomized controlled trials with abatacept in patients with severe LN treated concurrently with intravenous cyclophosphamide and steroids were negative.^{21,308,309}

EVALUATION OF TREATMENT RESPONSE

Definitions of remission differ among clinical studies and published guidelines. KDIGO defines a “complete remission” as reduction in urine PCR (from 24-hour urine collection) to <442 mg/g (50 mg/mmol) and stabilization of kidney function within 10% to 15% of baseline within 6 to 12 months of initiating treatment.³¹⁰ A novel treatment response definition added to the 2024 KDIGO update is

the criterion of “primary efficacy renal response,” which is defined by a UPCr <0.7 g/g (70 mg/mmol), eGFR that is no worse than 20% below the preflare value or ≥ 60 mL/min/1.73 m², and no use of rescue therapy for treatment failure.²⁰⁸ This proteinuria remission target of <0.7 g/g has also been described as a good predictor of long-term kidney outcomes in other glomerular diseases.³¹¹ Joint statements from the EULAR, European Renal Agency and European Dialysis and Transplant Association (ERA/EDTA) define a “complete clinical response” as proteinuria reduction to <0.5 to 0.7 g/day by 12 months.³¹² In addition, they suggest that improvement in proteinuria should be noted by 3 months, with at least a 50% reduction by 6 months. Given that proteinuria may take months to years to improve,²⁷⁸ assessments of other biomarkers and clinical symptoms factor into the treatment response during this time. Resolution of microscopic hematuria has only variably been included in definitions of treatment response, while remission of microhematuria likely correlates with reduction in inflammation, its association with improved long-term kidney outcomes has not been demonstrated in LN.¹²²

Persistent proteinuria after treatment induction and maintenance may be due to ongoing glomerular inflammation or glomerular scarring. Although serology and symptoms of SLE may aid in this assessment, there may be uncertainty when proteinuria is discordant to serology and symptoms. The role of repeat kidney biopsies remains controversial, particularly given the described discrepancy with clinical response for some patients.¹⁹⁹ A histologic activity index may take months to years to resolve, while chronic damage accrues with recurrent episodes of LN.⁷⁶ Data continue to emerge regarding the optimal timing of repeat kidney biopsies. Therefore there may be a greater utility of repeat biopsies when there hasn't been a clinical remission, there is concern about ongoing immunologic activity, and sufficient time has passed to allow resolution of histologic activity parameters after treatment was initiated.¹²² Ongoing studies will hopefully shed light on the sequence of immunologic, clinical, and histologic remission as has been characterized in other glomerular diseases, to allow either intensification or deescalation of immunosuppressive therapies.

In patients not able to achieve remission, verifying adherence remains pivotal, as outlined by the KDIGO guidelines. Nonadherence is typically defined by <80% of the prescribed doses taken by patients. Measurement of nonadherence may be challenging with self-administered questionnaires or even clinician assessments, and therefore more objective measures such as drug levels or drug dispensing records may be needed.³¹³ Patients with SLE and LN have high rates of nonadherence, which in turn leads to worse renal outcomes.³¹⁴⁻³¹⁷ Nonadherence to hydroxychloroquine has been associated with SLE flares and mortality.³¹⁸ If available, monitoring of hydroxychloroquine levels can serve as a measure of nonadherence. Many factors contribute to barriers to adherence, comprising those that are patient related, health care related, medication related, and disease related.^{319,320} Patient priorities and self-reported symptoms being perceived as less important by physicians have been identified as a recurring barrier.³²¹ Numerous interventions for improving adherence have been proposed

including improved communication among health care professionals and patients, patient education, regularly addressing adherence, involvement of caregivers, and well-coordinated transition from pediatric to adult care.³¹⁹ Specifically pertaining to communication and education, patients would like more weight and discussion given to self-reported symptoms and quality of life in medication decisions.³²¹ This counseling can be done in the context of the disease with appropriate and clear explanation of risks versus benefits of treatment. When performed regularly and with involvement of caregivers, this approach may improve rates of nonadherence. Many patients with lupus may not be aware that it is a potentially fatal disease and stop taking their medications at the first sign of a side effect, such as acne or hair growth. Education of patients about the seriousness of their illness to encourage adherence to their immunosuppression is important. Another way to try to improve adherence is to be mindful of the number of medications prescribed, particularly when patients have identified this to be an important issue. While it is reasonable to prescribe lipid-lowering medicine, RAS inhibition to lower proteinuria, and now the consideration of SGLT2 inhibitors, these maneuvers, unproven in lupus nephritis, must be weighed against the total burden of medications prescribed to these typically young patients and may be better reserved outside the induction immunosuppression window.

ANTIPHOSPHOLIPID SYNDROME

Antiphospholipid syndrome (APS) is a systemic autoimmune disease defined by thrombotic and obstetrical complications occurring in patients with persistent antiphospholipid antibodies. The Sydney criteria for APS, which were formulated in 2006, required one clinical criterion of a thrombosis or pregnancy-related complication, with at least one positive antiphospholipid antibody detectable on two occasions at least 12 weeks apart.³²² In 2023, the ACR and EULAR refined the APS classification criteria by allowing at least one positive antiphospholipid antibody (aPL) test within 3 years of identification of a clinical event; the clinical scoring based criteria are clustered into 6 clinical domains (macrovascular venous thromboembolism, macrovascular arterial thrombosis, microvascular, obstetric, cardiac valve, and hematologic) (Fig. 31.11).³²³ This now includes the previously labeled “noncriteria APS manifestations,” namely thrombocytopenia, heart valve disease, cutaneous, renal, and neurologic manifestations. The laboratory parameters are now classified under two domains: functional coagulation assays for the lupus anticoagulant and solid-phase enzyme-linked immunosorbent assays for IgG/IgM anticardiolipin and/or IgG/IgM anti-B2 glycoprotein I antibodies. Patients accumulating at least 3 points from the clinical and laboratory domains are classified as having APS. This new APS criterion has a specificity of 99% versus 86% for the Sapporo classification and a sensitivity of 84% versus the previous 99% (Fig. 31.11).³²³ Given the lower sensitivity with the newer classification, using these as diagnostic criteria in clinical practice may lead to inappropriately not anticoagulating patients who remain at risk for thrombotic events.³²⁴

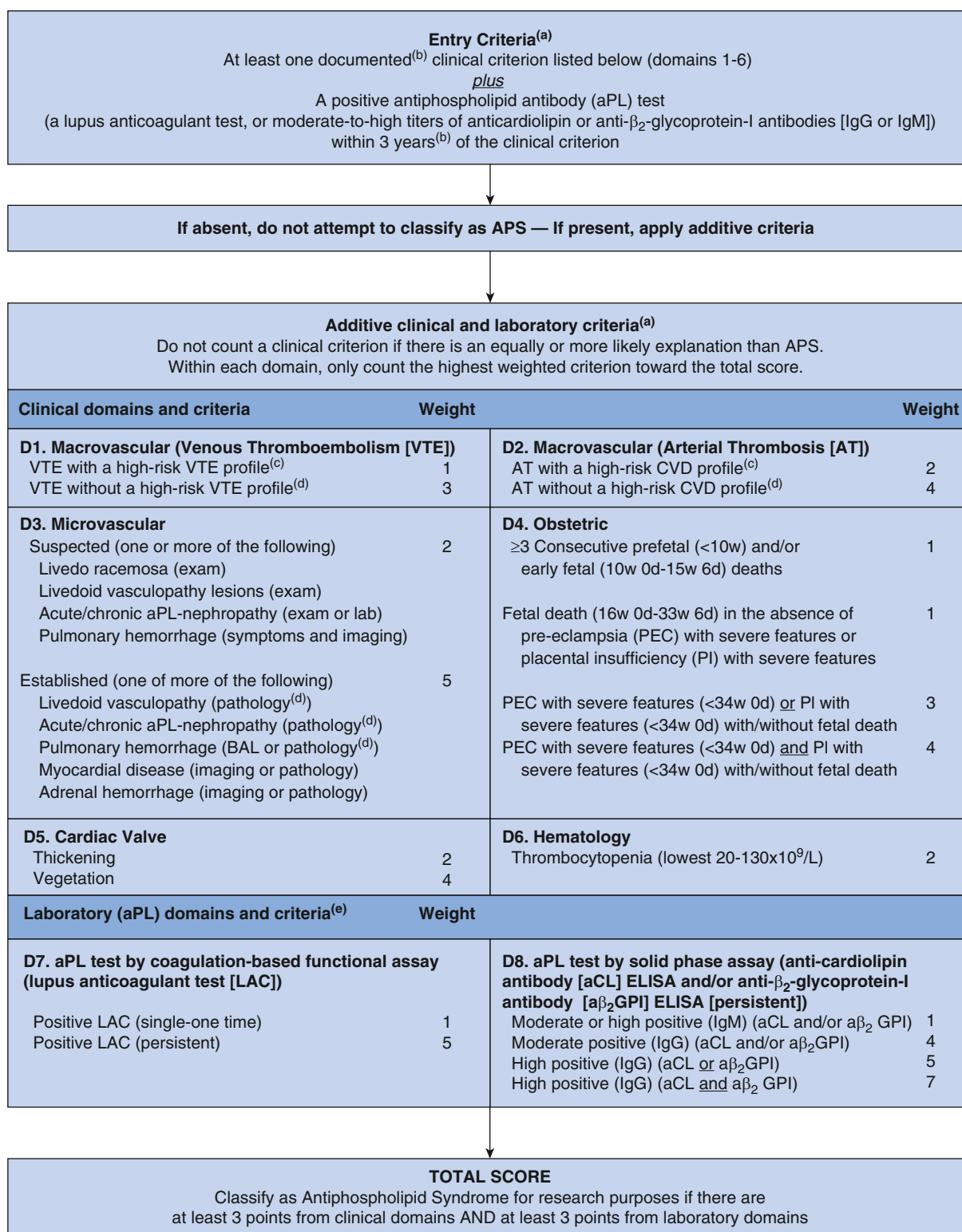
APS may be associated with glomerular disease, small and large vessel renal involvement, and coagulation problems in dialysis and renal transplant patients.^{292-294,325} Patients with APS have autoantibodies directed against plasma proteins bound to phospholipids. The antibodies include IgG and/or IgM anticardiolipin antibodies, antibodies to β_2 -glycoprotein I of the IgG or IgM isotype, and lupus anticoagulant activity.^{322,326,327} In some studies the presence of specific β_2 -glycoprotein I antibodies has been correlated with an increased risk of thrombotic events in patients with APS.³²⁸ APL antibodies may cause a false-positive VDRL. The prevalence of APL has been reported to be 1% to 5% in the general population.³²⁹ However, only a minority of these patients develop APS. In addition to having one of these autoantibodies, the diagnosis of APS must include one or more episodes of venous, arterial, or small vessel thrombosis or fetal morbidity.³³⁰ Thrombocytopenia and prolonged partial thromboplastin time are frequent laboratory findings.

EPIDEMIOLOGY

The incidence of APS is estimated to be around 5 cases per 100,000 people per year, and its prevalence is approximately 40 to 50 cases per 100,000 people.³²⁹ Among patients with APL antibodies, 30% to 55% have primary APS in which there is no associated autoimmune disease.^{292-294,322,326,327,330} APL antibodies are found in 25% to 75% of SLE patients, although most patients never experienced clinical features of the APS. A variable percentage (0%–23%) of patients in different series initially felt to have idiopathic APS have evolved over time into SLE-associated APS.³³¹ In an analysis of 29 published series with more than 1000 SLE patients, 34% were positive for the lupus anticoagulant and 44% for anticardiolipin antibodies.³³² Most studies have found a higher incidence of thrombotic events in SLE patients positive for APL antibodies. Lupus anticoagulants may carry a greater risk of thrombosis than anticardiolipin antibody, with the highest risk for those with triple positivity.^{333,334} A European study of almost 575 SLE patients found the prevalence of IgG anticardiolipin antibodies to be 23% and of IgM 14%.³³⁵ Patients with IgG antibodies had a clear association with thrombocytopenia and thromboses. A multicenter European analysis of 1000 SLE patients found thromboses in 7% of patients over 5 years. Patients with IgG anticardiolipin antibodies again had a higher incidence of thromboses, as did those with a lupus anticoagulant.³³⁶ APL antibodies are also found in up to 2% of normal individuals and in those with a variety of infections (commonly in patients with HIV or hepatitis C virus) and drug reactions, but these are not usually associated with the clinical features of the APS.³³⁷⁻³³⁹

PATHOGENESIS

Susceptible individuals may develop APL antibodies after exposure to infectious or other noxious agents. Among SLE patients there may be a genetic predisposition associated with HLA-DRB1 loci.³⁴⁰ However, despite the presence of APL antibodies, a “second hit” (such as pregnancy, contraceptive use, nephrotic syndrome, SLE, or hyperlipidemia) may

Fig. 31.11 2023 ACR/EULAR APS classification criteria.³²³

be necessary for thrombotic events. RNA-sequencing from kidney biopsy samples for patients with APS reveal increased expression of interferon, NET-related genes, and complement.³⁴¹ These results suggest innate immunity may be playing a role in the pathogenesis. The mechanism(s) of the procoagulant effect are likely multifactorial. APL antibodies exert procoagulant effects at multiple sites in the

clotting cascade including prothrombin, protein C, annexin V, factors VII and XII, platelets, serum proteases, and tissue factor procoagulant. They may also impair fibrinolysis through inhibition of factors such as tissue-type plasminogen activator. APL antibodies may also be procoagulant due to inhibition of the mechanistic target of rapamycin (mTOR) complex intracellular pathway.³⁴² mTOR, also previously

known as the mammalian target of rapamycin or the FK506-binding protein 12-rapamycin-associated protein 1 (FRAP1), is a kinase encoded by the MTOR gene in humans. mTOR belongs to the phosphatidylinositol 3-kinase-related kinase family of protein kinases. As noted earlier, neutrophil extracellular traps (NETs) are scaffolds of decondensed chromatin and are thought to be involved in triggering thrombosis through activation of coagulation factors, endothelium, platelets, and the complement system.³⁴³ Other studies support involvement of the classical, lectin, and alternative complement systems.³⁴⁴⁻³⁴⁷ Many patients have hypocomplementemia suggestive of complement protein consumption.³⁴⁴ The result of all these abnormalities is endothelial damage and intravascular coagulation. In cases of pregnancy-associated APS, the complications may be mediated predominantly by trophoblast dysfunction and complement activation rather than by thrombosis.³⁴⁸

CLINICAL MANIFESTATIONS

The clinical features of APS relate to thrombotic events and consequent ischemia. Among 1000 APS patients the most common features were deep vein thrombosis (32%), thrombocytopenia (22%), livedo reticularis (20%), stroke (13%), pulmonary embolism (9%), and fetal loss (9%).³³⁶ Patients may also experience pulmonary hypertension, cardiac involvement, memory impairment and other neurologic manifestations, fever, malaise, and constitutional symptoms.^{292-294,325,349} As mentioned earlier, patients who test positive for all three diagnostic tests (lupus anticoagulant, anticardiolipin antibodies, and β_2 -glycoprotein antibodies) are at higher risk for thromboembolic events. Catastrophic APS (CAPS), a rare event (occurring in <1% of APS patients), is associated with rapid development of thromboses in multiple organ systems and has a high fatality rate.^{330,350,351}

Renal involvement is generally an uncommon finding in patients with APS.^{352,353} Reports of renal involvement vary from as low as 3% to as high as 25% of patients with primary APS.^{336,354,355} Kidney involvement ranges from large vessels, such as renal artery stenosis or renal vein thrombosis, to small vessel involvement including glomerular thrombotic

microangiopathy (Fig. 31.12).^{292,325,356,357} Lesions involving the arteries and arterioles often have both an acute thrombotic component and a reactive or proliferative chronic one with intimal mucoid thickening, subendothelial fibrosis, and medial hyperplasia.^{357,358} Interstitial fibrosis and cortical atrophy may occur due to tissue ischemia. Glomerular lesions include glomerular capillary thrombosis with associated mesangiolysis, mesangial interposition and duplication of GBMs, and subendothelial accumulation of electron-lucent, flocculent material (“endotheliosis”), resembling the changes in other forms of glomerular thrombotic microangiopathy such as HUS and TTP.

Among APL-positive SLE patients, APL-nephropathy was found in two-thirds of those with APS and in one-third of those without APS. Although patients with APL-nephropathy had a higher frequency of hypertension and elevated serum creatinine levels at biopsy in this series, they did not have a higher frequency of progressive renal insufficiency, ESKD, or death at follow-up.³⁵⁹ This is in contrast to another series of more than 100 SLE patients, which found the presence of APL antibodies to be associated with both thrombotic events and a greater progression to renal failure.²⁹⁴ In patients with APL-nephropathy, renal biopsies with thrombotic microangiopathy may be misclassified as FSGS, membranous nephropathy, and membranoproliferative GN.³⁶⁰ However, some patients with APS may develop a number of other glomerular histologic patterns on LM examination including membranous nephropathy, minimal change/focal sclerosis, mesangial proliferative glomerulonephritis, and pauci-immune rapidly progressive glomerulonephritis (RPGN).³⁶¹

Kidney biopsies can be high risk in those presenting with APS because of thrombocytopenia or the need for continuous anticoagulation. Holding anticoagulation before and after a kidney biopsy, even if only for a few days, may be enough to activate systemic thrombosis in susceptible patients, and there is a risk of late bleeding with resumption of anticoagulation. Therefore a risk-benefit analysis should be undertaken before considering kidney biopsy.

The most frequent clinical renal findings are proteinuria, at times in the nephrotic range, active

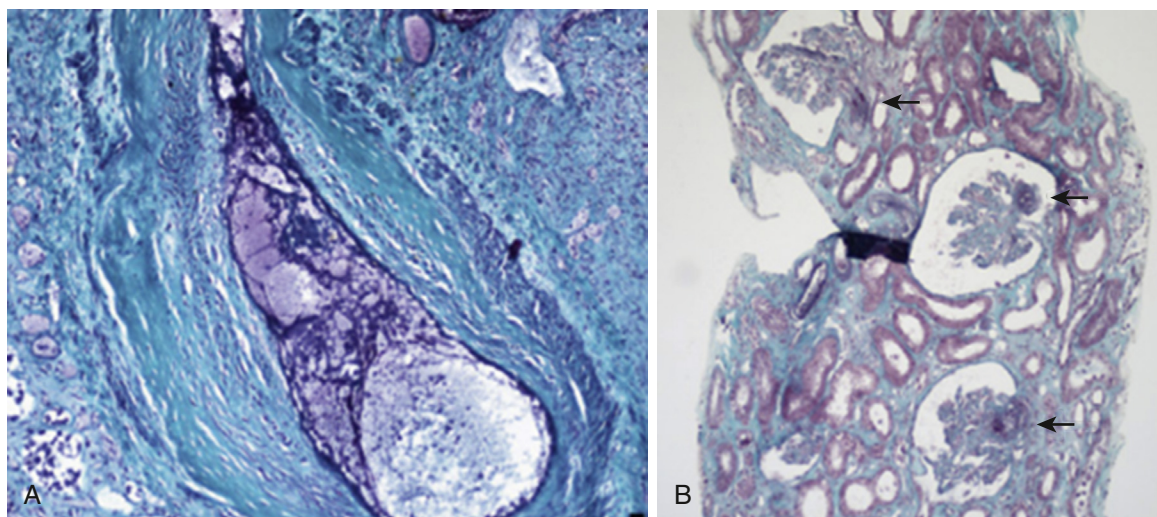


Fig. 31.12 Antiphospholipid nephropathy.³⁵³ (A) Severe fibrous intimal hyperplasia lesion in a native kidney for a patient with APS. (B) Thrombotic microangiopathy lesions in patients with APS (arrows).

urinary sediment, hypertension, and progressive renal dysfunction.^{292,293,356,358,360-362} Some patients present with an acute deterioration in renal function.³⁶⁰ With major renal arterial involvement, there may be renal infarction and renal vein thrombosis may be silent or present with sudden flank pain and a decrease in renal function. Renal artery stenosis has been reported with and without malignant hypertension.³⁶³⁻³⁶⁵

About 10% of biopsied patients with LN have glomerular microthromboses as the major histopathologic finding. Therapy of this glomerular lesion clearly differs from that of immune complex-mediated glomerulonephritis.⁹⁰ One study of 114 biopsied SLE patients found vasoocclusive lesions in one-third of biopsies, which correlated with both hypertension and increased serum creatinine.³⁶⁶ In SLE, features that correlate well with high titers of IgG APL antibodies are thrombocytopenia, the presence of a false-positive VDRL for syphilis, and a prolonged activated partial thromboplastin time.^{292,293,366} Neither the titer of anti-dsDNA antibodies nor the serum complement levels correlate well with the APL antibody levels. In SLE, high titers of IgG anticardiolipin antibody usually correlate well with the risk of thrombosis. However, in one study of 114 biopsied SLE patients, renal thrombi were related to lupus anticoagulant but not anticardiolipin antibodies.³⁶⁶ The clinical features of APS in SLE patients are identical to those of primary APS. An important study documents the prevalence of APL antibodies in 26% of 111 patients with LN followed for a mean of 173 months.²⁹⁴ Of the APL antibody-positive patients, 79% developed a thrombotic event or fetal loss, and the presence of antibodies was strongly correlated with the development of progressive CKD.

DIALYSIS AND TRANSPLANTATION

There is a high prevalence of APL antibodies (10% to 30%) in hemodialysis patients irrespective of patient age, sex, or duration of the dialysis.^{367,368} By contrast, patients with renal insufficiency and those on peritoneal dialysis have a much lower incidence of APL antibodies.²⁹² One study of hemodialysis patients found more patients with arteriovenous (AV) grafts than native fistulas to have an increased titer of IgG anticardiolipin antibody.^{368,369} There was a significant increase in the odds of having two or more episodes of AV graft thrombosis in patients with raised anticardiolipin titer. Whether AV grafts induce anticardiolipin antibodies or whether patients with anticardiolipin antibodies require AV grafts remains unclear.²⁹² In another study of 230 hemodialysis patients, titers of IgG anticardiolipin antibodies were elevated in 26% of the patients as opposed to elevated titers of IgM antibodies in only 4%.³⁶⁹ The mean time to AV graft failure was significantly shorter in the group with elevated IgG antibodies, and the use of warfarin increased graft survival in these patients.

In several studies, 20% to 60% of SLE patients with APL antibodies who received renal transplants had problems related to APS, such as venous thromboses, pulmonary emboli, or persistent thrombocytopenia.^{213,214,370,371} In one large study of non-SLE patients, 28% of 178 transplant patients had APL antibodies, which were associated with a threefold to fourfold increased risk of arterial and venous thromboses.³⁷⁰ However, another study of 337 renal transplant recipients found that 18% who were IgG

or IgM anticardiolipin antibody positive had no greater allograft loss or reduction in GFR than did patients who were anticardiolipin antibody negative.³⁷² Although most patients with APL antibodies who have tested positive for hepatitis C do not have evidence of increased thromboses and APS, when they receive a transplant, they appear to have a higher risk of thrombotic microangiopathy in the allograft.³⁷³ In many of these transplant studies, treatment with anticoagulation has proven successful in preventing recurrent thromboses and graft loss.^{213,214,371}

PREGNANCY

Given the maternal and fetal morbidity and mortality associated with APS, appropriate management in specialized centers is needed. Risk assessment before conception is required.³⁷⁴ In pregnant patients with APS, coumadin must be avoided. Both the EULAR and ACR guidelines recommend that pregnant patients commence heparin or low-molecular-weight heparin (LMWH) at therapeutic doses before conception or by 6 weeks' gestation, with those without a history of thrombosis receiving prophylactic doses and those with a thrombotic history receiving therapeutic doses.^{375,376}

TREATMENT

Optimal treatment of patients with APL antibodies and the APS remains to be defined.^{292,293,377} Many patients with APL antibodies do not experience thrombotic events. On the basis of limited data in asymptomatic patients with APL antibodies but no evidence of thrombotic events or the APS, low-dose aspirin as primary prophylaxis may be beneficial.³⁷⁸ EULAR 2023 recommendation updates for the management of APS suggest consideration of low-dose aspirin for asymptomatic patients with SLE or a high-risk profile (lupus anticoagulant, double/triple positivity, or persistently high APL titers).^{216,376}

Because patients with higher titers of IgG APL antibody have a greater incidence of thrombotic events, they may benefit from anticoagulation.^{335,336} In patients with full APS, anticoagulation with heparin followed by warfarin has proven more effective in preventing recurrent thrombosis than no therapy, aspirin, or low-dose anticoagulation.³⁷⁷ Studies in patients with predominantly venous events suggest no additional benefit with a target INR of 3 to 4 versus 2 to 3.^{377,379-381} It is unclear if this higher INR target would be of benefit for patients with arterial, as opposed to venous, thrombotic events. Of note, many patients did not routinely achieve the higher target INR and very few patients with arterial thrombosis were included. Preliminary studies suggest that the newer direct oral anticoagulants (DOACs) may be inferior to warfarin for the prevention of thrombotic events in patients with APS.³⁸² One meta-analysis suggests that the use of DOAC versus vitamin K antagonists is associated with increased odds of subsequent arterial thrombotic events (OR 5.43), especially stroke.³⁸³ The risk of major bleeding between the two was similar in this study. Therefore the 2023 EULAR guideline updates recommend that therapy with DOACs only be considered as second-line treatment if target INR 2 to 3 cannot be achieved despite good adherence to warfarin. In the event of recurrent thrombosis despite warfarin and achievement

of target INR, treatment considerations may include adding aspirin, increasing target INR to 3-4, or switching to low-molecular-weight heparin. The role of immunosuppression is uncertain in APS.^{292,293,337}

In patients with SLE, the anti-dsDNA antibody titer and serum complement may normalize with immunosuppression without a significant change in a high titer of IgG APL antibody.²⁹² In rare patients who cannot tolerate anticoagulation due to bleeding, who have thromboembolic events despite adequate anticoagulation, or who have catastrophic APS, plasmapheresis with corticosteroids and other immunosuppressives have been used with some success.^{384,385} It is uncertain whether hydroxychloroquine, used mostly in SLE patients, can prevent thromboembolic events in APS.^{378,386,387} There is insufficient and conflicting data on whether newer agents such as rituximab lower the levels of APLs or decrease the risk of thromboembolism.^{388,389} Eculizumab and the longer-acting ravulizumab have been reported in refractory cases of catastrophic APS.³⁹⁰ Outcomes may be improved with earlier administration of these anticomplement therapies.³⁹¹

Other therapies have also been described in APS. In recent years belimumab has been reported to reduce aPL levels in patients with SLE.³⁹²⁻³⁹⁴ Currently there is a pilot trial under way investigating belimumab for refractory or noncriteria manifestations of APS.³⁹⁵ HMG-CoA reductase inhibitors (statins) have demonstrated reductions in thrombus size for mice with APS, and mechanistic studies in humans demonstrate a favorable proinflammatory and prothrombotic response.³⁹⁶⁻³⁹⁸ However, efficacy of statins for reduction of thrombotic events for patients with APS has not been proven. The use of other treatments, such as intravenous γ -globulin and stem cell transplant, are reported only in isolated patients.^{399,400}

MIXED CONNECTIVE TISSUE DISEASE

Mixed connective tissue disease (MCTD) is characterized by the presence of serum anti-U1-ribonucleoprotein (RNP) antibodies with clinical features as seen in SLE, systemic sclerosis (SSc), RA, and polymyositis.^{110,401,402} They also typically have additional distinct serologic findings with a high ANA titer, often with a speckled pattern.^{402,403} Not all patients with clinical features of MCTD have a positive ENA, and not all ENA-positive patients have the clinical features of MCTD.⁴⁰³ U1-RNP antibodies are also found in 13% of SLE, 5% to 10% of patients with systemic sclerosis (SSc), and in healthy controls.⁴⁰⁴⁻⁴⁰⁶ Therefore it is necessary to initially rule out SLE, SSc, RA, or polymyositis before diagnosing MCTD. Because over time some patients fulfill diagnostic criteria for other connective tissue diseases, investigators have questioned whether MCTD is a distinct syndrome and have developed specific criteria to categorize patients as having MCTD.⁴⁰⁷ The term “undifferentiated autoimmune rheumatic and connective tissue disorder or overlap syndrome” has also been used but more often may be reserved for patients who have rheumatologic symptoms but do not have detectable anti-U1 RNP antibodies.^{403,408}

Predicted annual incidence of MCTD ranges from 0.2 to 1.9 per 100,000 adults.^{409,410} Estimates of prevalence range from 3.8 to 6.4 per 100,000 population.^{409,411} MCTD is much more common in females, with varying estimate ranges.^{409,412} The pathogenesis of MCTD is not well understood. The U1

small nuclear ribonucleoprotein particle (snRNP) plays a role in mRNA splicing; although autoantibodies directed against U1 snRNP may vary at the sites they bind for MCTD and SLE, there is often an overlap.⁴¹³ Epitope spreading has been proposed as a potential mechanism to account for changing clinical phenotypes.⁴¹⁴ Recently, an association between MCTD and DNA methylation of genes in the interferon pathway, similar to that reported in SLE and Sjögrens disease, has been described.⁴¹⁵

CLINICAL MANIFESTATIONS

In the early stages of MCTD, patients usually manifest nonspecific symptoms such as malaise, fatigue, myalgias, arthralgias, low-grade fever, and Raynaud phenomenon.⁴¹⁶ These symptoms are also seen in most cases of childhood MCTD.⁴¹⁷ Over time, features similar to other rheumatologic connective tissue diseases appear including arthralgias, deforming arthritis, myalgias and myositis, swollen hands and fingers, restrictive pulmonary disease and pulmonary hypertension, esophageal dysmotility, pericarditis and myocarditis, serositis, oral and nasal ulcers, digital ulcers and gangrene, discoid lupus-like lesions, malar rash, alopecia, photosensitivity, and lymphadenopathy.^{402,403,418} However, patients with MCTD, especially those documented to have anti-U1RNP antibodies, infrequently have major central nervous system disease or severe proliferative glomerulonephritis.^{402,403,418} Low-grade anemia, lymphopenia, and hypergammaglobulinemia are all common in MCTD.

The most widely used serologic test to confirm a diagnosis of MCTD is the ENA with anti-U1RNP antibodies.^{403,419} The diagnosis of MCTD is even firmer in those patients with IgG antibodies against an antigenic component of U1RNP, the 68-kD protein.^{419,420} Antibodies to other nuclear antigens have been found in MCTD, and some correlate better with some clinical features of specific rheumatologic diseases.⁴⁰² Antibodies against dsDNA, Sm antigen, and Ro are infrequently positive in MCTD, but up to 70% of patients will have a positive rheumatoid factor (RF).

The incidence of renal involvement has varied from 10% to 26% of adults and from 33% to 50% of children with MCTD.^{403,420} Many patients have mild clinical manifestations with only microhematuria and <500 mg proteinuria daily. However, heavier proteinuria, severe hypertension, and AKI reminiscent of “scleroderma renal crisis” may occur.^{418,421,422} Although the titer of anti-RNP does not correlate with renal involvement, the presence of serologic markers of active SLE (e.g., high anti-dsDNA antibody titers and anti-Sm antibody) is more common in those with renal disease.⁴⁰³ Low serum complement levels have not always correlated with the presence of renal involvement.⁴¹⁸ Children with MCTD more often have glomerular involvement with few clinical or urinary findings.⁴²³

PATHOLOGY

The pathology of MCTD is diverse with the glomerular lesions resembling the spectrum found in SLE and the vascular lesions resembling those in scleroderma. Glomerular disease is most common and is usually superimposed on a background of mesangial deposits and hypercellularity as in SLE.^{418,421,423-426} Up to 30% of biopsied patients have mesangial deposits of IgG and C3. Other patients have

focal proliferative glomerulonephritis with both mesangial and subendothelial deposits, but fibrinoid necrosis and crescents are rare. The most common pattern of glomerular involvement is membranous nephropathy reported in up to 35% of cases,^{421,423-425} with typical peripheral capillary wall granular IF staining for IgG, C3, and at times IgA and IgM. Some patients will have a mixed pattern of membranous plus mesangial proliferative GN.⁴²³ Renal biopsy findings may transform over time from one pattern of glomerular involvement to another, similar to patients with SLE. By ultrastructural analysis, LN-like findings have been reported including endothelial TRIs, deposits with “fingerprint” substructure and tubular basement membrane deposits.⁴¹⁸ In a review of 100 biopsied patients with MCTD, 12% had normal biopsies, 35% mesangial lesions, 10% proliferative glomerular lesions, and 36% membranous nephropathy.⁴²⁷ In addition, 15% to 25% of patients had interstitial disease and vascular lesions. In autopsy series, in which two-thirds of patients had clinical renal disease, a similar distribution of glomerular lesions was found.⁴²⁶ Other renal pathology findings in MCTD include secondary renal amyloidosis,⁴²⁸ vascular sclerosis ranging from intimal sclerosis to medial hyperplasia, and vascular lesions resembling those in scleroderma kidney with involvement of the interlobular arteries by intimal mucoid edema and fibrous sclerosis.⁴¹⁸

TREATMENT AND PROGNOSIS

Therapy of MCTD with corticosteroids is effective in treating the inflammatory features of joint disease and serositis. Steroids are less effective in treating scleroderma-like features such as cutaneous disease, esophageal involvement, and especially pulmonary hypertension. As patients with MCTD may be at risk for scleroderma renal crisis (SRC), glucocorticoids should be used with caution. Intravenous immunoglobulin and rituximab have been used to treat thrombocytopenia and hemolytic anemia.^{429,430} Treatment of the glomerular lesions is similar to that for LN.

Originally, MCTD was felt to have a good prognosis with low mortality and few patients developed other distinct connective tissue disorders. The longer that patients with MCTD have been followed, the greater the percentage who evolve into a specific connective tissue disorder.⁴⁰³ In some series, almost half of the patients with a short duration of follow-up were still felt to have true MCTD, but in those with longer follow-up, the percentage had dropped to 15% or less.^{403,418} Most patients evolve toward a picture of either SLE or systemic sclerosis, but some develop features of RA.^{407,418} and rates have been found to range from 15% to 30% at 10 to 12 years, with patients having more clinical features of scleroderma and polymyositis faring worse.^{403,418} One study of 161 MCTD patients followed for 8 years found 60% with unclassified MCTD, 17% with systemic sclerosis, 9% with SLE, 2.5% with RA, and 11.5% with undifferentiated connective tissue disease.⁴⁰⁷ A positive anti-dsDNA antibody predicted the development of SLE while hypomotility of the esophagus or sclerodactyly predicted the development of systemic sclerosis. The presence of anticardiolipin antibodies and anti- β_2 -glycoprotein antibodies increases the mortality risk. In one study, 5-, 10-, and 15-year survival rates were 98%, 96%, and 88%, respectively.⁴²⁹ The leading causes of mortality in MCTD are pulmonary hypertension and cardiovascular disease.^{403,431} Other causes include vascular

lesions of the coronary and other vessels, hypertensive scleroderma crisis, and chronic renal failure. Clearly, MCTD is not a benign disorder but rather a disease with the potential for significant morbidity and mortality.

OTHER SEROPOSITIVE CONNECTIVE TISSUE DISORDERS

SJOGREN SYNDROME

Primary Sjögren syndrome (pSS) is associated with lymphocytic and plasmacytic infiltrate in the salivary, parotid, and lacrimal glands. This results in the cardinal symptoms of xerostomia (dry mouth) and xerophthalmia (dry eyes). The immune process can affect other organs such as the kidneys. The incidence rates of pSS is 7 per 100,000 person-years, with a prevalence of 43 per 100,000 person-years, approximately.⁴³² The true prevalence of renal involvement in pSS is unknown but may range from only a few percent of cases to >30% as described in the Chinese population.⁴³³⁻⁴³⁵ Tubulointerstitial nephritis (TIN) is the most frequent form of renal disease that develops in about 75% of patients with pSS who have renal involvement, but glomerular lesions may also be seen in about a quarter of cases.⁴³⁶ Similar to TIN for other etiologies, the mainstay of management is glucocorticoids. Although data are limited, MMF with or without reduced-dose glucocorticoids have also been described.^{437,438} A less common renal presentation is membranoproliferative glomerulonephritis (MPGN) associated with cryoglobulinemia. The treatment approach is similar to other etiologies for mixed cryoglobulinemia, namely glucocorticoids and rituximab or cyclophosphamide if treatment is warranted. Plasmapheresis may be considered if life-threatening cryoglobulinemia develops. Rarely, ANCA-associated pauci-immune glomerulonephritis and immunoglobulin A nephropathy have also been reported in pSS.⁴³⁹

SYSTEMIC SCLEROSIS AND SCLERODERMA

Systemic sclerosis (SSc; scleroderma) is a multisystem disease characterized by vascular dysfunction and fibrosis. It is classified as limited or diffuse cutaneous SSc. In the limited form the fibrosis is restricted to the hands, arms, and face with Raynaud phenomenon. Diffuse cutaneous SSc is more severe and rapidly progressing and more likely to have other organ involvement including kidneys, esophagus, heart, and lungs. The reported incidence rates of SSc vary from 8 to 56 per million person-years, with a prevalence between 38 and 350 cases per million person-years.⁴⁴⁰ Up to 60% of patients with SSc may have some sort of kidney pathology.⁴⁴¹ The most serious kidney manifestation is SRC. This may occur in up to 15% of SSc cases.⁴⁴² The cause of SRC is unknown. Risk factors for SRC include diffuse skin involvement,⁴⁴³ glucocorticoid use (dose dependent),⁴⁴³ and presence of antibodies against ribonucleic acid (RNA) polymerase III.⁴⁴⁴ Other less common risk factors have also been described. Interestingly, anticentromere antibodies may be associated with a lower risk of SRC.⁴⁴⁵

SRC typically occurs within the first few years of SSc diagnosis. It often presents with new-onset hypertension with blood pressure >140/90 mm Hg.⁴⁴⁶ However, it is worth

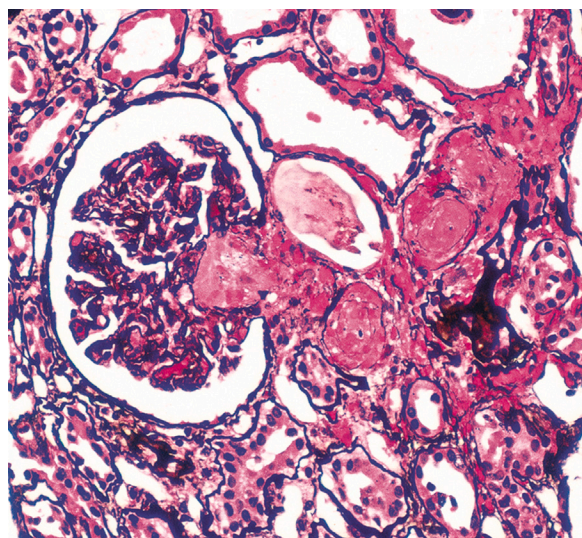


Fig. 31.13 Scleroderma renal crisis. Fibrinoid necrosis of the arteriole is present, with fibrin thrombi occluding the lumen and extending to the hilus of the glomerulus. The glomerulus itself shows only ischemic change with corrugation of the glomerular basement membrane and retraction of the tuft (Jones silver stain, $\times 200$). (From Fogo A, Kashgarian M. *Diagnostic atlas of renal pathology*. 3rd ed. Elsevier, 2017.)

mentioning that normotensive SRC has been described in up to 10% of patients.^{447,448} Microangiopathic hemolytic anemia may occur in up to 50% of cases.⁴⁴⁹ AKI inevitably develops. The urinalysis and urine sediment are typically bland.

The classic pathologic findings include vasculopathy with endothelial injury. The early stages reveal occasional fibrinoid necrosis of the afferent arterioles (Fig. 31.13). Mucoid changes of the arterioles/interlobular arteries with red blood cell fragments in the subacute stages. Finally, intimal thickening and fibrotic “onion-skinning” of the interlobular and arcuate renal arteries can occur in chronic stages.

The pillar of management remains angiotensin-converting enzyme inhibitors (ACEi). A Japanese retrospective study demonstrated that early initiation of ACEi within a couple of days of hospitalization resulted in a lower composite outcome of in-hospital mortality or hemodialysis-dependence at discharge.⁴⁵⁰ Dialysis may be required in up to half of patients, and of this group, more than 50% may recover off dialysis.⁴⁴⁵ Therefore monitoring for renal recovery when patients commence renal replacement therapy is prudent. Like other vascular diseases, this renal recovery may be gradual and occur over months to years. Once ESKD is confirmed, numerous studies suggest that patients with SRC on permanent dialysis have worse survival relative to those with other etiologies of ESKD.^{451,452} However, graft outcomes may not be necessarily worse for those who developed SRC.⁴⁵³

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KEY REFERENCES

- Parikh SV, Almaani S, Brodsky S, Rovin BH. Update on lupus nephritis: core curriculum 2020. *Am J Kidney Dis*. 2020;76(2):265–281. <https://doi.org/10.1053/j.ajkd.2019.10.017>.
- Barber MRW, Drenkard C, Falasinnu T, et al. Global epidemiology of systemic lupus erythematosus. *Nat Rev Rheumatol*. 2021;17(9):515–532. <https://doi.org/10.1038/s41584-021-00668-1>.
- Tian J, Zhang D, Yao X, Huang Y, Lu Q. Global epidemiology of systemic lupus erythematosus: a comprehensive systematic analysis and modelling study. *Ann Rheum Dis*. 2023;82(3):351–356. <https://doi.org/10.1136/ard-2022-223035>.
- Aringer M, Costenbader K, Daikh D, et al. 2019 European League against Rheumatism/American College of Rheumatology classification criteria for systemic lupus erythematosus. *Ann Rheum Dis*. 2019;78(9):1151–1159. <https://doi.org/10.1136/annrheumdis-2018-214819>.
- Lichtnekert J, Anders HJ. Lupus nephritis-related chronic kidney disease. *Nat Rev Rheumatol*. 2024;20(11):699–711. <https://doi.org/10.1038/s41584-024-01158-w>.
- Parodis I, Depascale R, Doria A, Anders HJ. When should targeted therapies be used in the treatment of lupus nephritis: early in the disease course or in refractory patients? *Autoimmun Rev*. 2024;23(1):103418. <https://doi.org/10.1016/j.autrev.2023.103418>.
- Bajema IM, Wilhelmus S, Alpers CE, et al. Revision of the International Society of Nephrology/Renal Pathology Society classification for lupus nephritis: clarification of definitions, and modified National Institutes of Health activity and chronicity indices. *Kidney Int*. 2018;93(4):789–796. <https://doi.org/10.1016/j.kint.2017.11.023>.
- Malvar A, Alberton V, Lococo B, et al. Remission of lupus nephritis: the trajectory of histological response in successfully treated patients. *Lupus Sci Med*. 2023;10(1). <https://doi.org/10.1136/lupus-2023-000932>.
- Sethi S, Madden BJ, Debiec H, et al. Exostosin 1/exostosin 2-associated membranous nephropathy. *J Am Soc Nephrol*. 2019;30(6):1123–1136. <https://doi.org/10.1681/ASN.2018080852>.
- Caza TN, Storey AJ, Hassen SI, et al. Discovery of seven novel putative antigens in membranous nephropathy and membranous lupus nephritis identified by mass spectrometry. *Kidney Int*. 2023;103(3):593–606. <https://doi.org/10.1016/j.kint.2023.01.001>.
- Santacruz JC, Pulido S, Arzuaga A, Mantilla MJ, Londono J. Lupus cystitis, from myth to reality: a narrative review. *Cureus*. 2021;13(12):e20409. <https://doi.org/10.7759/cureus.20409>.
- De Vriese AS, Sethi S, Fervenza FC. Lupus nephritis: redefining the treatment goals. *Kidney Int*. 2025;107(2):198–211. <https://doi.org/10.1016/j.kint.2024.10.018>.
- Mejia-Vilet JM, Zhang XL, Cruz C, et al. Urinary soluble CD163: a novel noninvasive biomarker of activity for lupus nephritis. *J Am Soc Nephrol*. 2020;31(6):1335–1347. <https://doi.org/10.1681/ASN.2019121285>.
- Fava A, Buyon J, Magder L, et al. Urine proteomic signatures of histological class, activity, chronicity, and treatment response in lupus nephritis. *JCI Insight*. 2024;9(2). <https://doi.org/10.1172/jci.insight.172569>.
- Gao C, Bian X, Wu L, et al. A nomogram predicting the histological activity of lupus nephritis from clinical parameters. *Nephrol Dial Transplant*. 2024;39(3):520–530. <https://doi.org/10.1093/ndt/gfad191>.
- Wind M, Fierro JJ, Bloemenkamp KWM, et al. Pregnancy outcome predictors in systemic lupus erythematosus: a systematic review and meta-analysis. *Lancet Rheumatol*. 2024;6(10):e667–e683. [https://doi.org/10.1016/S2665-9913\(24\)00160-7](https://doi.org/10.1016/S2665-9913(24)00160-7).
- Radin M, Cecchi I, Crisafulli F, et al. Complement levels during the first trimester predict disease flare and adverse pregnancy outcomes in systemic lupus erythematosus: a network meta-analysis on 532 pregnancies. *Autoimmun Rev*. 2023;22(12):103467.
- Gourley MF, Austin 3rd HA, Scott D, et al. Methylprednisolone and cyclophosphamide, alone or in combination, in patients with lupus nephritis. A randomized, controlled trial. *Ann Intern Med*.

- 1996;125(7):549–557. <https://doi.org/10.7326/0003-4819-125-7-199610010-00003>.
188. Houssiau FA, Vasconcelos C, D'Cruz D, et al. Early response to immunosuppressive therapy predicts good renal outcome in lupus nephritis: lessons from long-term followup of patients in the Euro-Lupus Nephritis Trial. *Arthritis Rheum.* 2004;50(12):3934–3940. <https://doi.org/10.1002/art.20666>.
208. Kidney Disease: Improving Global Outcomes Lupus Nephritis Work G. KDIGO 2024 clinical practice guideline for the management of LUPUS NEPHRITIS. *Kidney Int.* 2024;105(1S):S1–S69. <https://doi.org/10.1016/j.kint.2023.09.002>.
216. Fanouriakis A, Kostopoulou M, Andersen J, et al. EULAR recommendations for the management of systemic lupus erythematosus: 2023 update. *Ann Rheum Dis.* 2024;83(1):15–29. <https://doi.org/10.1136/ard-2023-224762>.
220. Rovin BH, Solomons N, Pendergraft 3rd WF, et al. A randomized, controlled double-blind study comparing the efficacy and safety of dose-ranging voclosporin with placebo in achieving remission in patients with active lupus nephritis. *Kidney Int.* 2019;95(1):219–231. <https://doi.org/10.1016/j.kint.2018.08.025>.
223. Figueroa-Parra G, Bautista-Vargas M, Navarro-Mendoza E, Duarte-Garcia A. Optimal glucocorticoid therapy in lupus nephritis. *Nephrol Dial Transplant.* 2024;gfae294. <https://doi.org/10.1093/ndt/gfae294>.
224. Bose B, Silverman ED, Bargman JM. Ten common mistakes in the management of lupus nephritis. *Am J Kidney Dis.* 2014;63(4):667–676. <https://doi.org/10.1053/j.ajkd.2013.10.056>.
225. Austin 3rd HA, Klippel JH, Balow JE, et al. Therapy of lupus nephritis. Controlled trial of prednisone and cytotoxic drugs. *N Engl J Med.* 1986;314(10):614–619. <https://doi.org/10.1056/NEJM198603063141004>.
239. Isenberg D, Appel GB, Contreras G, et al. Influence of race/ethnicity on response to lupus nephritis treatment: the ALMS study. *Rheumatology.* 2010;49(1):128–140. <https://doi.org/10.1093/rheumatology/kep346>.
240. Walsh M, Solomons N, Lisk L, Jayne DR. Mycophenolate mofetil or intravenous cyclophosphamide for lupus nephritis with poor kidney function: a subgroup analysis of the Aspreva Lupus Management Study. *Am J Kidney Dis.* 2013;61(5):710–715. <https://doi.org/10.1053/j.ajkd.2012.11.042>.
241. Tang Z, Yang G, Yu C, et al. Effects of mycophenolate mofetil for patients with crescentic lupus nephritis. *Nephrology.* 2008;13(8):702–707. <https://doi.org/10.1111/j.1440-1797.2008.00975.x>.
245. Liu Z, Zhang H, Liu Z, et al. Multitarget therapy for induction treatment of lupus nephritis: a randomized trial. *Ann Intern Med.* 2015;162(1):18–26. <https://doi.org/10.7326/M14-1030>.
246. Zhang H, Liu Z, Zhou M, et al. Multitarget therapy for maintenance treatment of lupus nephritis. *J Am Soc Nephrol.* 2017;28(12):3671–3678. <https://doi.org/10.1681/ASN.2017030263>.
247. Rovin BH, Teng YKO, Ginzler EM, et al. Efficacy and safety of voclosporin versus placebo for lupus nephritis (AURORA 1): a double-blind, randomised, multicentre, placebo-controlled, phase 3 trial. *Lancet.* 2021;397(10289):2070–2080. [https://doi.org/10.1016/S0140-6736\(21\)00578-X](https://doi.org/10.1016/S0140-6736(21)00578-X).
248. Arriens C, Teng YKO, Ginzler EM, et al. Update on the efficacy and safety profile of voclosporin: an integrated analysis of clinical trials in lupus nephritis. *Arthritis Care Res.* 2023;75(7):1399–1408. <https://doi.org/10.1002/acr.25007>.
250. van Gelder T. How cyclosporine reduces mycophenolic acid exposure by 40% while other calcineurin inhibitors do not. *Kidney Int.* 2021;100(6):1185–1189. <https://doi.org/10.1016/j.kint.2021.06.036>.
253. Furie R, Rovin BH, Houssiau F, et al. Two-year, randomized, controlled trial of belimumab in lupus nephritis. *N Engl J Med.* 2020;383(12):1117–1128. <https://doi.org/10.1056/NEJMoa2001180>.
254. Rovin BH, Furie R, Teng YKO, et al. A secondary analysis of the Belimumab International Study in Lupus Nephritis trial examined effects of belimumab on kidney outcomes and preservation of kidney function in patients with lupus nephritis. *Kidney Int.* 2022;101(2):403–413. <https://doi.org/10.1016/j.kint.2021.08.027>.
257. Perez-Arias AA, Marquez-Macedo SE, Pena-Vizcarra OR, et al. The influence of repeated flares in response to therapy and prognosis in lupus nephritis. *Nephrol Dial Transplant.* 2023;38(4):884–893. <https://doi.org/10.1093/ndt/gfac304>.
259. Houssiau FA, D'Cruz D, Sangle S, et al. Azathioprine versus mycophenolate mofetil for long-term immunosuppression in lupus nephritis: results from the MAINTAIN Nephritis Trial. *Ann Rheum Dis.* 2010;69(12):2083–2089. <https://doi.org/10.1136/ard.2010.131995>.
260. Dooley MA, Jayne D, Ginzler EM, et al. Mycophenolate versus azathioprine as maintenance therapy for lupus nephritis. *N Engl J Med.* 2011;365(20):1886–1895. <https://doi.org/10.1056/NEJMoa1014460>.
300. Mackensen A, Muller F, Mougiakakos D, et al. Anti-CD19 CAR T cell therapy for refractory systemic lupus erythematosus. *Nat Med.* 2022;28(10):2124–2132. <https://doi.org/10.1038/s41591-022-02017-5>.
301. Muller F, Taubmann J, Bucci L, et al. CD19 CAR T-cell therapy in autoimmune disease—a case series with follow-up. *N Engl J Med.* 2024;390(8):687–700. <https://doi.org/10.1056/NEJMoa2308917>.
302. Mullard A. CAR T cell therapies raise hopes—and questions—for lupus and autoimmune disease. *Nat Rev Drug Discov.* 2023;22(11):859–861. <https://doi.org/10.1038/d41573-023-00166-x>.
303. Parodis I, Gatto M, Sjowall C. B cells in systemic lupus erythematosus: targets of new therapies and surveillance tools. *Front Med.* 2022;9:952304. <https://doi.org/10.3389/fmed.2022.952304>.
312. Fanouriakis A, Kostopoulou M, Cheema K, et al. 2019 update of the joint European League against rheumatism and European renal association-European dialysis and transplant association (EULAR/ERA-EDTA) recommendations for the management of lupus nephritis. *Ann Rheum Dis.* 2020;79(6):713–723. <https://doi.org/10.1136/annrheumdis-2020-216924>.
321. Sloan M, Lever E, Gordon C, et al. Medication decision-making and adherence in lupus: patient-physician discordance and the impact of previous 'adverse medical experiences'. *Rheumatology.* 2022;61(4):1417–1429. <https://doi.org/10.1093/rheumatology/keab534>.
376. Tektonidou MG, Andreoli L, Limper M, et al. EULAR recommendations for the management of antiphospholipid syndrome in adults. *Ann Rheum Dis.* 2019;78(10):1296–1304. <https://doi.org/10.1136/annrheumdis-2019-215213>.
352. Scheen M, Adedjouma A, Esteve E, et al. Kidney disease in antiphospholipid antibody syndrome: risk factors, pathophysiology and management. *Autoimmun Rev.* 2022;21(5):103072. <https://doi.org/10.1016/j.autrev.2022.103072>.
383. Khairani CD, Bejjani A, Piazza G, et al. Direct oral anticoagulants vs vitamin K antagonists in patients with antiphospholipid syndromes: meta-analysis of randomized trials. *J Am Coll Cardiol.* 2023;81(1):16–30. <https://doi.org/10.1016/j.jacc.2022.10.008>.
393. Emmi G, Bettiol A, Palterer B, et al. Belimumab reduces antiphospholipid antibodies in SLE patients independently of hydroxychloroquine treatment. *Autoimmun Rev.* 2019;18(3):312–314. <https://doi.org/10.1016/j.autrev.2018.11.002>.
417. Terminiello A, Marrani E, Pagnini I, et al. Childhood mixed connective tissue disease at disease onset: evidence from a systematic review. *Autoimmun Rev.* 2024;23(4):103513. <https://doi.org/10.1016/j.autrev.2023.103513>.

REFERENCES

- Parikh SV, Almaani S, Brodsky S, Rovin BH. Update on lupus nephritis: core curriculum 2020. *Am J Kidney Dis.* 2020;76(2):265–281. <https://doi.org/10.1053/j.ajkd.2019.10.017>.
- Gasparotto M, Gatto M, Binda V, Doria A, Moroni G. Lupus nephritis: clinical presentations and outcomes in the 21st century. *Rheumatology.* 2020;59(suppl5):v39–v51. <https://doi.org/10.1093/rheumatology/keaa381>.
- Barber MRW, Drenkard C, Falasinnu T, et al. Global epidemiology of systemic lupus erythematosus. *Nat Rev Rheumatol.* 2021;17(9):515–532. <https://doi.org/10.1038/s41584-021-00668-1>.
- Tian J, Zhang D, Yao X, Huang Y, Lu Q. Global epidemiology of systemic lupus erythematosus: a comprehensive systematic analysis and modelling study. *Ann Rheum Dis.* 2023;82(3):351–356. <https://doi.org/10.1136/ard-2022-223035>.
- Fatoye F, Gebrye T, Mbada C. Global and regional prevalence and incidence of systemic lupus erythematosus in low-and-middle income countries: a systematic review and meta-analysis. *Rheumatol Int.* 2022;42(12):2097–2107. <https://doi.org/10.1007/s00296-022-05183-4>.
- Izmirly PM, Ferucci ED, Somers EC, et al. Incidence rates of systemic lupus erythematosus in the USA: estimates from a meta-analysis of the Centers for Disease Control and Prevention national lupus registries. *Lupus Sci Med.* 2021;8(1). <https://doi.org/10.1136/lupus-2021-000614>.
- Pons-Estel GJ, Alarcon GS, Scofield L, Reinlib L, Cooper GS. Understanding the epidemiology and progression of systemic lupus erythematosus. *Semin Arthritis Rheum.* 2010;39(4):257–268. <https://doi.org/10.1016/j.semarthrit.2008.10.007>.
- Appel GB, Contreras G, Dooley MA, et al. Mycophenolate mofetil versus cyclophosphamide for induction treatment of lupus nephritis. *J Am Soc Nephrol.* 2009;20(5):1103–1112. <https://doi.org/10.1681/ASN.2008101028>.
- Barr RG, Seliger S, Appel GB, et al. Prognosis in proliferative lupus nephritis: the role of socio-economic status and race/ethnicity. *Nephrol Dial Transplant.* 2003;18(10):2039–2046. <https://doi.org/10.1093/ndt/gfg345>.
- Contreras G, Lenz O, Pardo V, et al. Outcomes in African Americans and Hispanics with lupus nephritis. *Kidney Int.* 2006;69(10):1846–1851. <https://doi.org/10.1038/sj.ki.5000243>.
- Osio-Salido E, Manapat-Reyes H. Epidemiology of systemic lupus erythematosus in Asia. *Lupus.* 2010;19(12):1365–1373. <https://doi.org/10.1177/0961203310374305>.
- Burgos PI, McGwin Jr G, Pons-Estel GJ, Reveille JD, Alarcon GS, Vila LM. US patients of Hispanic and African ancestry develop lupus nephritis early in the disease course: data from LUMINA, a multiethnic US cohort (LUMINA LXXIV). *Ann Rheum Dis.* 2011;70(2):393–394. <https://doi.org/10.1136/ard.2010.131482>.
- Nee R, Martinez-Orsorio J, Yuan CM, et al. Survival disparity of African American versus non-African American patients with ESRD due to SLE. *Am J Kidney Dis.* 2015;66(4):630–637. <https://doi.org/10.1053/j.ajkd.2015.04.011>.
- Alarcon GS, McGwin Jr G, Petri M, et al. Baseline characteristics of a multiethnic lupus cohort: profile. *Lupus.* 2002;11(2):95–101. <https://doi.org/10.1191/0961203302lu155oa>.
- Bastian HM, Roseman JM, McGwin Jr G, et al. Systemic lupus erythematosus in three ethnic groups. XII. Risk factors for lupus nephritis after diagnosis. *Lupus.* 2002;11(3):152–160. <https://doi.org/10.1191/0961203302lu158oa>.
- Feldman CH, Hiraki LT, Liu J, et al. Epidemiology and sociodemographics of systemic lupus erythematosus and lupus nephritis among US adults with Medicaid coverage, 2000–2004. *Arthritis Rheum.* 2013;65(3):753–763. <https://doi.org/10.1002/art.37795>.
- Rahman A, Isenberg DA. Systemic lupus erythematosus. *N Engl J Med.* 2008;358(9):929–939. <https://doi.org/10.1056/NEJM-ra071297>.
- Zandman-Goddard G, Solomon M, Rosman Z, Peeva E, Shoenfeld Y. Environment and lupus-related diseases. *Lupus.* 2012;21(3):241–250. <https://doi.org/10.1177/0961203311426568>.
- Harley IT, Kaufman KM, Langefeld CD, Harley JB, Kelly JA. Genetic susceptibility to SLE: new insights from fine mapping and genome-wide association studies. *Nat Rev Genet.* 2009;10(5):285–290. <https://doi.org/10.1038/nrg2571>.
- Munroe ME, James JA. Genetics of lupus nephritis: clinical implications. *Semin Nephrol.* 2015;35(5):396–409. <https://doi.org/10.1016/j.semnephrol.2015.08.002>.
- Appel GBJD, Rovin BH. In: Johnson RFJFJ, ed. *Lupus Nephritis. Mosby-Elsevier.* Comprehensive Clinical Nephrology; 2014.
- Salmon JE, Millard S, Schachter LA, et al. Fc gamma RIIA alleles are heritable risk factors for lupus nephritis in African Americans. *J Clin Invest.* 1996;97(5):1348–1354. <https://doi.org/10.1172/JCI118552>.
- Niu Z, Zhang P, Tong Y. Value of HLA-DR genotype in systemic lupus erythematosus and lupus nephritis: a meta-analysis. *Int J Rheum Dis.* 2015;18(1):17–28. <https://doi.org/10.1111/1756-185X.12528>.
- Lu X, Chen X, Forney C, et al. Global discovery of lupus genetic risk variant allelic enhancer activity. *Nat Commun.* 2021;12(1):1611. <https://doi.org/10.1038/s41467-021-21854-5>.
- Harley N, Ihle B. Wegener's granulomatosis—use of cyclosporin-A: a case report. *Aust N Z J Med.* 1990;20(1):71–73. <https://doi.org/10.1111/j.1445-5994.1990.tb00376.x>.
- Chung SA, Brown EE, Williams AH, et al. Lupus nephritis susceptibility loci in women with systemic lupus erythematosus. *J Am Soc Nephrol.* 2014;25(12):2859–2870. <https://doi.org/10.1681/ASN.2013050446>.
- Lin CP, Adrianto I, Lessard CJ, et al. Role of MYH9 and APOL1 in African and non-African populations with lupus nephritis. *Gene Immun.* 2012;13(3):232–238. <https://doi.org/10.1038/gene.2011.82>.
- Freedman BI, Langefeld CD, Andringa KK, et al. End-stage renal disease in African Americans with lupus nephritis is associated with APOL1. *Arthritis Rheumatol.* 2014;66(2):390–396. <https://doi.org/10.1002/art.38220>.
- Waldman M, Madaio MP. Pathogenic autoantibodies in lupus nephritis. *Lupus.* 2005;14(1):19–24. <https://doi.org/10.1191/0961203305lu2054oa>.
- Aringer M, Costenbader K, Daikh D, et al. 2019 European League against Rheumatism/American College of Rheumatology classification criteria for systemic lupus erythematosus. *Ann Rheum Dis.* 2019;78(9):1151–1159. <https://doi.org/10.1136/annrheumdis-2018-214819>.
- Crow MK. Developments in the clinical understanding of lupus. *Arthritis Res Ther.* 2009;11(5):245. <https://doi.org/10.1186/ar2762>.
- Crispin JC, Liossis SN, Kis-Toth K, et al. Pathogenesis of human systemic lupus erythematosus: recent advances. *Trends Mol Med.* 2010;16(2):47–57. <https://doi.org/10.1016/j.molmed.2009.12.005>.
- Mortensen ES, Fenton KA, Rekvig OP. Lupus nephritis: the central role of nucleosomes revealed. *Am J Pathol.* 2008;172(2):275–283. <https://doi.org/10.2353/ajpath.2008.070563>.
- Bhat P, Radhakrishnan J. B lymphocytes and lupus nephritis: new insights into pathogenesis and targeted therapies. *Kidney Int.* 2008;73(3):261–268. <https://doi.org/10.1038/sj.ki.5002663>.
- Sherer Y, Gorstein A, Fritzler MJ, Shoenfeld Y. Autoantibody explosion in systemic lupus erythematosus: more than 100 different antibodies found in SLE patients. *Semin Arthritis Rheum.* 2004;34(2):501–537. <https://doi.org/10.1016/j.semarthrit.2004.07.002>.
- Alperin JM, Ortiz-Fernandez L, Sawalha AH. Monogenic lupus: a developing paradigm of disease. *Front Immunol.* 2018;9:2496. <https://doi.org/10.3389/fimmu.2018.02496>.
- Appel GB, Silva FG, Pirani CL, Meltzer JI, Estes D. Renal involvement in systemic lupus erythematosus (SLE): a study of 56 patients emphasizing histologic classification. *Medicine (Baltimore).* 1978;57(5):371–410. <https://doi.org/10.1097/00005792-197809000-00001>.
- Akkaya M, Kwak K, Pierce SK. B cell memory: building two walls of protection against pathogens. *Nat Rev Immunol.* 2020;20(4):229–238. <https://doi.org/10.1038/s41577-019-0244-2>.
- Lichtnekert J, Anders HJ. Lupus nephritis-related chronic kidney disease. *Nat Rev Rheumatol.* 2024;20(11):699–711. <https://doi.org/10.1038/s41584-024-01158-w>.
- Figgett WA, Vincent FB, Saulep-Easton D, Mackay F. Roles of ligands from the TNF superfamily in B cell development, function, and regulation. *Semin Immunol.* 2014;26(3):191–202. <https://doi.org/10.1016/j.smim.2014.06.001>.

41. Parodis I, Depascale R, Doria A, Anders HJ. When should targeted therapies be used in the treatment of lupus nephritis: early in the disease course or in refractory patients? *Autoimmun Rev.* 2024;23(1):103418. <https://doi.org/10.1016/j.autrev.2023.103418>.
42. Ng KP, Manson JJ, Rahman A, Isenberg DA. Association of antinucleosome antibodies with disease flare in serologically active clinically quiescent patients with systemic lupus erythematosus. *Arthritis Rheum.* 2006;55(6):900–904. <https://doi.org/10.1002/art.22356>.
43. V DA, ed. *Renal Disease in Systemic Lupus Erythematosus, Mixed Connective Tissue Disease, Sjogren's Syndrome, and Rheumatoid Arthritis 7% Pathology Of the Kidney*. Philadelphia; Jennette C OLSM, et al., eds. *Pathology Of the Kidney*.
44. D'AGATI V. Renal disease in systemic lupus erythematosus, mixed connective tissue disease, Sjogren syndrome, and rheumatoid arthritis. *Heptinstall's Pathology of the Kidney*. 1998:541–624.
45. Wang S, Broder A, Shao D, et al. Urine proteomics link complement activation with interstitial fibrosis/tubular atrophy in lupus nephritis patients. *Semin Arthritis Rheum.* 2023;63:152263. <https://doi.org/10.1016/j.semarthrit.2023.152263>.
46. Liu T, Yang M, Xia Y, et al. Microarray-based analysis of renal complement components reveals a therapeutic target for lupus nephritis. *Arthritis Res Ther.* 2021;23(1):223. <https://doi.org/10.1186/s13075-021-02605-9>.
47. Bethunaickan R, Berthier CC, Zhang W, Kretzler M, Davidson A. Comparative transcriptional profiling of 3 murine models of SLE nephritis reveals both unique and shared regulatory networks. *PLoS One.* 2013;8(10):e77489. <https://doi.org/10.1371/journal.pone.0077489>.
48. Parikh SV, Malvar A, Song H, et al. Characterising the immune profile of the kidney biopsy at lupus nephritis flare differentiates early treatment responders from non-responders. *Lupus Sci Med.* 2015;2(1):e000112. <https://doi.org/10.1136/lupus-2015-000112>.
49. Birmingham DJ, Bitter JE, Ndukwu EG, et al. Relationship of circulating anti-C3b and anti-C1q IgG to lupus nephritis and its flare. *Clin J Am Soc Nephrol.* 2016;11(1):47–53. <https://doi.org/10.2215/CJN.03990415>.
50. Li QY, Li HY, Fu G, Yu F, Wu Y, Zhao MH. Autoantibodies against C-reactive protein influence complement activation and clinical course in lupus nephritis. *J Am Soc Nephrol.* 2017;28(10):3044–3054. <https://doi.org/10.1681/ASN.2016070735>.
51. Hakkim A, Furnrohr BG, Amann K, et al. Impairment of neutrophil extracellular trap degradation is associated with lupus nephritis. *Proc Natl Acad Sci U S A.* 2010;107(21):9813–9818. <https://doi.org/10.1073/pnas.0909927107>.
52. Villanueva E, Yalavarthi S, Berthier CC, et al. Netting neutrophils induce endothelial damage, infiltrate tissues, and expose immunostimulatory molecules in systemic lupus erythematosus. *J Immunol.* 2011;187(1):538–552. <https://doi.org/10.4049/jimmunol.1100450>.
53. Espeli M, Bokers S, Giannico G, et al. Local renal autoantibody production in lupus nephritis. *J Am Soc Nephrol.* 2011;22(2):296–305. <https://doi.org/10.1681/ASN.2010050515>.
54. Bao L, Haas M, Quigg RJ. Complement factor H deficiency accelerates development of lupus nephritis. *J Am Soc Nephrol.* 2011;22(2):285–295. <https://doi.org/10.1681/ASN.2010060647>.
55. Behara VY, Whittier WL, Korbet SM, Schwartz MM, Martens M, Lewis EJ. Pathogenetic features of severe segmental lupus nephritis. *Nephrol Dial Transplant.* 2010;25(1):153–159. <https://doi.org/10.1093/ndt/gfp424>.
56. Sprangers BAG. In: Lewis ESMKS, ed. *Renal Vascular Involvement in SLE*. Oxford Press; 2010. Lupus Nephritis.
57. Hu W, Chen Y, Wang S, et al. Clinical-morphological features and outcomes of lupus podocytopathy. *Clin J Am Soc Nephrol.* 2016;11(4):585–592. <https://doi.org/10.2215/CJN.06720615>.
58. Bomback AS, Appel GB. Updates on the treatment of lupus nephritis. *J Am Soc Nephrol.* 2010;21(12):2028–2035. <https://doi.org/10.1681/ASN.2010050472>.
59. Hogan J, Appel GB. Update on the treatment of lupus nephritis. *Curr Opin Nephrol Hypertens.* 2013;22(2):224–230. <https://doi.org/10.1097/MNH.0b013e32835d921c>.
60. Appel GBD. *Lupus Nephritis - Pathology and Pathogenesis 7% 7*. In: Wallace DHBH, ed. *Dubois' Lupus Erythematosus*; 2007.
61. Appel GB, Cohen DJ, Pirani CL, Meltzer JI, Estes D. Long-term follow-up of patients with lupus nephritis. A study based on the classification of the World Health Organization. *Am J Med.* 1987;83(5):877–885. [https://doi.org/10.1016/0002-9343\(87\)90645-0](https://doi.org/10.1016/0002-9343(87)90645-0).
62. Churg A, Brallas M, Cronin SR, Churg J. Formes frustes of Churg-Strauss syndrome. *Chest.* 1995;108(2):320–323. <https://doi.org/10.1378/chest.108.2.320>.
63. Weening JJ, D'Agati VD, Schwartz MM, et al. The classification of glomerulonephritis in systemic lupus erythematosus revisited. *Kidney Int.* 2004;65(2):521–530. <https://doi.org/10.1111/j.1523-1755.2004.00443.x>.
64. Markowitz GS, D'Agati VD. The ISN/RPS 2003 classification of lupus nephritis: an assessment at 3 years. *Kidney Int.* 2007;71(6):491–495. <https://doi.org/10.1038/sj.ki.5002118>.
65. Bajema IM, Wilhelmus S, Alpers CE, et al. Revision of the International Society of Nephrology/Renal Pathology Society classification for lupus nephritis: clarification of definitions, and modified National Institutes of Health activity and chronicity indices. *Kidney Int.* 2018;93(4):789–796. <https://doi.org/10.1016/j.kint.2017.11.023>.
66. Magil AB, Puterman ML, Ballon HS, et al. Prognostic factors in diffuse proliferative lupus glomerulonephritis. *Kidney Int.* 1988;34(4):511–517. <https://doi.org/10.1038/ki.1988.211>.
67. Ponticelli C, Zucchelli P, Moroni G, Cagnoli L, Banfi G, Pasquali S. Long-term prognosis of diffuse lupus nephritis. *Clin Nephrol.* 1987;28(6):263–271.
68. Nasr SH, D'Agati VD, Park HR, et al. Necrotizing and crescentic lupus nephritis with antineutrophil cytoplasmic antibody seropositivity. *Clin J Am Soc Nephrol.* 2008;3(3):682–690. <https://doi.org/10.2215/CJN.04391007>.
69. Vandepapeliere J, Aydin S, Cosyns JP, Depresseux G, Jadoul M, Housiau FA. Prognosis of proliferative lupus nephritis subsets in the Louvain Lupus Nephritis inception Cohort. *Lupus.* 2014;23(2):159–165. <https://doi.org/10.1177/0961203313514623>.
70. Rijnink EC, Teng YKO, Wilhelmus S, et al. Clinical and histopathologic characteristics associated with renal outcomes in lupus nephritis. *Clin J Am Soc Nephrol.* 2017;12(5):734–743. <https://doi.org/10.2215/CJN.10601016>.
71. Donadio Jr JV, Burgess JH, Holley KE. Membranous lupus nephropathy: a clinicopathologic study. *Medicine (Baltim).* 1977;56(6):527–536. <https://doi.org/10.1097/00005792-197711000-00007>.
72. Sloan RP, Schwartz MM, Korbet SM, Borok RZ. Long-term outcome in systemic lupus erythematosus membranous glomerulonephritis. Lupus Nephritis Collaborative Study Group. *J Am Soc Nephrol.* 1996;7(2):299–305. <https://doi.org/10.1681/ASN.V72299>.
73. Pasquali S, Banfi G, Zucchelli A, Moroni G, Ponticelli C, Zucchelli P. Lupus membranous nephropathy: long-term outcome. *Clin Nephrol.* 1993;39(4):175–182.
74. Jennette JC, Iskandar SS, Dalldorf FG. Pathologic differentiation between lupus and nonlupus membranous glomerulopathy. *Kidney Int.* 1983;24(3):377–385. <https://doi.org/10.1038/ki.1983.170>.
75. Kudose S, Santoriello D, Bomback AS, Stokes MB, D'Agati VD, Markowitz GS. Sensitivity and specificity of pathologic findings to diagnose lupus nephritis. *Clin J Am Soc Nephrol.* 2019;14(11):1605–1615. <https://doi.org/10.2215/CJN.01570219>.
76. Malvar A, Alberton V, Lococo B, et al. Remission of lupus nephritis: the trajectory of histological response in successfully treated patients. *Lupus Sci Med.* 2023;10(1). <https://doi.org/10.1136/lupus-2023-000932>.
77. D'Agati V, Appel GB. Renal pathology of human immunodeficiency virus infection. *Semin Nephrol.* 1998;18(4):406–421.
78. Sethi S, Madden BJ, Debiec H, et al. Exostosin 1/exostosin 2-associated membranous nephropathy. *J Am Soc Nephrol.* 2019;30(6):1123–1136. <https://doi.org/10.1681/ASN.2018080852>.
79. Ravindran A, Casal Moura M, Fervenza FC, et al. Patients with membranous lupus nephritis, exostosin-positivity and exostosin-negativity represent two different phenotypes. *J Am Soc Nephrol.* 2021;32(3):695–706. <https://doi.org/10.1681/ASN.2020081181>.
80. Caza TN, Storey AJ, Hassen SI, et al. Discovery of seven novel putative antigens in membranous nephropathy and membranous lupus nephritis identified by mass spectrometry. *Kidney Int.* 2023;103(3):593–606. <https://doi.org/10.1016/j.kint.2023.01.001>.

81. Austin 3rd HA, Boumpas DT, Vaughan EM, Balow JE. Predicting renal outcomes in severe lupus nephritis: contributions of clinical and histologic data. *Kidney Int.* 1994;45(2):544–550. <https://doi.org/10.1038/ki.1994.70>.
82. Valeri A, Radhakrishnan J, Estes D, et al. Intravenous pulse cyclophosphamide treatment of severe lupus nephritis: a prospective five-year study. *Clin Nephrol.* 1994;42(2):71–78.
83. Radhakrishnan J, Kunis CL, D'Agati V, Appel GB. Cyclosporine treatment of lupus membranous nephropathy. *Clin Nephrol.* 1994;42(3):147–154.
84. Park MH, D'Agati V, Appel GB, Pirani CL. Tubulointerstitial disease in lupus nephritis: relationship to immune deposits, interstitial inflammation, glomerular changes, renal function, and prognosis. *Nephron.* 1986;44(4):309–319. <https://doi.org/10.1159/000184012>.
85. Hill GS, Delahousse M, Nochy D, Mandet C, Bariety J. Proteinuria and tubulointerstitial lesions in lupus nephritis. *Kidney Int.* 2001;60(5):1893–1903. <https://doi.org/10.1046/j.1523-1755.2001.00017.x>.
86. Daniel L, Sichez H, Giorgi R, et al. Tubular lesions and tubular cell adhesion molecules for the prognosis of lupus nephritis. *Kidney Int.* 2001;60(6):2215–2221. <https://doi.org/10.1046/j.1523-1755.2001.00055.x>.
87. D'Agati VD, Appel GB, Estes D, Knowles 2nd DM, Pirani CL. Monoclonal antibody identification of infiltrating mononuclear leukocytes in lupus nephritis. *Kidney Int.* 1986;30(4):573–581. <https://doi.org/10.1038/ki.1986.223>.
88. Dhingra S, Qureshi R, Abdellatif A, Gaber LW, Truong LD. Tubulointerstitial nephritis in systemic lupus erythematosus: innocent bystander or ominous presage. *Histol Histopathol.* 2014;29(5):553–565. <https://doi.org/10.14670/HH-29.10.553>.
89. Banfi G, Bertani T, Boeri V, et al. Renal vascular lesions as a marker of poor prognosis in patients with lupus nephritis. Gruppo Italiano per lo Studio della Nefrite Lupica (GISNEL). *Am J Kidney Dis.* 1991;18(2):240–248. [https://doi.org/10.1016/s0272-6386\(12\)80885-7](https://doi.org/10.1016/s0272-6386(12)80885-7).
90. Appel GB, Pirani CL, D'Agati V. Renal vascular complications of systemic lupus erythematosus. *J Am Soc Nephrol.* 1994;4(8):1499–1515. <https://doi.org/10.1681/ASN.V481499>.
91. Abdellatif AA, Waris S, Lakhani A, Kadikoy H, Haque W, Truong LD. True vasculitis in lupus nephritis. *Clin Nephrol.* 2010;74(2):106–112. <https://doi.org/10.5414/cnp74106>.
92. Kraft SW, Schwartz MM, Korbet SM, Lewis EJ. Glomerular podocytopathy in patients with systemic lupus erythematosus. *J Am Soc Nephrol.* 2005;16(1):175–179. <https://doi.org/10.1681/ASN.2004050350>.
93. Shea-Simonds P, Cairns TD, Roufosse C, Cook T, Vyse TJ. Lupus podocytopathy. *Rheumatology.* 2009;48(12):1616–1618. <https://doi.org/10.1093/rheumatology/kep256>.
94. Salvatore SP, Barisoni LM, Herzenberg AM, Chander PN, Nicklele V, Seshan SV. Collapsing glomerulopathy in 19 patients with systemic lupus erythematosus or lupus-like disease. *Clin J Am Soc Nephrol.* 2012;7(6):914–925. <https://doi.org/10.2215/CJN.11751111>.
95. Larsen CP, Beggs ML, Saeed M, Walker PD. Apolipoprotein L1 risk variants associate with systemic lupus erythematosus-associated collapsing glomerulopathy. *J Am Soc Nephrol.* 2013;24(5):722–725. <https://doi.org/10.1681/ASN.2012121180>.
96. Markowitz GS, D'Agati VD. Classification of lupus nephritis. *Curr Opin Nephrol Hypertens.* 2009;18(3):220–225. <https://doi.org/10.1097/mnh.0b013e328327b379>.
97. Font J, Cervera R, Lopez-Soto A, Darnell A, Ingelmo M. Mixed membranous and proliferative glomerulonephritis in primary Sjogren's syndrome. *Br J Rheumatol.* 1989;28(6):548–550. <https://doi.org/10.1093/rheumatology/28.6.548>.
98. Santacruz JC, Pulido S, Arzuaga A, Mantilla MJ, Londono J. Lupus cystitis, from myth to reality: a narrative review. *Cureus.* 2021;13(12):e20409. <https://doi.org/10.7759/cureus.20409>.
99. Chen MY, Lee KL, Hsu PN, Wu CS, Wu CH. Is there an ethnic difference in the prevalence of lupus cystitis? A report of six cases. *Lupus.* 2004;13(4):263–269. <https://doi.org/10.1191/0961203304lu527cr>.
100. Koike T, Takabayashi K. Lupus cystitis in the Japanese. *Intern Med.* 1996;35(2):87–88. <https://doi.org/10.2169/internalmedicine.35.87>.
101. Abe Y, Kawamata N, Makiyama A, Ashizawa T, Hayashi T, Tamura N. In lupus cystitis, is the urinary tract dilated or obstructed? *Immun Inflamm Dis.* 2023;11(2):e777. <https://doi.org/10.1002/iid3.777>.
102. Shimizu M, Shimbo A, Yamazaki S, Mori M. Concurrent lupus enteritis and cystitis. *Pediatr Int.* 2021;63(9):1142–1143. <https://doi.org/10.1111/ped.14642>.
103. Wang G, Zhuo N, Tian F, Li J, Wen Z. Two complications in patients with systemic lupus erythematosus: lupus cystitis and lupus enteritis. *Arch Med Sci.* 2022;18(3):822–824. <https://doi.org/10.5114/aoms/146547>.
104. Mukhopadhyay S, Jana S, Roy MK, et al. Lupus cystitis: an unusual presentation of systemic lupus erythematosus. *Indian J Nephrol.* 2014;24(5):308–311. <https://doi.org/10.4103/0971-4065.133010>.
105. Maruoka H, Honda S, Takeo M, Koga T, Fukuda T, Aizawa H. Tacrolimus treatment for refractory lupus cystitis. *Mod Rheumatol.* 2006;16(4):264–266. <https://doi.org/10.1007/s10165-006-0494-2>.
106. Xu L, Song X, Dai Q, et al. Successful treatment of refractory cystitis associated with systemic lupus erythematosus with Belimumab. *Int Immunopharmacol.* 2021;96:107574. <https://doi.org/10.1016/j.intimp.2021.107574>.
107. Mannik M, Merrill CE, Stamps LD, Wener MH. Multiple autoantibodies form the glomerular immune deposits in patients with systemic lupus erythematosus. *J Rheumatol.* 2003;30(7):1495–1504.
108. Foster MH, Cizman B, Madaio MP. Nephritogenic autoantibodies in systemic lupus erythematosus: immunochemical properties, mechanisms of immune deposition, and genetic origins. *Lab Invest.* 1993;69(5):494–507.
109. Johnson RJ, Feehally J, Floege J. *Comprehensive Clinical Nephrology E-Book.* Elsevier Health Sciences; 2014.
110. Sharp GC, Irvin WS, Tan EM, Gould RG, Holman HR. Mixed connective tissue disease—an apparently distinct rheumatic disease syndrome associated with a specific antibody to an extractable nuclear antigen (ENA). *Am J Med.* 1972;52(2):148–159. [https://doi.org/10.1016/0002-9343\(72\)90064-2](https://doi.org/10.1016/0002-9343(72)90064-2).
111. Buyon JP, Clancy RM. Maternal autoantibodies and congenital heart block: mediators, markers, and therapeutic approach. *Semin Arthritis Rheum.* 2003;33(3):140–154. <https://doi.org/10.1016/j.semarthrit.2003.09.002>.
112. Laitman RS, Glicklich D, Sablay LB, Grayzel AI, Barland P, Bank N. Effect of long-term normalization of serum complement levels on the course of lupus nephritis. *Am J Med.* 1989;87(2):132–138. [https://doi.org/10.1016/s0002-9343\(89\)80687-4](https://doi.org/10.1016/s0002-9343(89)80687-4).
113. Walport MJ. Complement. Second of two parts. *N Engl J Med.* 2001;344(15):1140–1144. <https://doi.org/10.1056/NEJM200104123441506>.
114. Greisman SG, Redecha PB, Kimberly RP, Christian CL. Differences among immune complexes: association of C1q in SLE immune complexes with renal disease. *J Immunol.* 1987;138(3):739–745.
115. Noel LH, Droz D, Rothfield NF. Clinical and serologic significance of cutaneous deposits of immunoglobulins, C3, and C1q in SLE patients with nephritis. *Clin Immunol Immunopathol.* 1978;10(4):381–388. [https://doi.org/10.1016/0090-1229\(78\)90149-6](https://doi.org/10.1016/0090-1229(78)90149-6).
116. Rovin BH, Zhang X. Biomarkers for lupus nephritis: the quest continues. *Clin J Am Soc Nephrol.* 2009;4(11):1858–1865. <https://doi.org/10.2215/CJN.03530509>.
117. Reichlin M. Clinical and immunological significance of antibodies to Ro and La in systemic lupus erythematosus. *Arthritis Rheum.* 1982;25(7):767–771. <https://doi.org/10.1002/art.1780250710>.
118. Kowal C, Degiorgio LA, Lee JY, et al. Human lupus autoantibodies against NMDA receptors mediate cognitive impairment. *Proc Natl Acad Sci U S A.* 2006;103(52):19854–19859. <https://doi.org/10.1073/pnas.0608397104>.
119. Morioka T, Woitas R, Fujigaki Y, Batsford SR, Vogt A. Histone mediated glomerular deposition of small size DNA anti-DNA complex. *Kidney Int.* 1994;45(4):991–997. <https://doi.org/10.1038/ki.1994.134>.
120. D'Cruz DP, Houssiau FA, Ramirez G, et al. Antibodies to endothelial cells in systemic lupus erythematosus: a potential marker for nephritis and vasculitis. *Clin Exp Immunol.* 1991;85(2):254–261. <https://doi.org/10.1111/j.1365-2249.1991.tb05714.x>.
121. Zand L, Fervenza FC, Coppo R. Microscopic hematuria as a risk factor for IgAN progression: considering this biomarker in selecting and monitoring patients. *Clin Kidney J.* 2023;16(suppl 2):ii19–ii27. <https://doi.org/10.1093/cjk/sfad232>.

122. De Vriese AS, Sethi S, Fervenza FC. Lupus nephritis: redefining the treatment goals. *Kidney Int.* 2025;107(2):198–211. <https://doi.org/10.1016/j.kint.2024.10.018>.
123. Gold HT, El Shahawy O, Izmirly PM, Masson M, Cohen B, Buyon JP. Evaluation and randomised controlled trial of home urinalysis testing in patients with SLE at elevated risk for developing lupus nephritis: a study protocol. *Lupus Sci Med.* 2024;11(2). <https://doi.org/10.1136/lupus-2024-001390>.
124. Vodehnal S, Mohan C. Urinary biomarkers for active Lupus Nephritis that have survived independent validation across cohorts. *Kidney Int.* 2024;106(6):1135–1145. <https://doi.org/10.1016/j.kint.2024.09.007>.
125. Zhang T, Li H, Vanarsa K, et al. Association of urine sCD163 with proliferative lupus nephritis, fibrinoid necrosis, cellular crescents and intrarenal M2 macrophages. *Front Immunol.* 2020;11:671. <https://doi.org/10.3389/fimmu.2020.00671>.
126. Mejia-Vilet JM, Zhang XL, Cruz C, et al. Urinary soluble CD163: a novel noninvasive biomarker of activity for lupus nephritis. *J Am Soc Nephrol.* 2020;31(6):1335–1347. <https://doi.org/10.1681/ASN.2019121285>.
127. Gupta R, Yadav A, Aggarwal A. Urinary soluble CD163 is a good biomarker for renal disease activity in lupus nephritis. *Clin Rheumatol.* 2021;40(3):941–948. <https://doi.org/10.1007/s10067-020-05343-6>.
128. Fava A, Buyon J, Magder L, et al. Urine proteomic signatures of histological class, activity, chronicity, and treatment response in lupus nephritis. *JCI Insight.* 2024;9(2). <https://doi.org/10.1172/jci.insight.172569>.
129. Tonsawan P, Sawanyawisuth K. Clinical comparisons between previously diagnosed SLE and newly diagnosed SLE by kidney biopsy. *Auto Immun Highlights.* 2020;11(1):18. <https://doi.org/10.1186/s13317-020-00140-2>.
130. Austin 3rd HA, Illei GG, Braun MJ, Balow JE. Randomized, controlled trial of prednisone, cyclophosphamide, and cyclosporine in lupus membranous nephropathy. *J Am Soc Nephrol.* 2009;20(4):901–911. <https://doi.org/10.1681/ASN.2008060665>.
131. Steiman AJ, Gladman DD, Ibanez D, Urowitz MB. Prolonged serologically active clinically quiescent systemic lupus erythematosus: frequency and outcome. *J Rheumatol.* 2010;37(9):1822–1827. <https://doi.org/10.3899/jrheum.100007>.
132. Gao C, Bian X, Wu L, et al. A nomogram predicting the histological activity of lupus nephritis from clinical parameters. *Nephrol Dial Transplant.* 2024;39(3):520–530. <https://doi.org/10.1093/ndt/gfad191>.
133. RL R. In: Wallace DJ HBH, ed. *Drug Induced Lupus 7% 7. Dubois' Lupus Erythematosus*; 2007.
134. Yung RL, Johnson KJ, Richardson BC. New concepts in the pathogenesis of drug-induced lupus. *Lab Invest.* 1995;73(6):746–759.
135. Crowson AN, Magro CM. Diltiazem and subacute cutaneous lupus erythematosus-like lesions. *N Engl J Med.* 1995;333(21):1429. <https://doi.org/10.1056/NEJM199511233332117>.
136. Gordon MM, Porter D. Minocycline induced lupus: case series in the West of Scotland. *J Rheumatol.* 2001;28(5):1004–1006.
137. Debandt M, Vittecoq O, Descamps V, Le Loet X, Meyer O. Anti-TNF-alpha-induced systemic lupus syndrome. *Clin Rheumatol.* 2003;22(1):56–61. <https://doi.org/10.1007/s10067-002-0654-5>.
138. Shakoor N, Michalska M, Harris CA, Block JA. Drug-induced systemic lupus erythematosus associated with etanercept therapy. *Lancet.* 2002;359(9306):579–580. [https://doi.org/10.1016/S0140-6736\(02\)07714-0](https://doi.org/10.1016/S0140-6736(02)07714-0).
139. Shapiro KS, Pinn VW, Harrington JT, Levey AS. Immune complex glomerulonephritis in hydralazine-induced SLE. *Am J Kidney Dis.* 1984;3(4):270–274. [https://doi.org/10.1016/s0272-6386\(84\)80044-x](https://doi.org/10.1016/s0272-6386(84)80044-x).
140. Short AK, Lockwood CM. Antigen specificity in hydralazine associated ANCA positive systemic vasculitis. *QJM.* 1995;88(11):775–783.
141. Burlingame RW, Rubin RL. Drug-induced anti-histone auto-antibodies display two patterns of reactivity with substructures of chromatin. *J Clin Invest.* 1991;88(2):680–690. <https://doi.org/10.1172/JCI115353>.
142. Bundhun PK, Soogund MZ, Huang F. Impact of systemic lupus erythematosus on maternal and fetal outcomes following pregnancy: a meta-analysis of studies published between years 2001–2016. *J Autoimmun.* 2017;79:17–27. <https://doi.org/10.1016/j.jaut.2017.02.009>.
143. Buyon JP, Kim MY, Guerra MM, et al. Predictors of pregnancy outcomes in patients with lupus: a cohort study. *Ann Intern Med.* 2015;163(3):153–163. <https://doi.org/10.7326/M14-2235>.
144. Wind M, Fierro JJ, Bloemenkamp KWM, et al. Pregnancy outcome predictors in systemic lupus erythematosus: a systematic review and meta-analysis. *Lancet Rheumatol.* 2024;6(10):e667–e683. [https://doi.org/10.1016/S2665-9913\(24\)00160-7](https://doi.org/10.1016/S2665-9913(24)00160-7).
145. Saavedra MA, Sanchez A, Morales S, Navarro-Zarza JE, Angeles U, Jara LJ. Primigravida is associated with flare in women with systemic lupus erythematosus. *Lupus.* 2015;24(2):180–185. <https://doi.org/10.1177/0961203314552116>.
146. Shen G, Swaminathan M, Huang I, et al. Racial disparities in pregnancy outcomes among women with rheumatic diseases: a systematic literature review. *Semin Arthritis Rheum.* 2023;60:152193. <https://doi.org/10.1016/j.semarthrit.2023.152193>.
147. Kwok LW, Tam LS, Zhu T, Leung YY, Li E. Predictors of maternal and fetal outcomes in pregnancies of patients with systemic lupus erythematosus. *Lupus.* 2011;20(8):829–836. <https://doi.org/10.1177/0961203310397967>.
148. Smyth A, Oliveira GH, Lahr BD, Bailey KR, Norby SM, Garovic VD. A systematic review and meta-analysis of pregnancy outcomes in patients with systemic lupus erythematosus and lupus nephritis. *Clin J Am Soc Nephrol.* 2010;5(11):2060–2068. <https://doi.org/10.2215/CJN.00240110>.
149. Davis-Porada J, Kim MY, Guerra MM, et al. Low frequency of flares during pregnancy and post-partum in stable lupus patients. *Arthritis Res Ther.* 2020;22(1):52. <https://doi.org/10.1186/s13075-020-2139-9>.
150. Radin M, Cecchi I, Crisafulli F, et al. Complement levels during the first trimester predict disease flare and adverse pregnancy outcomes in systemic lupus erythematosus: a network meta-analysis on 532 pregnancies. *Autoimmun Rev.* 2023;22(12):103467. <https://doi.org/10.1016/j.autrev.2023.103467>.
151. Hayslett JP. Maternal and fetal complications in pregnant women with systemic lupus erythematosus. *Am J Kidney Dis.* 1991;17(2):123–126. [https://doi.org/10.1016/s0272-6386\(12\)81115-2](https://doi.org/10.1016/s0272-6386(12)81115-2).
152. Julkunen H, Kaaja R, Palosuo T, Gronhagen-Riska C, Teramo K. Pregnancy in lupus nephropathy. *Acta Obstet Gynecol Scand.* 1993;72(4):258–263. <https://doi.org/10.3109/00016349309068034>.
153. Clowse ME, Magder L, Witter F, Petri M. Hydroxychloroquine in lupus pregnancy. *Arthritis Rheum.* 2006;54(11):3640–3647. <https://doi.org/10.1002/art.22159>.
154. Izmirly P, Saxena A, Buyon JP. Progress in the pathogenesis and treatment of cardiac manifestations of neonatal lupus. *Curr Opin Rheumatol.* 2017;29(5):467–472. <https://doi.org/10.1097/BOR.0000000000000414>.
155. Huybrechts KF, Bateman BT, Zhu Y, et al. Hydroxychloroquine early in pregnancy and risk of birth defects. *Am J Obstet Gynecol.* 2021;224(3):290 e1–e290. <https://doi.org/10.1016/j.ajog.2020.09.007>.
156. Bermas BL, Chambers C. Hydroxychloroquine early in pregnancy and risk of birth defects: don't throw out the baby with the bathwater. *Am J Obstet Gynecol.* 2021;224(5):548–549. <https://doi.org/10.1016/j.ajog.2020.12.1218>.
157. Michalski AM, Richardson SD, Browne ML, et al. Sex ratios among infants with birth defects, national birth defects prevention study, 1997–2009. *Am J Med Genet.* 2015;167A(5):1071–1081. <https://doi.org/10.1002/ajmg.a.36865>.
158. Kao JH, Lan TY, Lu CH, et al. Pregnancy outcomes in patients treated with belimumab: report from real-world experience. *Semin Arthritis Rheum.* 2021;51(5):963–968. <https://doi.org/10.1016/j.semarthrit.2021.06.005>.
159. Petri M, Landy H, Clowse MEB, et al. Belimumab use during pregnancy: a summary of birth defects and pregnancy loss from belimumab clinical trials, a pregnancy registry and postmarketing reports. *Ann Rheum Dis.* 2023;82(2):217–225. <https://doi.org/10.1136/ard-2022-222505>.
160. Petri M, Howard D, Repke J. Frequency of lupus flare in pregnancy. The hopkins lupus pregnancy center experience. *Arthritis Rheum.* 1991;34(12):1538–1545. <https://doi.org/10.1002/art.1780341210>.
161. Moroni G, Quaglini S, Banfi G, et al. Pregnancy in lupus nephritis. *Am J Kidney Dis.* 2002;40(4):713–720. <https://doi.org/10.1053/ajkd.2002.35678>.

162. Mehta B, Luo Y, Xu J, et al. Trends in maternal and fetal outcomes among pregnant women with systemic lupus erythematosus in the United States: a cross-sectional analysis. *Ann Intern Med.* 2019;171(3):164–171. <https://doi.org/10.7326/M19-0120>.
163. McNeil HP, Chesterman CN, Krilis SA. Immunology and clinical importance of antiphospholipid antibodies. *Adv Immunol.* 1991;49:193–280. [https://doi.org/10.1016/s0065-2776\(08\)60777-4](https://doi.org/10.1016/s0065-2776(08)60777-4).
164. de Fremont GM, Salmon JE, Costedoat-Chalumeau N. Pregnancy outcome predictors in systemic lupus erythematosus: prospective for brighter perspectives. *Lancet Rheumatol.* 2024;6(10):e654–e655. [https://doi.org/10.1016/S2665-9913\(24\)00184-X](https://doi.org/10.1016/S2665-9913(24)00184-X).
165. Hocaoglu M, Valenzuela-Almada MO, Dabit JY, et al. Incidence, prevalence, and mortality of lupus nephritis: a population-based study over four decades using the lupus midwest network. *Arthritis Rheumatol.* 2023;75(4):567–573. <https://doi.org/10.1002/art.42375>.
166. Chrysostomou C, Faustini F, Gunnarsson I, et al. Long-term outcomes of lupus nephritis in comparison to other CKD etiologies. *Kidney Int Rep.* 2025;10(1):157–168. <https://doi.org/10.1016/j.ekir.2024.10.021>.
167. Hermansen M-L, Lindhardsen J, Torp-Pedersen C, Faurschou M, Jacobsen S. The risk of cardiovascular morbidity and cardiovascular mortality in systemic lupus erythematosus and lupus nephritis: a Danish nationwide population-based cohort study. *Rheumatology.* 2017;kew475. <https://doi.org/10.1093/rheumatology/kew475>.
168. Gustafsson JT, Herlitz Lindberg M, Gunnarsson I, et al. Excess atherosclerosis in systemic lupus erythematosus, A matter of renal involvement: case control study of 281 SLE patients and 281 individually matched population controls. *PLoS One.* 2017;12(4):e0174572. <https://doi.org/10.1371/journal.pone.0174572>.
169. Sarnak MJ, Levey AS, Schoolwerth AC, et al. Kidney disease as a risk factor for development of cardiovascular disease: a statement from the American heart association councils on kidney in cardiovascular disease, high blood pressure Research, clinical cardiology, and epidemiology and prevention. *Circulation.* 2003;108(17):2154–2169. <https://doi.org/10.1161/01.CIR.0000095676.90936.80>.
170. Thong KM, Chan TM. Infectious complications in lupus nephritis treatment: a systematic review and meta-analysis. *Lupus.* 2019;28(3):334–346. <https://doi.org/10.1177/0961203319829817>.
171. Korbet SM, Lewis EJ, Schwartz MM, Reichlin M, Evans J, Rohde RD. Factors predictive of outcome in severe lupus nephritis. Lupus Nephritis Collaborative Study Group. *Am J Kidney Dis.* 2000;35(5):904–914. [https://doi.org/10.1016/s0272-6386\(00\)70262-9](https://doi.org/10.1016/s0272-6386(00)70262-9).
172. Chen YE, Korbet SM, Katz RS, Schwartz MM, Lewis EJ, Collaborative Study G. Value of a complete or partial remission in severe lupus nephritis. *Clin J Am Soc Nephrol.* 2008;3(1):46–53. <https://doi.org/10.2215/CJN.03280807>.
173. Hanly JG, O’Keeffe AG, Su L, et al. The frequency and outcome of lupus nephritis: results from an international inception cohort study. *Rheumatology.* 2016;55(2):252–262. <https://doi.org/10.1093/rheumatology/kev311>.
174. Moroni G, Vercelloni PG, Quaglini S, et al. Changing patterns in clinical-histological presentation and renal outcome over the last five decades in a cohort of 499 patients with lupus nephritis. *Ann Rheum Dis.* 2018;77(9):1318–1325. <https://doi.org/10.1136/annrheumdis-2017-212732>.
175. Nossent HC, Henzen-Logmans SC, Vroom TM, Berden JH, Swaak TJ. Contribution of renal biopsy data in predicting outcome in lupus nephritis. Analysis of 116 patients. *Arthritis Rheum.* 1990;33(7):970–977. <https://doi.org/10.1002/art.1780330708>.
176. Esdaile JM, Federgreen W, Quintal H, Suissa S, Hayslett JP, Kashgarian M. Predictors of one year outcome in lupus nephritis: the importance of renal biopsy. *Q J Med.* 1991;81(295):907–918.
177. Schwartz MM, Lan SP, Bernstein J, Hill GS, Holley K, Lewis EJ. Role of pathology indices in the management of severe lupus glomerulonephritis. Lupus Nephritis Collaborative Study Group. *Kidney Int.* 1992;42(3):743–748. <https://doi.org/10.1038/ki.1992.342>.
178. Balow JE, Boumpas DT, Fessler BJ, Austin 3rd HA. Management of lupus nephritis. *Kidney Int Suppl.* 1996;53:S88–S92.
179. Najafi CC, Korbet SM, Lewis EJ, et al. Significance of histologic patterns of glomerular injury upon long-term prognosis in severe lupus glomerulonephritis. *Kidney Int.* 2001;59(6):2156–2163. <https://doi.org/10.1046/j.1523-1755.2001.00730.x>.
180. Steinberg AD, Steinberg SC. Long-term preservation of renal function in patients with lupus nephritis receiving treatment that includes cyclophosphamide versus those treated with prednisone only. *Arthritis Rheum.* 1991;34(8):945–950. <https://doi.org/10.1002/art.1780340803>.
181. Gourley MF, Austin 3rd HA, Scott D, et al. Methylprednisolone and cyclophosphamide, alone or in combination, in patients with lupus nephritis. A randomized, controlled trial. *Ann Intern Med.* 1996;125(7):549–557. <https://doi.org/10.7326/0003-4819-125-7-199610010-00003>.
182. Boumpas DT, Austin 3rd HA, Vaughn EM, et al. Controlled trial of pulse methylprednisolone versus two regimens of pulse cyclophosphamide in severe lupus nephritis. *Lancet.* 1992;340(8822):741–745. [https://doi.org/10.1016/0140-6736\(92\)92292-n](https://doi.org/10.1016/0140-6736(92)92292-n).
183. Dooley MA, Hogan S, Jennette C, Falk R. Cyclophosphamide therapy for lupus nephritis: poor renal survival in black Americans. Glomerular Disease Collaborative Network. *Kidney Int.* 1997;51(4):1188–1195. <https://doi.org/10.1038/ki.1997.162>.
184. Hom G, Graham RR, Modrek B, et al. Association of systemic lupus erythematosus with C8orf13-BLK and ITGAM-ITGAX. *N Engl J Med.* 2008;358(9):900–909. <https://doi.org/10.1056/NEJMoa0707865>.
185. Bakir AA, Levy PS, Dunea G. The prognosis of lupus nephritis in African-Americans: a retrospective analysis. *Am J Kidney Dis.* 1994;24(2):159–171. [https://doi.org/10.1016/s0272-6386\(12\)80177-6](https://doi.org/10.1016/s0272-6386(12)80177-6).
186. Tejani A, Nicastrì AD, Chen CK, Fikrig S, Gurumurthy K. Lupus nephritis in black and Hispanic children. *Am J Dis Child.* 1983;137(5):481–483. <https://doi.org/10.1001/archpedi.1983.02140310059017>.
187. Barette S, Calvez V, Mouquet C, et al. Clinical features and contribution of virological findings to the management of Kaposi sarcoma in organ-allograft recipients. *Arch Dermatol.* 2000;136(12):1452–1458. <https://doi.org/10.1001/archderm.136.12.1452>.
188. Houssiau FA, Vasconcelos C, D’Cruz D, et al. Early response to immunosuppressive therapy predicts good renal outcome in lupus nephritis: lessons from long-term followup of patients in the Euro-Lupus Nephritis Trial. *Arthritis Rheum.* 2004;50(12):3934–3940. <https://doi.org/10.1002/art.20666>.
189. Moroni G, Quaglini S, Maccario M, Banfi G, Ponticelli C. “Nephritic flares” are predictors of bad long-term renal outcome in lupus nephritis. *Kidney Int.* 1996;50(6):2047–2053. <https://doi.org/10.1038/ki.1996.528>.
190. Mosca M, Bencivelli W, Neri R, et al. Renal flares in 91 SLE patients with diffuse proliferative glomerulonephritis. *Kidney Int.* 2002;61(4):1502–1509. <https://doi.org/10.1046/j.1523-1755.2002.00280.x>.
191. Sprangers B, Monahan M, Appel GB. Diagnosis and treatment of lupus nephritis flares—an update. *Nat Rev Nephrol.* 2012;8(12):709–717. <https://doi.org/10.1038/nrneph.2012.220>.
192. Ioannidis JP, Boki KA, Katsorida ME, et al. Remission, relapse, and re-mission of proliferative lupus nephritis treated with cyclophosphamide. *Kidney Int.* 2000;57(1):258–264. <https://doi.org/10.1046/j.1523-1755.2000.00832.x>.
193. Pablos JL, Gutierrez-Millet V, Gomez-Reino JJ. Remission of lupus nephritis with cyclophosphamide and late relapses following therapy withdrawal. *Scand J Rheumatol.* 1994;23(3):142–144. <https://doi.org/10.3109/03009749409103047>.
194. Ciruelo E, de la Cruz J, Lopez I, Gomez-Reino JJ. Cumulative rate of relapse of lupus nephritis after successful treatment with cyclophosphamide. *Arthritis Rheum.* 1996;39(12):2028–2034. <https://doi.org/10.1002/art.1780391212>.
195. Parikh SV, Nagaraja HN, Hebert L, Rovin BH. Renal flare as a predictor of incident and progressive CKD in patients with lupus nephritis. *Clin J Am Soc Nephrol.* 2014;9(2):279–284. <https://doi.org/10.2215/CJN.05040513>.
196. Hill GS, Delahousse M, Nochy D, et al. Outcome of relapse in lupus nephritis: roles of reversal of renal fibrosis and response of inflammation to therapy. *Kidney Int.* 2002;61(6):2176–2186. <https://doi.org/10.1046/j.1523-1755.2002.00357.x>.
197. Conlon PJ, Fischer CA, Levesque MC, et al. Clinical, biochemical and pathological predictors of poor response to intravenous cyclophosphamide in patients with proliferative lupus nephritis. *Clin Nephrol.* 1996;46(3):170–175.

198. Schwartz MM. The Holy Grail: pathological indices in lupus nephritis. *Kidney Int.* 2000;58(3):1354–1355. <https://doi.org/10.1046/j.1523-1755.2000.00293.x>.
199. Malvar A, Pirruccio P, Alberton V, et al. Histologic versus clinical remission in proliferative lupus nephritis. *Nephrol Dial Transplant.* 2017;32(8):1338–1344. <https://doi.org/10.1093/ndt/gfv296>.
200. Zickert A, Sundelin B, Svenungsson E, Gunnarsson I. Role of early repeated renal biopsies in lupus nephritis. *Lupus Sci Med.* 2014;1(1):e000018. <https://doi.org/10.1136/lupus-2014-000018>.
201. Kharouf F, Li Q, Whittall Garcia LP, Jauhal A, Gladman DD, Touma Z. Short and long-term outcomes of patients with pure membranous lupus nephritis compared to patients with proliferative disease. *Rheumatology.* 2025;64(4):1912–1922. <https://doi.org/10.1093/rheumatology/keae436>.
202. Almaani S, Meara A, Rovin BH. Update on lupus nephritis. *Clin J Am Soc Nephrol.* 2017;12(5):825–835. <https://doi.org/10.2215/CJN.05780616>.
203. Parikh SV, Almaani S, Brodsky S, Rovin BH. Update on lupus nephritis: core curriculum 2020. *Am J Kidney Dis.* 2020;76(2):265–281. <https://doi.org/10.1053/j.ajkd.2019.10.017>.
204. Chagnac A, Kiberd BA, Farinas MC, et al. Outcome of the acute glomerular injury in proliferative lupus nephritis. *J Clin Invest.* 1989;84(3):922–930. <https://doi.org/10.1172/JCI114254>.
205. Ward MM. Cardiovascular and cerebrovascular morbidity and mortality among women with end-stage renal disease attributable to lupus nephritis. *Am J Kidney Dis.* 2000;36(3):516–525. <https://doi.org/10.1053/ajkd.2000.9792>.
206. Cheigh JS, Stenzel KH. End-stage renal disease in systemic lupus erythematosus. *Am J Kidney Dis.* 1993;21(1):2–8. [https://doi.org/10.1016/s0272-6386\(12\)80712-8](https://doi.org/10.1016/s0272-6386(12)80712-8).
207. Ward MM. Outcomes of renal transplantation among patients with end-stage renal disease caused by lupus nephritis. *Kidney Int.* 2000;57(5):2136–2143. <https://doi.org/10.1046/j.1523-1755.2000.00065.x>.
208. Kidney Disease: Improving Global Outcomes Lupus Nephritis Work G. KDIGO 2024 clinical practice guideline for the management of LUPUS NEPHRITIS. *Kidney Int.* 2024;105(1S):S1–S69. <https://doi.org/10.1016/j.kint.2023.09.002>.
209. Lochhead KM, Pirsch JD, D'Alessandro AM, et al. Risk factors for renal allograft loss in patients with systemic lupus erythematosus. *Kidney Int.* 1996;49(2):512–517. <https://doi.org/10.1038/ki.1996.73>.
210. Stone JH, Millward CL, Olson JL, Amend WJ, Criswell LA. Frequency of recurrent lupus nephritis among ninety-seven renal transplant patients during the cyclosporine era. *Arthritis Rheum.* 1998;41(4):678–686. [https://doi.org/10.1002/1529-0131\(199804\)41:4<678::AID-ART15>3.0.CO;2-7](https://doi.org/10.1002/1529-0131(199804)41:4<678::AID-ART15>3.0.CO;2-7).
211. Contreras G, Mattiazzi A, Guerra G, et al. Recurrence of lupus nephritis after kidney transplantation. *J Am Soc Nephrol.* 2010;21(7):1200–1207. <https://doi.org/10.1681/ASN.2009101093>.
212. Norby GE, Strom EH, Midtvedt K, et al. Recurrent lupus nephritis after kidney transplantation: a surveillance biopsy study. *Ann Rheum Dis.* 2010;69(8):1484–1487. <https://doi.org/10.1136/ard.2009.122796>.
213. Radhakrishnan J, Williams GS, Appel GB, Cohen DJ. Renal transplantation in anticardiolipin antibody-positive lupus erythematosus patients. *Am J Kidney Dis.* 1994;23(2):286–289. [https://doi.org/10.1016/s0272-6386\(12\)80986-3](https://doi.org/10.1016/s0272-6386(12)80986-3).
214. Stone JH, Amend WJ, Criswell LA. Antiphospholipid antibody syndrome in renal transplantation: occurrence of clinical events in 96 consecutive patients with systemic lupus erythematosus. *Am J Kidney Dis.* 1999;34(6):1040–1047. [https://doi.org/10.1016/S0272-6386\(99\)70009-0](https://doi.org/10.1016/S0272-6386(99)70009-0).
215. Morris HK, Canetta PA, Appel GB. Impact of the ALMS and MAINTAIN trials on the management of lupus nephritis. *Nephrol Dial Transplant.* 2013;28(6):1371–1376. <https://doi.org/10.1093/ndt/gfs447>.
216. Fanouriakis A, Kostopoulou M, Andersen J, et al. EULAR recommendations for the management of systemic lupus erythematosus: 2023 update. *Ann Rheum Dis.* 2024;83(1):15–29. <https://doi.org/10.1136/ard-2023-224762>.
217. Pena-Vizcarra OR, Zavala-Miranda MF, Juarez-Cuevas B, et al. Effect of antimalarials on clinical outcomes in lupus nephritis. *Rheumatology.* 2024;63(8):2230–2238. <https://doi.org/10.1093/rheumatology/kead576>.
218. Gladman DD, Urowitz MB, Rahman P, Ibanez D, Tam LS. Accrual of organ damage over time in patients with systemic lupus erythematosus. *J Rheumatol.* 2003;30(9):1955–1959.
219. Tselios K, Gladman DD, Al-Sheikh H, Su J, Urowitz MB. Medium versus high initial prednisone dose for remission induction in lupus nephritis: a propensity score-matched analysis. *Arthritis Care Res.* 2022;74(9):1451–1458. <https://doi.org/10.1002/acr.24592>.
220. Rovin BH, Solomons N, Pendergraft 3rd WF, et al. A randomized, controlled double-blind study comparing the efficacy and safety of dose-ranging voclosporin with placebo in achieving remission in patients with active lupus nephritis. *Kidney Int.* 2019;95(1):219–231. <https://doi.org/10.1016/j.kint.2018.08.025>.
221. Rovin BH, Furie R, Latinis K, et al. Efficacy and safety of rituximab in patients with active proliferative lupus nephritis: the Lupus Nephritis Assessment with Rituximab study. *Arthritis Rheum.* 2012;64(4):1215–1226. <https://doi.org/10.1002/art.34359>.
222. Kallas R, Li J, Petri M. Predictors of osteonecrosis in systemic lupus erythematosus: a prospective cohort study. *Arthritis Care Res.* 2022;74(7):1122–1132. <https://doi.org/10.1002/acr.24541>.
223. Figueroa-Parra G, Bautista-Vargas M, Navarro-Mendoza E, Duarte-Garcia A. Optimal glucocorticoid therapy in lupus nephritis. *Nephrol Dial Transplant.* 2024;gfae294. <https://doi.org/10.1093/ndt/gfae294>.
224. Bose B, Silverman ED, Bargman JM. Ten common mistakes in the management of lupus nephritis. *Am J Kidney Dis.* 2014;63(4):667–676. <https://doi.org/10.1053/j.ajkd.2013.10.056>.
225. Austin 3rd HA, Klippel JH, Balow JE, et al. Therapy of lupus nephritis. Controlled trial of prednisone and cytotoxic drugs. *N Engl J Med.* 1986;314(10):614–619. <https://doi.org/10.1056/NEJM198603063141004>.
226. Anders HJ, Appel GB. Lupus nephritis: implications of the new ACR lupus nephritis guidelines. *Nat Rev Nephrol.* 2012;8(9):500–501. <https://doi.org/10.1038/nrneph.2012.162>.
227. Illei GG, Austin HA, Crane M, et al. Combination therapy with pulse cyclophosphamide plus pulse methylprednisolone improves long-term renal outcome without adding toxicity in patients with lupus nephritis. *Ann Intern Med.* 2001;135(4):248–257. <https://doi.org/10.7326/0003-4819-135-4-200108210-00009>.
228. Rathi M, Goyal A, Jaryal A, et al. Comparison of low-dose intravenous cyclophosphamide with oral mycophenolate mofetil in the treatment of lupus nephritis. *Kidney Int.* 2016;89(1):235–242. <https://doi.org/10.1038/ki.2015.318>.
229. Group AT. Treatment of lupus nephritis with abatacept: the abatacept and cyclophosphamide combination efficacy and safety study. *Arthritis Rheumatol.* 2014;66(11):3096–3104. <https://doi.org/10.1002/art.38790>.
230. Boumpas DT, Austin 3rd HA, Vaughan EM, Yarboro CH, Klippel JH, Balow JE. Risk for sustained amenorrhea in patients with systemic lupus erythematosus receiving intermittent pulse cyclophosphamide therapy. *Ann Intern Med.* 1993;119(5):366–369. <https://doi.org/10.7326/0003-4819-119-5-199309010-00003>.
231. Koyama H, Wada T, Nishizawa Y, Iwanaga T, Aoki Y. Cyclophosphamide-induced ovarian failure and its therapeutic significance in patients with breast cancer. *Cancer.* 1977;39(4):1403–1409. [https://doi.org/10.1002/1097-0142\(197704\)39:4<1403::aid-cnrcr2820390408>3.0.co;2-8](https://doi.org/10.1002/1097-0142(197704)39:4<1403::aid-cnrcr2820390408>3.0.co;2-8).
232. Tamirou F, Husson SN, Gruson D, Debieve F, Lauwerys BR, Houssiau FA. Brief report: the euro-lupus low-dose intravenous cyclophosphamide regimen does not impact the ovarian reserve, as measured by serum levels of anti-mullerian hormone. *Arthritis Rheumatol.* 2017;69(6):1267–1271. <https://doi.org/10.1002/art.40079>.
233. Rivkees SA, Crawford JD. The relationship of gonadal activity and chemotherapy-induced gonadal damage. *JAMA.* 1988;259(14):2123–2125.
234. Heijl C, Westman K, Hoglund P, Mohammad AJ. Malignancies in patients with antineutrophil cytoplasmic antibody-associated vasculitis: a population-based cohort study. *J Rheumatol.* 2020;47(8):1229–1237. <https://doi.org/10.3899/jrheum.181438>.
235. Appel AS, Appel GB. An update on the use of mycophenolate mofetil in lupus nephritis and other primary glomerular diseases. *Nat Clin Pract Nephrol.* 2009;5(3):132–142. <https://doi.org/10.1038/ncpneph1036>.
236. Chan TM, Li FK, Tang CS, et al. Efficacy of mycophenolate mofetil in patients with diffuse proliferative lupus

- nephritis. Hong Kong-Guangzhou Nephrology Study Group. *N Engl J Med*. 2000;343(16):1156–1162. <https://doi.org/10.1056/NEJM200010193431604>.
237. Chan TM, Tse KC, Tang CS, Mok MY, Li FK. Hong Kong Nephrology Study G. Long-term study of mycophenolate mofetil as continuous induction and maintenance treatment for diffuse proliferative lupus nephritis. *J Am Soc Nephrol*. 2005;16(4):1076–1084. <https://doi.org/10.1681/ASN.2004080686>.
 238. Ginzler EM, Dooley MA, Aranow C, et al. Mycophenolate mofetil or intravenous cyclophosphamide for lupus nephritis. *N Engl J Med*. 2005;353(21):2219–2228. <https://doi.org/10.1056/NEJMoa043731>.
 239. Isenberg D, Appel GB, Contreras G, et al. Influence of race/ethnicity on response to lupus nephritis treatment: the ALMS study. *Rheumatology*. 2010;49(1):128–140. <https://doi.org/10.1093/rheumatology/kep346>.
 240. Walsh M, Solomons N, Lisk L, Jayne DR. Mycophenolate mofetil or intravenous cyclophosphamide for lupus nephritis with poor kidney function: a subgroup analysis of the Aspreva Lupus Management Study. *Am J Kidney Dis*. 2013;61(5):710–715. <https://doi.org/10.1053/j.ajkd.2012.11.042>.
 241. Tang Z, Yang G, Yu C, et al. Effects of mycophenolate mofetil for patients with crescentic lupus nephritis. *Nephrology*. 2008;13(8):702–707. <https://doi.org/10.1111/j.1440-1797.2008.00975.x>.
 242. Parikh SV, Rovin BH. Current and emerging therapies for lupus nephritis. *J Am Soc Nephrol*. 2016;27(10):2929–2939. <https://doi.org/10.1681/ASN.2016040415>.
 243. Rafael-Vidal C, Altabas I, Perez N, Mourino Rodriguez C, Pego-Reigosa JM, Garcia S. Calcineurin and systemic lupus erythematosus: the rationale for using calcineurin inhibitors in the treatment of lupus nephritis. *Int J Mol Sci*. 2021;22(3). <https://doi.org/10.3390/ijms22031263>.
 244. Bao H, Liu ZH, Xie HL, Hu WX, Zhang HT, Li LS. Successful treatment of class V+IV lupus nephritis with multitarget therapy. *J Am Soc Nephrol*. 2008;19(10):2001–2010. <https://doi.org/10.1681/ASN.2007121272>.
 245. Liu Z, Zhang H, Liu Z, et al. Multitarget therapy for induction treatment of lupus nephritis: a randomized trial. *Ann Intern Med*. 2015;162(1):18–26. <https://doi.org/10.7326/M14-1030>.
 246. Zhang H, Liu Z, Zhou M, et al. Multitarget therapy for maintenance treatment of lupus nephritis. *J Am Soc Nephrol*. 2017;28(12):3671–3678. <https://doi.org/10.1681/ASN.2017030263>.
 247. Rovin BH, Teng YKO, Ginzler EM, et al. Efficacy and safety of voclosporin versus placebo for lupus nephritis (AURORA 1): a double-blind, randomised, multicentre, placebo-controlled, phase 3 trial. *Lancet*. 2021;397(10289):2070–2080. [https://doi.org/10.1016/S0140-6736\(21\)00578-X](https://doi.org/10.1016/S0140-6736(21)00578-X).
 248. Arriens C, Teng YKO, Ginzler EM, et al. Update on the efficacy and safety profile of voclosporin: an integrated analysis of clinical trials in lupus nephritis. *Arthritis Care Res*. 2023;75(7):1399–1408. <https://doi.org/10.1002/acr.25007>.
 249. McArn AC, Nixon AR, Jarrell KL. Voclosporin: a novel calcineurin inhibitor for the treatment of lupus nephritis. *Ann Pharmacother*. 2022;10600280221075331. <https://doi.org/10.1177/10600280221075331>.
 250. van Gelder T. How cyclosporine reduces mycophenolic acid exposure by 40% while other calcineurin inhibitors do not. *Kidney Int*. 2021;100(6):1185–1189. <https://doi.org/10.1016/j.kint.2021.06.036>.
 251. Thurman JM. BLISS in the treatment of lupus nephritis. *Clin J Am Soc Nephrol*. 2021;16(6):969–971. <https://doi.org/10.2215/CJN.17991120>.
 252. Furie R, Petri M, Zamani O, et al. A phase III, randomized, placebo-controlled study of belimumab, a monoclonal antibody that inhibits B lymphocyte stimulator, in patients with systemic lupus erythematosus. *Arthritis Rheum*. 2011;63(12):3918–3930. <https://doi.org/10.1002/art.30613>.
 253. Furie R, Rovin BH, Houssiau F, et al. Two-year, randomized, controlled trial of belimumab in lupus nephritis. *N Engl J Med*. 2020;383(12):1117–1128. <https://doi.org/10.1056/NEJMoa2001180>.
 254. Rovin BH, Furie R, Teng YKO, et al. A secondary analysis of the Belimumab International Study in Lupus Nephritis trial examined effects of belimumab on kidney outcomes and preservation of kidney function in patients with lupus nephritis. *Kidney Int*. 2022;101(2):403–413. <https://doi.org/10.1016/j.kint.2021.08.027>.
 255. Struempfer H, Chen C, Cai W. Population pharmacokinetics of belimumab following intravenous administration in patients with systemic lupus erythematosus. *J Clin Pharmacol*. 2013;53(7):711–720. <https://doi.org/10.1002/jcph.104>.
 256. Anders HJ, Furie R, Malvar A, et al. Effect of belimumab on kidney-related outcomes in patients with lupus nephritis: post hoc subgroup analyses of the phase 3 BLISS-LN trial. *Nephrol Dial Transplant*. 2023;38(12):2733–2742. <https://doi.org/10.1093/ndt/gfad167>.
 257. Perez-Arias AA, Marquez-Macedo SE, Pena-Vizcarra OR, et al. The influence of repeated flares in response to therapy and prognosis in lupus nephritis. *Nephrol Dial Transplant*. 2023;38(4):884–893. <https://doi.org/10.1093/ndt/gfac304>.
 258. Contreras G, Pardo V, Leclercq B, et al. Sequential therapies for proliferative lupus nephritis. *N Engl J Med*. 2004;350(10):971–980. <https://doi.org/10.1056/NEJMoa031855>.
 259. Houssiau FA, D'Cruz D, Sangle S, et al. Azathioprine versus mycophenolate mofetil for long-term immunosuppression in lupus nephritis: results from the MAINTAIN Nephritis Trial. *Ann Rheum Dis*. 2010;69(12):2083–2089. <https://doi.org/10.1136/ard.2010.131995>.
 260. Dooley MA, Jayne D, Ginzler EM, et al. Mycophenolate versus azathioprine as maintenance therapy for lupus nephritis. *N Engl J Med*. 2011;365(20):1886–1895. <https://doi.org/10.1056/NEJMoa1014460>.
 261. Jourde-Chiche N, Costedoat-Chalumeau N, Baumstarck K, et al. Weaning of maintenance immunosuppressive therapy in lupus nephritis (WIN-Lupus): results of a multicentre randomised controlled trial. *Ann Rheum Dis*. 2022;81(10):1420–1427. <https://doi.org/10.1136/annrheumdis-2022-222435>.
 262. Malvar A, Alberton V, Lococo B, et al. Kidney biopsy-based management of maintenance immunosuppression is safe and may ameliorate flare rate in lupus nephritis. *Kidney Int*. 2020;97(1):156–162. <https://doi.org/10.1016/j.kint.2019.07.018>.
 263. Joo YB, Kang YM, Kim HA, et al. Outcome and predictors of renal survival in patients with lupus nephritis: comparison between cyclophosphamide and mycophenolate mofetil. *Int J Rheum Dis*. 2018;21(5):1031–1039. <https://doi.org/10.1111/1756-185X.13274>.
 264. Yap DYH, Tang C, Ma MKM, et al. Longterm data on disease flares in patients with proliferative lupus nephritis in recent Years. *J Rheumatol*. 2017;44(9):1375–1383. <https://doi.org/10.3899/jrheum.170226>.
 265. Pakhotan R, Gladman DD, Su J, Urowitz MB. Sustained complete renal remission is a predictor of reduced mortality, chronic kidney disease and end-stage renal disease in lupus nephritis. *Lupus*. 2018;27(3):468–474. <https://doi.org/10.1177/0961203317726376>.
 266. Vincenti F, Cohen SD, Appel G. Novel B cell therapeutic targets in transplantation and immune-mediated glomerular diseases. *Clin J Am Soc Nephrol*. 2010;5(1):142–151. <https://doi.org/10.2215/CJN.04580709>.
 267. Merrill JT, Neuwelt CM, Wallace DJ, et al. Efficacy and safety of rituximab in moderately-to-severely active systemic lupus erythematosus: the randomized, double-blind, phase II/III systemic lupus erythematosus evaluation of rituximab trial. *Arthritis Rheum*. 2010;62(1):222–233. <https://doi.org/10.1002/art.27233>.
 268. Lightstone L. The landscape after LUNAR: rituximab's crater-filled path. *Arthritis Rheum*. 2012;64(4):962–965. <https://doi.org/10.1002/art.34362>.
 269. Ng KP, Leandro MJ, Edwards JC, Ehrenstein MR, Cambridge G, Isenberg DA. Repeated B cell depletion in treatment of refractory systemic lupus erythematosus. *Ann Rheum Dis*. 2006;65(7):942–945. <https://doi.org/10.1136/ard.2005.044487>.
 270. Terrier B, Amoura Z, Ravaud P, et al. Safety and efficacy of rituximab in systemic lupus erythematosus: results from 136 patients from the French AutoImmunity and Rituximab registry. *Arthritis Rheum*. 2010;62(8):2458–2466. <https://doi.org/10.1002/art.27541>.
 271. Condon MB, Ashby D, Pepper RJ, et al. Prospective observational single-centre cohort study to evaluate the effectiveness of treating lupus nephritis with rituximab and mycophenolate mofetil but no

- oral steroids. *Ann Rheum Dis.* 2013;72(8):1280–1286. <https://doi.org/10.1136/annrheumdis-2012-202844>.
272. Moroni G, Maccario M, Banfi G, Quaglini S, Ponticelli C. Treatment of membranous lupus nephritis. *Am J Kidney Dis.* 1998;31(4):681–686. <https://doi.org/10.1053/ajkd.1998.v31.pm9531186>.
 273. Mok CC, Ying KY, Lau CS, et al. Treatment of pure membranous lupus nephropathy with prednisone and azathioprine: an open-label trial. *Am J Kidney Dis.* 2004;43(2):269–276. <https://doi.org/10.1053/j.ajkd.2003.10.029>.
 274. Spetie DN, Tang Y, Rovin BH, et al. Mycophenolate therapy of SLE membranous nephropathy. *Kidney Int.* 2004;66(6):2411–2415. <https://doi.org/10.1111/j.1523-1755.2004.66030.x>.
 275. Karim MY, Pisoni CN, Ferro L, et al. Reduction of proteinuria with mycophenolate mofetil in predominantly membranous lupus nephropathy. *Rheumatology.* 2005;44(10):1317–1321. <https://doi.org/10.1093/rheumatology/kei019>.
 276. Radhakrishnan J, Moutzouris DA, Ginzler EM, Solomons N, Siempos II, Appel GB. Mycophenolate mofetil and intravenous cyclophosphamide are similar as induction therapy for class V lupus nephritis. *Kidney Int.* 2010;77(2):152–160. <https://doi.org/10.1038/ki.2009.412>.
 277. Ward F, Bargman JM. Membranous lupus nephritis: the same, but different. *Am J Kidney Dis.* 2016;68(6):954–966. <https://doi.org/10.1053/j.ajkd.2016.07.026>.
 278. Touma Z, Urowitz MB, Ibanez D, Gladman DD. Time to recovery from proteinuria in patients with lupus nephritis receiving standard treatment. *J Rheumatol.* 2014;41(4):688–697. <https://doi.org/10.3899/jrheum.130005>.
 279. Moroni G, Quaglini S, Gravellone L, et al. Membranous nephropathy in systemic lupus erythematosus: long-term outcome and prognostic factors of 103 patients. *Semin Arthritis Rheum.* 2012;41(5):642–651. <https://doi.org/10.1016/j.semarthrit.2011.08.002>.
 280. Uramoto KM, Michet Jr CJ, Thumboo J, Sunku J, O'Fallon WM, Gabriel SE. Trends in the incidence and mortality of systemic lupus erythematosus, 1950–1992. *Arthritis Rheum.* 1999;42(1):46–50. [https://doi.org/10.1002/1529-0131\(199901\)42:1<46::AID-ANR6>3.0.CO;2-2](https://doi.org/10.1002/1529-0131(199901)42:1<46::AID-ANR6>3.0.CO;2-2).
 281. Roman MJ, Shanker BA, Davis A, et al. Prevalence and correlates of accelerated atherosclerosis in systemic lupus erythematosus. *N Engl J Med.* 2003;349(25):2399–2406. <https://doi.org/10.1056/NEJMoa035471>.
 282. Bruce IN, Urowitz MB, Gladman DD, Ibanez D, Steiner G. Risk factors for coronary heart disease in women with systemic lupus erythematosus: the Toronto Risk Factor Study. *Arthritis Rheum.* 2003;48(11):3159–3167. <https://doi.org/10.1002/art.11296>.
 283. Zhao XY, Li SS, He YX, et al. SGLT2 inhibitors alleviated podocyte damage in lupus nephritis by decreasing inflammation and enhancing autophagy. *Ann Rheum Dis.* 2023;82(10):1328–1340. <https://doi.org/10.1136/ard-2023-224242>.
 284. Wang H, Li T, Sun F, et al. Safety and efficacy of the SGLT2 inhibitor dapagliflozin in patients with systemic lupus erythematosus: a phase I/II trial. *RMD Open.* 2022;8(2). <https://doi.org/10.1136/rmdopen-2022-002686>.
 285. Morales E, Galindo M. SGLT2 inhibitors in lupus nephropathy, a new therapeutic strategy for nephroprotection. *Ann Rheum Dis.* 2022;81(9):1337–1338. <https://doi.org/10.1136/annrheumdis-2022-222512>.
 286. Morales E, Sandino J, Galindo M. Lupus nephropathy beyond immunosuppression: searching for nephro and cardioprotection. *Front Nephrol.* 2023;3:1105676. <https://doi.org/10.3389/fneph.2023.1105676>.
 287. Lv J, Wong MG, Hladunewich MA, et al. Effect of oral methylprednisolone on decline in kidney function or kidney failure in patients with IgA nephropathy: the TESTING randomized clinical trial. *JAMA.* 2022;327(19):1888–1898. <https://doi.org/10.1001/jama.2022.5368>.
 288. Jiang Y, Duarte-Garcia AA, Putman MS, Gazeley DJ. Incidence of pneumocystis jirovecii pneumonia and prophylaxis-associated adverse events among patients with systemic lupus erythematosus. *J Rheumatol.* 2025;52(1):47–51. <https://doi.org/10.3899/jrheum.2023-1038>.
 289. Boone B, Lazaroff SM, Wheless L, Wolfe RM, Barnado A. Rates of Pneumocystis jirovecii pneumonia and prophylaxis prescribing patterns in a large electronic health record cohort of patients with systemic lupus erythematosus. *Semin Arthritis Rheum.* 2022;57:152106. <https://doi.org/10.1016/j.semarthrit.2022.152106>.
 290. Munthanunach P, Ngamjanyaporn P, Pisitkun P, Rotjanapan P. The role of trimethoprim/sulfamethoxazole in preventing opportunistic infections in systemic lupus erythematosus patients receiving low-level immunosuppressive treatment: an open-label, randomized, controlled trial. *Clin Exp Med.* 2024;24(1):241. <https://doi.org/10.1007/s10238-024-01503-z>.
 291. Zhou S, Aitken SL. Prophylaxis against pneumocystis jirovecii pneumonia in adults. *JAMA.* 2023;330(2):182–183. <https://doi.org/10.1001/jama.2023.9844>.
 292. Joseph RE, Radhakrishnan J, Appel GB. Antiphospholipid antibody syndrome and renal disease. *Curr Opin Nephrol Hypertens.* 2001;10(2):175–181. <https://doi.org/10.1097/00041552-200103000-00004>.
 293. Nzerue CM, Hewan-Lowe K, Pierangeli S, Harris EN. “Black swan in the kidney”: renal involvement in the antiphospholipid antibody syndrome. *Kidney Int.* 2002;62(3):733–744. <https://doi.org/10.1046/j.1523-1755.2002.00500.x>.
 294. Moroni G, Ventura D, Riva P, et al. Antiphospholipid antibodies are associated with an increased risk for chronic renal insufficiency in patients with lupus nephritis. *Am J Kidney Dis.* 2004;43(1):28–36. <https://doi.org/10.1053/j.ajkd.2003.09.011>.
 295. Furie RA, Aroca G, Cascino MD, et al. B-cell depletion with obinutuzumab for the treatment of proliferative lupus nephritis: a randomised, double-blind, placebo-controlled trial. *Ann Rheum Dis.* 2022;81(1):100–107. <https://doi.org/10.1136/annrheumdis-2021-220920>.
 296. Furie RA, Rovin BH, Garg JP, et al. Efficacy And safety of obinutuzumab in active lupus nephritis. *N Engl J Med.* 0(0). <https://doi.org/10.1056/NEJMoa2410965>.
 297. Mysler EF, Spindler AJ, Guzman R, et al. Efficacy and safety of ocrelizumab in active proliferative lupus nephritis: results from a randomized, double-blind, phase III study. *Arthritis Rheum.* 2013;65(9):2368–2379. <https://doi.org/10.1002/art.38037>.
 298. Ginzler EM, Wax S, Rajeswaran A, et al. Atacicept in combination with MMF and corticosteroids in lupus nephritis: results of a prematurely terminated trial. *Arthritis Res Ther.* 2012;14(1):R33. <https://doi.org/10.1186/ar3738>.
 299. Furie R, Khamashta M, Merrill JT, et al. Anifrolumab, an anti-interferon-alpha receptor monoclonal antibody, in moderate-to-severe systemic lupus erythematosus. *Arthritis Rheumatol.* 2017;69(2):376–386. <https://doi.org/10.1002/art.39962>.
 300. Mackensen A, Muller F, Mougiakakos D, et al. Anti-CD19 CAR T cell therapy for refractory systemic lupus erythematosus. *Nat Med.* 2022;28(10):2124–2132. <https://doi.org/10.1038/s41591-022-02017-5>.
 301. Muller F, Taubmann J, Bucci L, et al. CD19 CAR T-cell therapy in autoimmune disease—a case series with follow-up. *N Engl J Med.* 2024;390(8):687–700. <https://doi.org/10.1056/NEJMoa2308917>.
 302. Mullard A. CAR T cell therapies raise hopes - and questions - for lupus and autoimmune disease. *Nat Rev Drug Discov.* 2023;22(11):859–861. <https://doi.org/10.1038/d41573-023-00166-x>.
 303. Parodis I, Gatto M, Sjowall C. B cells in systemic lupus erythematosus: targets of new therapies and surveillance tools. *Front Med.* 2022;9:952304. <https://doi.org/10.3389/fmed.2022.952304>.
 304. Parodis I, Long X, Karlsson MCI, Huang X. B cell tolerance and targeted therapies in SLE. *J Clin Med.* 2023;12(19). <https://doi.org/10.3390/jcm12196268>.
 305. Sharpe AH, Abbas AK. T-cell costimulation—biology, therapeutic potential, and challenges. *N Engl J Med.* 2006;355(10):973–975. <https://doi.org/10.1056/NEJMp068087>.
 306. Davis Jr JC, Totoritis MC, Rosenberg J, Sklenar TA, Wofsy D. Phase I clinical trial of a monoclonal antibody against CD40-ligand (IDEC-131) in patients with systemic lupus erythematosus. *J Rheumatol.* 2001;28(1):95–101.
 307. Kawai T, Andrews D, Colvin RB, Sachs DH, Cosimi AB. Thromboembolic complications after treatment with monoclonal antibody against CD40 ligand. *Nat Med.* 2000;6(2):114. <https://doi.org/10.1038/72162>.
 308. Wofsy D, Hillson JL, Diamond B. Abatacept for lupus nephritis: alternative definitions of complete response support conflicting

- conclusions. *Arthritis Rheum.* 2012;64(11):3660–3665. <https://doi.org/10.1002/art.34624>.
309. Rovin B. The access trial. *J Am Soc Nephrol.* 2013.
 310. Rovin BH, Adler SG, Barratt J, et al. Executive summary of the KDIGO 2021 guideline for the management of glomerular diseases. *Kidney Int.* 2021;100(4):753–779. <https://doi.org/10.1016/j.kint.2021.05.015>.
 311. Mariani LH, Trachtman H, Thompson A, et al. Proteinuria as an endpoint in clinical trials of focal segmental glomerulosclerosis. *Am J Kidney Dis.* 2025;85(5):610–617. <https://doi.org/10.1053/j.ajkd.2024.08.011>.
 312. Fanouriakis A, Kostopoulou M, Cheema K, et al. 2019 update of the joint European League against rheumatism and European renal association-European dialysis and transplant association (EULAR/ERA-EDTA) recommendations for the management of lupus nephritis. *Ann Rheum Dis.* 2020;79(6):713–723. <https://doi.org/10.1136/annrheumdis-2020-216924>.
 313. Costedoat-Chalumeau N, Tamiou F, Piette JC. Treatment adherence in systemic lupus erythematosus and rheumatoid arthritis: time to focus on this important issue. *Rheumatology.* 2018;57(9):1507–1509. <https://doi.org/10.1093/rheumatology/kex337>.
 314. Bruce IN, Gladman DD, Urowitz MB. Factors associated with refractory renal disease in patients with systemic lupus erythematosus: the role of patient nonadherence. *Arthritis Care Res.* 2000;13(6):406–408.
 315. Costedoat-Chalumeau N, Pouchot J, Guettrot-Imbert G, et al. Adherence to treatment in systemic lupus erythematosus patients. *Best Pract Res Clin Rheumatol.* 2013;27(3):329–340. <https://doi.org/10.1016/j.berh.2013.07.001>.
 316. Marengo MF, Waimann CA, de Achaval S, et al. Measuring therapeutic adherence in systemic lupus erythematosus with electronic monitoring. *Lupus.* 2012;21(11):1158–1165. <https://doi.org/10.1177/0961203312447868>.
 317. Petri M, Perez-Gutthann S, Longenecker JC, Hochberg M. Morbidity of systemic lupus erythematosus: role of race and socioeconomic status. *Am J Med.* 1991;91(4):345–353. [https://doi.org/10.1016/0002-9343\(91\)90151-m](https://doi.org/10.1016/0002-9343(91)90151-m).
 318. Nguyen Y, Blanchet B, Urowitz MB, et al. Association between severe nonadherence to hydroxychloroquine and systemic lupus erythematosus flares, damage, and mortality in 660 patients from the SLICC inception cohort. *Arthritis Rheumatol.* 2023;75(12):2195–2206. <https://doi.org/10.1002/art.42645>.
 319. Emamikia S, Gentline C, Enman Y, Parodis I. How can we enhance adherence to medications in patients with systemic lupus erythematosus? Results from a qualitative study. *J Clin Med.* 2022;11(7). <https://doi.org/10.3390/jcm11071857>.
 320. Hardy C, Gladman DD, Su J, Rozenbojm N, Urowitz MB. Barriers to medication adherence and degree of nonadherence in a systemic lupus erythematosus (SLE) outpatient population. *Rheumatol Int.* 2021;41(8):1457–1464. <https://doi.org/10.1007/s00296-021-04898-0>.
 321. Sloan M, Lever E, Gordon C, et al. Medication decision-making and adherence in lupus: patient-physician discordance and the impact of previous 'adverse medical experiences.' *Rheumatology.* 2022;61(4):1417–1429. <https://doi.org/10.1093/rheumatology/keab534>.
 322. Miyakis S, Lockshin MD, Atsumi T, et al. International consensus statement on an update of the classification criteria for definite antiphospholipid syndrome (APS). *J Thromb Haemostasis.* 2006;4(2):295–306. <https://doi.org/10.1111/j.1538-7836.2006.01753.x>.
 323. Barbhaiya M, Zuily S, Naden R, et al. The 2023 ACR/EULAR antiphospholipid syndrome classification criteria. *Arthritis Rheumatol.* 2023;75(10):1687–1702. <https://doi.org/10.1002/art.42624>.
 324. Zen M, Tonello M, Carta F, et al. Insights into the 2023 ACR/EULAR antiphospholipid syndrome classification criteria: findings from a cohort of 205 patients with primary APS. *Rheumatology.* 2024;1–6. <https://doi.org/10.1093/rheumatology/keae665>.
 325. Ruiz-Irastorza G, Crowther M, Branch W, Khamashta MA. Antiphospholipid syndrome. *Lancet.* 2010;376(9751):1498–1509. [https://doi.org/10.1016/S0140-6736\(10\)60709-X](https://doi.org/10.1016/S0140-6736(10)60709-X).
 326. Giannakopoulos B, Passam F, Ioannou Y, Krilis SA. How we diagnose the antiphospholipid syndrome. *Blood.* 2009;113(5):985–994. <https://doi.org/10.1182/blood-2007-12-129627>.
 327. Kaul M, Erkan D, Sammaritano L, Lockshin MD. Assessment of the 2006 revised antiphospholipid syndrome classification criteria. *Ann Rheum Dis.* 2007;66(7):927–930. <https://doi.org/10.1136/ard.2006.067314>.
 328. Detkov, Gil-Aguado A, Lavilla P, Cuesta MV, Fontan G, Pascual-Salcedo D. Do antibodies to beta2-glycoprotein 1 contribute to the better characterization of the antiphospholipid syndrome? *Lupus.* 1999;8(6):430–438. <https://doi.org/10.1177/096120339900800604>.
 329. Cervera R. Antiphospholipid syndrome. *Thromb Res.* 2017;151(suppl 1):S43–S47. [https://doi.org/10.1016/S0049-3848\(17\)30066-X](https://doi.org/10.1016/S0049-3848(17)30066-X).
 330. Cervera R, Serrano R, Pons-Estel GJ, et al. Morbidity and mortality in the antiphospholipid syndrome during a 10-year period: a multicentre prospective study of 1000 patients. *Ann Rheum Dis.* 2015;74(6):1011–1018. <https://doi.org/10.1136/annrheumdis-2013-204838>.
 331. Veres K, Szodoray P, Szekanez Z, et al. Clinical and immunoserological characteristics of the transition from primary to overlap antiphospholipid syndrome. *Lupus.* 2010;19(13):1520–1526. <https://doi.org/10.1177/0961203310374336>.
 332. Love PE, Santoro SA. Antiphospholipid antibodies: anticardiolipin and the lupus anticoagulant in systemic lupus erythematosus (SLE) and in non-SLE disorders. Prevalence and clinical significance. *Ann Intern Med.* 1990;112(9):682–698. <https://doi.org/10.7326/0003-4819-112-9-682>.
 333. Galli M, Luciani D, Bertolini G, Barbu T. Lupus anticoagulants are stronger risk factors for thrombosis than anticardiolipin antibodies in the antiphospholipid syndrome: a systematic review of the literature. *Blood.* 2003;101(5):1827–1832. <https://doi.org/10.1182/blood-2002-02-0441>.
 334. Pengo V, Ruffatti A, Legnani C, et al. Clinical course of high-risk patients diagnosed with antiphospholipid syndrome. *J Thromb Haemostasis.* 2010;8(2):237–242. <https://doi.org/10.1111/j.1538-7836.2009.03674.x>.
 335. Sebastiani GD, Galeazzi M, Tincani A, et al. Anticardiolipin and anti-beta2GPI antibodies in a large series of European patients with systemic lupus erythematosus. Prevalence and clinical associations. European Concerted Action on the Immunogenetics of SLE. *Scand J Rheumatol.* 1999;28(6):344–351.
 336. Cervera R, Piette JC, Font J, et al. Antiphospholipid syndrome: clinical and immunologic manifestations and patterns of disease expression in a cohort of 1,000 patients. *Arthritis Rheum.* 2002;46(4):1019–1027. <https://doi.org/10.1002/art.10187>.
 337. Lockshin MD. Answers to the antiphospholipid-antibody syndrome? *N Engl J Med.* 1995;332(15):1025–1027. <https://doi.org/10.1056/NEJM199504133321510>.
 338. Harada M, Fujisawa Y, Sakisaka S, et al. High prevalence of anticardiolipin antibodies in hepatitis C virus infection: lack of effects on thrombocytopenia and thrombotic complications. *J Gastroenterol.* 2000;35(4):272–277. <https://doi.org/10.1007/s005350050345>.
 339. Anki R, Bonmarchand M, Coutellier A, Herson S, Karmochkine M. Antiphospholipid antibodies are an epiphenomenon in HIV-infected patients. *AIDS.* 1999;13(10):1282–1283. <https://doi.org/10.1097/00002030-199907090-00024>.
 340. Lundstrom E, Gustafsson JT, Jonsen A, et al. HLA-DRB1*04/*13 alleles are associated with vascular disease and antiphospholipid antibodies in systemic lupus erythematosus. *Ann Rheum Dis.* 2013;72(6):1018–1025. <https://doi.org/10.1136/annrheumdis-2012-201760>.
 341. Tektonidou MG, Verrou KM, Gakiopoulou H, et al. Kidney whole-transcriptome profiling in primary antiphospholipid syndrome reveals complement, interferons and NETs-related gene expression. *Rheumatology.* 2024;63(11):3184–3190. <https://doi.org/10.1093/rheumatology/keae397>.
 342. Canaud G, Bienaime F, Tabarin F, et al. Inhibition of the mTORC pathway in the antiphospholipid syndrome. *N Engl J Med.* 2014;371(4):303–312. <https://doi.org/10.1056/NEJMoa1312890>.
 343. Laridan E, Martinod K, De Meyer SF. Neutrophil extracellular traps in arterial and venous thrombosis. *Semin Thromb Hemost.* 2019;45(1):86–93. <https://doi.org/10.1055/s-0038-1677040>.
 344. Oku K, Atsumi T, Bohgaki M, et al. Complement activation in patients with primary antiphospholipid syndrome. *Ann Rheum Dis.* 2009;68(6):1030–1035. <https://doi.org/10.1136/ard.2008.090670>.

345. Munakata Y, Saito T, Matsuda K, Seino J, Shibata S, Sasaki T. Detection of complement-fixing antiphospholipid antibodies in association with thrombosis. *Thromb Haemost*. 2000;83(5):728–731.
346. Breen KA, Seed P, Parmar K, Moore GW, Stuart-Smith SE, Hunt BJ. Complement activation in patients with isolated antiphospholipid antibodies or primary antiphospholipid syndrome. *Thromb Haemost*. 2012;107(3):423–429. <https://doi.org/10.1160/TH11-08-0554>.
347. Durigutto P, Macor P, Pozzi N, et al. Complement activation and thrombin generation by MBL bound to beta2-glycoprotein I. *J Immunol*. 2020;205(5):1385–1392. <https://doi.org/10.4049/jimmunol.2000570>.
348. Tedesco F, Borghi MO, Gerosa M, et al. Pathogenic role of complement in antiphospholipid syndrome and therapeutic implications. *Front Immunol*. 2018;9:1388. <https://doi.org/10.3389/fimmu.2018.01388>.
349. Sacre K, Brihaye B, Hyafil F, et al. Asymptomatic myocardial ischemic disease in antiphospholipid syndrome: a controlled cardiac magnetic resonance imaging study. *Arthritis Rheum*. 2010;62(7):2093–2100. <https://doi.org/10.1002/art.27488>.
350. Cervera R. Update on the diagnosis, treatment, and prognosis of the catastrophic antiphospholipid syndrome. *Curr Rheumatol Rep*. 2010;12(1):70–76. <https://doi.org/10.1007/s11926-009-0073-6>.
351. Hanouna G, Morel N, Le Thi Huong D, et al. Catastrophic antiphospholipid syndrome and pregnancy: an experience of 13 cases. *Rheumatology*. 2013;52(9):1635–1641. <https://doi.org/10.1093/rheumatology/ket167>.
352. Scheen M, Adedjouma A, Esteve E, et al. Kidney disease in antiphospholipid antibody syndrome: risk factors, pathophysiology and management. *Autoimmun Rev*. 2022;21(5):103072. <https://doi.org/10.1016/j.autrev.2022.103072>.
353. Bienaime F, Legendre C, Terzi F, Canaud G. Antiphospholipid syndrome and kidney disease. *Kidney Int*. 2017;91(1):34–44. <https://doi.org/10.1016/j.kint.2016.06.026>.
354. Sangle SR, D'Cruz DP, Jan W, et al. Renal artery stenosis in the antiphospholipid (Hughes) syndrome and hypertension. *Ann Rheum Dis*. 2003;62(10):999–1002. <https://doi.org/10.1136/ard.62.10.999>.
355. Sinico RA, Cavazzana I, Nuzzo M, et al. Renal involvement in primary antiphospholipid syndrome: retrospective analysis of 160 patients. *Clin J Am Soc Nephrol*. 2010;5(7):1211–1217. <https://doi.org/10.2215/CJN.00460110>.
356. Nochy D, Daugas E, Droz D, et al. The intrarenal vascular lesions associated with primary antiphospholipid syndrome. *J Am Soc Nephrol*. 1999;10(3):507–518. <https://doi.org/10.1681/ASN.V103507>.
357. Venkata Seshan S, Barisoni L, Smith S. Renal disease in anti-phospholipid syndrome (APLS). *J Am Soc Nephrol*. 1996;7:1343.
358. Nochy D, Daugas E, Huong DL, Piette JC, Hill G. Kidney involvement in the antiphospholipid syndrome. *J Autoimmun*. 2000;15(2):127–132. <https://doi.org/10.1006/jaut.2000.0423>.
359. Drigo B, Pijl AS, Duyts H, et al. Shifting carbon flow from roots into associated microbial communities in response to elevated atmospheric CO₂. *Proc Natl Acad Sci U S A*. 2010;107(24):10938–10942. <https://doi.org/10.1073/pnas.0912421107>.
360. Saracino A, Ramunni A, Pannarale G, Coratelli P. Kidney disease associated with primary antiphospholipid syndrome: clinical signs and histopathological features in an case experience of five cases. *Clin Nephrol*. 2005;63(6):471–476. <https://doi.org/10.5414/cnp63471>.
361. Fakhouri F, Noel LH, Zuber J, et al. The expanding spectrum of renal diseases associated with antiphospholipid syndrome. *Am J Kidney Dis*. 2003;41(6):1205–1211. [https://doi.org/10.1016/s0272-6386\(03\)00352-4](https://doi.org/10.1016/s0272-6386(03)00352-4).
362. Tektonidou MG, Sotsiou F, Nakopoulou L, Vlachoyiannopoulos PG, Moutsopoulos HM. Antiphospholipid syndrome nephropathy in patients with systemic lupus erythematosus and antiphospholipid antibodies: prevalence, clinical associations, and long-term outcome. *Arthritis Rheum*. 2004;50(8):2569–2579. <https://doi.org/10.1002/art.20433>.
363. Godfrey T, Khamashta MA, Hughes GR. Antiphospholipid syndrome and renal artery stenosis. *QJM*. 2000;93(2):127–129. <https://doi.org/10.1093/qjmed/93.2.127>.
364. Remondino GI, Mysler E, Pissano MN, et al. A reversible bilateral renal artery stenosis in association with antiphospholipid syndrome. *Lupus*. 2000;9(1):65–67. <https://doi.org/10.1177/09612033000900112>.
365. Riccialdelli L, Arnaldi G, Giacchetti G, Pantanetti P, Mantero F. Hypertension due to renal artery occlusion in a patient with antiphospholipid syndrome. *Am J Hypertens*. 2001;14(1):62–65. [https://doi.org/10.1016/s0895-7061\(00\)00296-x](https://doi.org/10.1016/s0895-7061(00)00296-x).
366. Daugas E, Nochy D, Huong DLT, et al. Antiphospholipid syndrome nephropathy in systemic lupus erythematosus. *J Am Soc Nephrol*. 2002;13(1):42–52. <https://doi.org/10.1681/ASN.V13142>.
367. Prakash R, Miller 3rd CC, Suki WN. Anticardiolipin antibody in patients on maintenance hemodialysis and its association with recurrent arteriovenous graft thrombosis. *Am J Kidney Dis*. 1995;26(2):347–352. [https://doi.org/10.1016/0272-6386\(95\)90656-8](https://doi.org/10.1016/0272-6386(95)90656-8).
368. Brunet P, Aillaud MF, San Marco M, et al. Antiphospholipids in hemodialysis patients: relationship between lupus anticoagulant and thrombosis. *Kidney Int*. 1995;48(3):794–800. <https://doi.org/10.1038/ki.1995.352>.
369. ARJ V. Anticardiolipin antibodies in hemodialysis patients. *J Am Soc Nephrol*. 1996;7:1423.
370. Ducloux D, Pellet E, Fournier V, et al. Prevalence and clinical significance of antiphospholipid antibodies in renal transplant recipients. *Transplantation*. 1999;67(1):90–93. <https://doi.org/10.1097/00007890-199901150-00015>.
371. Friedman GS, Meier-Kriesche HU, Kaplan B, et al. Hypercoagulable states in renal transplant candidates: impact of anticoagulation upon incidence of renal allograft thrombosis. *Transplantation*. 2001;72(6):1073–1078. <https://doi.org/10.1097/00007890-200109270-00016>.
372. Forman JP, Lin J, Pascual M, Denton MD, Tolkooff-Rubin N. Significance of anticardiolipin antibodies on short and long term allograft survival and function following kidney transplantation. *Am J Transplant*. 2004;4(11):1786–1791. <https://doi.org/10.1046/j.1600-6143.2004.00602.x>.
373. Baid S, Pascual M, Williams Jr WW, et al. Renal thrombotic microangiopathy associated with anticardiolipin antibodies in hepatitis C-positive renal allograft recipients. *J Am Soc Nephrol*. 1999;10(1):146–153. <https://doi.org/10.1681/ASN.V101146>.
374. Branch DW, Lim MY. How I diagnose and treat antiphospholipid syndrome in pregnancy. *Blood*. 2024;143(9):757–768. <https://doi.org/10.1182/blood.2023020727>.
375. Sammaritano LR, Bermas BL, Chakravarty EE, et al. 2020 American College of Rheumatology guideline for the management of reproductive health in rheumatic and musculoskeletal diseases. *Arthritis Rheumatol*. 2020;72(4):529–556. <https://doi.org/10.1002/art.41191>.
376. Tektonidou MG, Andreoli L, Limper M, et al. EULAR recommendations for the management of antiphospholipid syndrome in adults. *Ann Rheum Dis*. 2019;78(10):1296–1304. <https://doi.org/10.1136/annrheumdis-2019-215213>.
377. Khamashta MA, Cuadrado MJ, Mujic F, Taub NA, Hunt BJ, Hughes GR. The management of thrombosis in the antiphospholipid-antibody syndrome. *N Engl J Med*. 1995;332(15):993–997. <https://doi.org/10.1056/NEJM199504133321504>.
378. Alarcon-Segovia D, Boffa MC, Branch W, et al. Prophylaxis of the antiphospholipid syndrome: a consensus report. *Lupus*. 2003;12(7):499–503. <https://doi.org/10.1191/0961203303lu388oa>.
379. Crowther MA, Ginsberg JS, Julian J, et al. A comparison of two intensities of warfarin for the prevention of recurrent thrombosis in patients with the antiphospholipid antibody syndrome. *N Engl J Med*. 2003;349(12):1133–1138. <https://doi.org/10.1056/NEJMoa035241>.
380. Finazzi G, Marchioli R, Brancaccio V, et al. A randomized clinical trial of high-intensity warfarin vs. conventional antithrombotic therapy for the prevention of recurrent thrombosis in patients with the antiphospholipid syndrome (WAPS). *J Thromb Haemostasis*. 2005;3(5):848–853. <https://doi.org/10.1111/j.1538-7836.2005.01340.x>.
381. Rosove MH, Brewer PM. Antiphospholipid thrombosis: clinical course after the first thrombotic event in 70 patients. *Ann Intern*

- Med.* 1992;117(4):303–308. <https://doi.org/10.7326/0003-4819-117-4-303>.
382. Adelhelm JBH, Christensen R, Balbi GGM, Voss A. Therapy with direct oral anticoagulants for secondary prevention of thromboembolic events in the antiphospholipid syndrome: a systematic review and meta-analysis of randomised trials. *Lupus Sci Med.* 2023;10(2). <https://doi.org/10.1136/lupus-2023-001018>.
 383. Khairani CD, Bejjani A, Piazza G, et al. Direct oral anticoagulants vs vitamin K antagonists in patients with antiphospholipid syndromes: meta-analysis of randomized trials. *J Am Coll Cardiol.* 2023;81(1):16–30. <https://doi.org/10.1016/j.jacc.2022.10.008>.
 384. Kobayashi S, Tamura N, Tsuda H, Mokuno C, Hashimoto H, Hirose S. Immunoabsorbent plasmapheresis for a patient with antiphospholipid syndrome during pregnancy. *Ann Rheum Dis.* 1992;51(3):399–401. <https://doi.org/10.1136/ard.51.3.399>.
 385. do Sameiro Faria M, Mota C, Barbot J, et al. Haemolytic uraemic syndrome, cardiomyopathy, cutaneous vasculopathy and antiphospholipid activity. *Nephrol Dial Transplant.* 2000;15(11):1891–1892. <https://doi.org/10.1093/ndt/15.11.1891>.
 386. Jung H, Bobba R, Su J, et al. The protective effect of antimalarial drugs on thrombovascular events in systemic lupus erythematosus. *Arthritis Rheum.* 2010;62(3):863–868. <https://doi.org/10.1002/art.27289>.
 387. Ruiz-Irastorza G, Ramos-Casals M, Brito-Zeron P, Khamashta MA. Clinical efficacy and side effects of antimalarials in systemic lupus erythematosus: a systematic review. *Ann Rheum Dis.* 2010;69(1):20–28. <https://doi.org/10.1136/ard.2008.101766>.
 388. Erkan D, Vega J, Ramon G, Kozora E, Lockshin MD. A pilot open-label phase II trial of rituximab for non-criteria manifestations of antiphospholipid syndrome. *Arthritis Rheum.* 2013;65(2):464–471. <https://doi.org/10.1002/art.37759>.
 389. Kumar D, Roubey RA. Use of rituximab in the antiphospholipid syndrome. *Curr Rheumatol Rep.* 2010;12(1):40–44. <https://doi.org/10.1007/s11926-009-0074-5>.
 390. Antovic A, Faustini F, Karandyszovska N, et al. Eculizumab and ravulizumab in the treatment of refractory antiphospholipid syndrome—a first case report. *Rheumatology.* 2024;63(12):e339–e340. <https://doi.org/10.1093/rheumatology/keae353>.
 391. Carrara C, Mataj B, Gastoldi S, Ruggerenti P, Sciascia S, Roccatello D. Case report: timing of eculizumab treatment in catastrophic antiphospholipid syndrome. *Front Immunol.* 2024;15:1460317. <https://doi.org/10.3389/fimmu.2024.1460317>.
 392. Sciascia S, Rubini E, Radin M, Cecchi I, Rossi D, Roccatello D. Anticardiolipin and anti-beta 2 glycoprotein-I antibodies disappearance in patients with systemic lupus erythematosus and antiphospholipid syndrome while on belimumab. *Ann Rheum Dis.* 2018;77(11):1694–1695. <https://doi.org/10.1136/annrheumdis-2018-213496>.
 393. Emmi G, Bettiol A, Palterer B, et al. Belimumab reduces antiphospholipid antibodies in SLE patients independently of hydroxychloroquine treatment. *Autoimmun Rev.* 2019;18(3):312–314. <https://doi.org/10.1016/j.autrev.2018.11.002>.
 394. Bettiol A, Pregnotato F, Sciascia S, Emmi G, Prisco D, Meroni PL. Association of subcutaneous belimumab and long-term antimalarial treatment reduces antiphospholipid antibodies levels in systemic lupus erythematosus: post-hoc analysis of a randomised placebo-controlled trial—comment on: ‘Effect of belimumab treatment on antiphospholipid antibody levels: post-hoc analysis based on two randomised placebo-controlled trials in systemic lupus erythematosus’ by Chatzidionysiou et al. *Ann Rheum Dis.* 2022;81(8):e140. <https://doi.org/10.1136/annrheumdis-2020-218377>.
 395. Sciascia S, Radin M, Cecchi I, et al. Open-label, prospective, phase II descriptive pilot trial of belimumab therapy for refractory and/or non-criteria manifestations of antiphospholipid syndrome: study protocol. *Clin Exp Rheumatol.* 2023;41(3):597–604. <https://doi.org/10.55563/clinexprheumatol/qa2yb4>.
 396. Ferrara DE, Liu X, Espinola RG, et al. Inhibition of the thrombogenic and inflammatory properties of antiphospholipid antibodies by fluvastatin in an in vivo animal model. *Arthritis Rheum.* 2003;48(11):3272–3279. <https://doi.org/10.1002/art.11449>.
 397. Lopez-Pedreria C, Ruiz-Limon P, Aguirre MA, et al. Global effects of fluvastatin on the prothrombotic status of patients with antiphospholipid syndrome. *Ann Rheum Dis.* 2011;70(4):675–682. <https://doi.org/10.1136/ard.2010.135525>.
 398. Erkan D, Willis R, Murthy VL, et al. A prospective open-label pilot study of fluvastatin on proinflammatory and prothrombotic biomarkers in antiphospholipid antibody positive patients. *Ann Rheum Dis.* 2014;73(6):1176–1180. <https://doi.org/10.1136/annrheumdis-2013-203622>.
 399. Shapira I, Andrade D, Allen SL, Salmon JE. Brief report: induction of sustained remission in recurrent catastrophic antiphospholipid syndrome via inhibition of terminal complement with eculizumab. *Arthritis Rheum.* 2012;64(8):2719–2723. <https://doi.org/10.1002/art.34440>.
 400. Hsiao GR, Wolf RE, Kimpel DL. Intravenous immunoglobulin to prevent recurrent thrombosis in the antiphospholipid syndrome. *J Clin Rheumatol.* 2001;7(5):336–339. <https://doi.org/10.1097/00124743-200110000-00017>.
 401. Alarcon-Segovia D, Cardiel MH. Comparison between 3 diagnostic criteria for mixed connective tissue disease. Study of 593 patients. *J Rheumatol.* 1989;16(3):328–334.
 402. Maldonado ME, Perez M, Pignac-Kobinger J, et al. Clinical and immunologic manifestations of mixed connective tissue disease in a Miami population compared to a Midwestern US Caucasian population. *J Rheumatol.* 2008;35(3):429–437.
 403. Hoffman RW, Greidinger EL. Mixed connective tissue disease. *Curr Opin Rheumatol.* 2000;12(5):386–390. <https://doi.org/10.1097/00002281-200009000-00006>.
 404. Dima A, Jurcut C, Baicus C. The impact of anti-U1-RNP positivity: systemic lupus erythematosus versus mixed connective tissue disease. *Rheumatol Int.* 2018;38(7):1169–1178. <https://doi.org/10.1007/s00296-018-4059-4>.
 405. Abbasa S, Seror R, Henry J, et al. Anti-RNP positivity in primary Sjogren’s syndrome is associated with a more active disease and a more frequent muscular and pulmonary involvement. *RMD Open.* 2019;5(2):e001033. <https://doi.org/10.1136/rmdopen-2019-001033>.
 406. Li X, Liu X, Cui J, et al. Epidemiological survey of antinuclear antibodies in healthy population and analysis of clinical characteristics of positive population. *J Clin Lab Anal.* 2019;33(8):e22965. <https://doi.org/10.1002/jcla.22965>.
 407. Cappelli S, Bellando Randone S, Martinovic D, et al. “To be or not to be,” ten years after: evidence for mixed connective tissue disease as a distinct entity. *Semin Arthritis Rheum.* 2012;41(4):589–598. <https://doi.org/10.1016/j.semarthrit.2011.07.010>.
 408. Gendi NS, Welsh KI, Van Venrooij WJ, Vancheeswaran R, Gilroy J, Black CM. HLA type as a predictor of mixed connective tissue disease differentiation. Ten-year clinical and immunogenetic follow-up of 46 patients. *Arthritis Rheum.* 1995;38(2):259–266. <https://doi.org/10.1002/art.1780380216>.
 409. Gunnarsson R, Molberg O, Gilboe IM, Gran JT, Group PS. The prevalence and incidence of mixed connective tissue disease: a national multicentre survey of Norwegian patients. *Ann Rheum Dis.* 2011;70(6):1047–1051. <https://doi.org/10.1136/ard.2010.143792>.
 410. Ungprasert P, Crowson CS, Chowdhary VR, Ernste FC, Moders KG, Matteson EL. Epidemiology of mixed connective tissue disease, 1985–2014: a population-based study. *Arthritis Care Res.* 2016;68(12):1843–1848. <https://doi.org/10.1002/acr.22872>.
 411. Ferucci ED, Johnston JM, Gordon C, Helmick CG, Lim SS. Prevalence of mixed connective tissue disease in a population-based registry of American Indian/Alaska native people in 2007. *Arthritis Care Res.* 2017;69(8):1271–1275. <https://doi.org/10.1002/acr.23135>.
 412. Burdett MA, Hoffman RW, Deutscher SL, Wang GS, Johnson JC, Sharp GC. Long-term outcome in mixed connective tissue disease: longitudinal clinical and serologic findings. *Arthritis Rheum.* 1999;42(5):899–909. [https://doi.org/10.1002/1529-0131\(199905\)42:5<899::AID-ANR8>3.0.CO;2-L](https://doi.org/10.1002/1529-0131(199905)42:5<899::AID-ANR8>3.0.CO;2-L).
 413. Somarelli JA, Mesa A, Rodriguez R, et al. Epitope mapping of the U1 small nuclear ribonucleoprotein particle in patients with systemic lupus erythematosus and mixed connective tissue disease. *Lupus.* 2011;20(3):274–289. <https://doi.org/10.1177/0961203310387180>.
 414. Escola-Verge L, Pinal-Fernandez I, Fernandez-Codina A, et al. Mixed connective tissue disease and Epitope spreading: an historical cohort study. *J Clin Rheumatol.* 2017;23(3):155–159. <https://doi.org/10.1097/RHU.0000000000000500>.
 415. Carnero-Montoro E, Barturen G, Povedano E, et al. Epigenome-wide comparative study reveals key differences between mixed connective tissue disease and related systemic autoimmune

- diseases. *Front Immunol*. 2019;10:1880. <https://doi.org/10.3389/fimmu.2019.01880>.
416. Grader-Beck T, Wigley FM. Raynaud's phenomenon in mixed connective tissue disease. *Rheum Dis Clin N Am*. 2005;31(3):465–481. <https://doi.org/10.1016/j.rdc.2005.04.006>, vi.
 417. Terminiello A, Marrani E, Pagnini I, et al. Childhood mixed connective tissue disease at disease onset: evidence from a systematic review. *Autoimmun Rev*. 2024;23(4):103513. <https://doi.org/10.1016/j.autrev.2023.103513>.
 418. Kitridou RC, Akmal M, Turkel SB, Ehresmann GR, Quismorio Jr FP, Massry SG. Renal involvement in mixed connective tissue disease: a longitudinal clinicopathologic study. *Semin Arthritis Rheum*. 1986;16(2):135–145. [https://doi.org/10.1016/0049-0172\(86\)90047-8](https://doi.org/10.1016/0049-0172(86)90047-8).
 419. Hoffman RW, Cassidy JT, Takeda Y, Smith-Jones EI, Wang GS, Sharp GC. U1-70-kd autoantibody-positive mixed connective tissue disease in children. A longitudinal clinical and serologic analysis. *Arthritis Rheum*. 1993;36(11):1599–1602. <https://doi.org/10.1002/art.1780361115>.
 420. Appelboom T, Kahn MF, Mairesse N. Antibodies to small ribonucleoprotein and to 73-kD heat shock protein: two distinct markers of mixed connective tissue disease. *Clin Exp Immunol*. 1995;100(3):486–488. <https://doi.org/10.1111/j.1365-2249.1995.tb03727.x>.
 421. Fuller TJ, Richman AV, Auerbach D, Alexander RW, Lottenberg R, Longley S. Immune-complex glomerulonephritis in a patient with mixed connective tissue disease. *Am J Med*. 1977;62(5):761–764. [https://doi.org/10.1016/0002-9343\(77\)90880-4](https://doi.org/10.1016/0002-9343(77)90880-4).
 422. Celikbilek M, Elsurur R, Afsar B, Ozdemir HB, Sezer S, Ozdemir NF. Mixed connective tissue disease: a case with scleroderma renal crisis following abortion. *Clin Rheumatol*. 2007;26(9):1545–1547. <https://doi.org/10.1007/s10067-006-0442-8>.
 423. Ito S, Nakamura T, Kurosawa R, et al. Glomerulonephritis in children with mixed connective tissue disease. *Clin Nephrol*. 2006;66(3):160–165. <https://doi.org/10.5414/cnp66160>.
 424. Palferman TG, McIntosh CS, Kershaw M. Mixed connective tissue disease with associated glomerulonephritis and hypocomplementaemia. *Postgrad Med J*. 1980;56(653):177–179. <https://doi.org/10.1136/pgmj.56.653.177>.
 425. Kobayashi S, Nagase M, Kimura M, Ohyama K, Ikeya M, Honda N. Renal involvement in mixed connective tissue disease. Report of 5 cases. *Am J Nephrol*. 1985;5(4):282–289. <https://doi.org/10.1159/000166947>.
 426. Sawai T, Murakami K, Kurasono Y. Morphometric analysis of the kidney lesions in mixed connective tissue disease (MCTD). *Tohoku J Exp Med*. 1994;174(2):141–154. <https://doi.org/10.1620/tjem.174.141>.
 427. MCTD BR. In: *The Kidney Inb Collagen Vacular Disease*. 1993.
 428. Kessler E, Halpern M, Chagnac A, Zevin D, Hammel I, Ben Bassat M. Unusual renal deposit in mixed connective tissue disease. *Arch Pathol Lab Med*. 1992;116(3):261–264.
 429. Ulmer A, Kotter I, Pfaff A, Fierlbeck G. Efficacy of pulsed intravenous immunoglobulin therapy in mixed connective tissue disease. *J Am Acad Dermatol*. 2002;46(1):123–127. <https://doi.org/10.1067/mjd.2001.118539>.
 430. Wang CY, Guo SH, Wang LP, Wang XY, Shen HL. Clinical efficacy of low-dose rituximab on hematological abnormalities in patients with connective tissue disease. *Int J Clin Pharmacol Ther*. 2019;57(10):500–505. <https://doi.org/10.5414/CP203453>.
 431. Hajas A, Szodray P, Nakken B, et al. Clinical course, prognosis, and causes of death in mixed connective tissue disease. *J Rheumatol*. 2013;40(7):1134–1142. <https://doi.org/10.3899/jrheum.121272>.
 432. Qin B, Wang J, Yang Z, et al. Epidemiology of primary Sjogren's syndrome: a systematic review and meta-analysis. *Ann Rheum Dis*. 2015;74(11):1983–1989. <https://doi.org/10.1136/annrheumdis-2014-205375>.
 433. Malladi AS, Sack KE, Shiboski SC, et al. Primary Sjogren's syndrome as a systemic disease: a study of participants enrolled in an international Sjogren's syndrome registry. *Arthritis Care Res*. 2012;64(6):911–918. <https://doi.org/10.1002/acr.21610>.
 434. Maripuri S, Grande JP, Osborn TG, et al. Renal involvement in primary Sjogren's syndrome: a clinicopathologic study. *Clin J Am Soc Nephrol*. 2009;4(9):1423–1431. <https://doi.org/10.2215/CJN.00980209>.
 435. Lin DF, Yan SM, Zhao Y, et al. Clinical and prognostic characteristics of 573 cases of primary Sjogren's syndrome. *Chin Med J (Engl)*. 2010;123(22):3252–3257.
 436. Jasiek M, Karras A, Le Guern V, et al. A multicentre study of 95 biopsy-proven cases of renal disease in primary Sjogren's syndrome. *Rheumatology*. 2017;56(3):362–370. <https://doi.org/10.1093/rheumatology/kew376>.
 437. Evans RD, Laing CM, Ciurtin C, Walsh SB. Tubulointerstitial nephritis in primary Sjogren syndrome: clinical manifestations and response to treatment. *BMC Musculoskelet Disord*. 2016;17:2. <https://doi.org/10.1186/s12891-015-0858-x>.
 438. Koratala A, Reeves WH, Segal MS. Tubulointerstitial nephritis in sjogren syndrome treated with mycophenolate mofetil. *J Clin Rheumatol*. 2017;23(7):402–403. <https://doi.org/10.1097/RHU.0000000000000595>.
 439. Kidder D, Rutherford E, Kipgen D, Fleming S, Geddes C, Stewart GA. Kidney biopsy findings in primary Sjogren syndrome. *Nephrol Dial Transplant*. 2015;30(8):1363–1369. <https://doi.org/10.1093/ndt/gfv042>.
 440. Ingegnoli F, Ughi N, Mihai C. Update on the epidemiology, risk factors, and disease outcomes of systemic sclerosis. *Best Pract Res Clin Rheumatol*. 2018;32(2):223–240. <https://doi.org/10.1016/j.berh.2018.08.005>.
 441. D'Angelo WA, Fries JF, Masi AT, Shulman LE. Pathologic observations in systemic sclerosis (scleroderma). A study of fifty-eight autopsy cases and fifty-eight matched controls. *Am J Med*. 1969;46(3):428–440. [https://doi.org/10.1016/0002-9343\(69\)90044-8](https://doi.org/10.1016/0002-9343(69)90044-8).
 442. Muangchan C, Canadian Scleroderma Research G, Baron M, Pope J. The 15% rule in scleroderma: the frequency of severe organ complications in systemic sclerosis. A systematic review. *J Rheumatol*. 2013;40(9):1545–1556. <https://doi.org/10.3899/jrheum.121380>.
 443. Teixeira L, Mouthon L, Mahr A, et al. Mortality and risk factors of scleroderma renal crisis: a French retrospective study of 50 patients. *Ann Rheum Dis*. 2008;67(1):110–116. <https://doi.org/10.1136/ard.2006.066985>.
 444. Gordon SM, Stitt RS, Nee R, et al. Risk factors for future scleroderma renal crisis at systemic sclerosis diagnosis. *J Rheumatol*. 2019;46(1):85–92. <https://doi.org/10.3899/jrheum.171186>.
 445. Penn H, Howie AJ, Kingdon EJ, et al. Scleroderma renal crisis: patient characteristics and long-term outcomes. *QJM*. 2007;100(8):485–494. <https://doi.org/10.1093/qjmed/hcm052>.
 446. Woodworth TG, Suliman YA, Li W, Furst DE, Clements P. Scleroderma renal crisis and renal involvement in systemic sclerosis. *Nat Rev Nephrol*. 2016;12(11):678–691. <https://doi.org/10.1038/nrneph.2016.124>.
 447. Helfrich DJ, Banner B, Steen VD, Medsger Jr TA. Normotensive renal failure in systemic sclerosis. *Arthritis Rheum*. 1989;32(9):1128–1134. <https://doi.org/10.1002/anr.1780320911>.
 448. Atencah SE, Robertson R, Ukoha N, Idolor ON, Pippin J. Systemic sclerosis with a normotensive scleroderma renal crisis: a diagnostic dilemma. *Cureus*. 2024;16(7):e64167. <https://doi.org/10.7759/cureus.64167>.
 449. Hudson M, Baron M, Tatibouet S, Furst DE, Khanna D. International scleroderma renal crisis study I. Exposure to ACE inhibitors prior to the onset of scleroderma renal crisis—results from the international scleroderma renal crisis survey. *Semin Arthritis Rheum*. 2014;43(5):666–672. <https://doi.org/10.1016/j.semarthrit.2013.09.008>.
 450. Ida T, Ikeda K, Ohbe H, et al. Early initiation of angiotensin-converting enzyme inhibitor in patients with scleroderma renal crisis: a nationwide inpatient database study. *Rheumatology*. 2024;63(6):1507–1511. <https://doi.org/10.1093/rheumatology/kead343>.
 451. Steen VD, Medsger Jr TA. Long-term outcomes of scleroderma renal crisis. *Ann Intern Med*. 2000;133(8):600–603. <https://doi.org/10.7326/0003-4819-133-8-200010170-00010>.
 452. Abbott KC, Trespalacios FC, Welch PG, Agodoa LY. Scleroderma at end stage renal disease in the United States: patient characteristics and survival. *J Nephrol*. 2002;15(3):236–240.
 453. Hruskova Z, Pippas M, Stel VS, et al. Characteristics and outcomes of patients with systemic sclerosis (scleroderma) requiring renal replacement therapy in Europe: results from the ERA-EDTA registry. *Am J Kidney Dis*. 2019;73(2):184–193. <https://doi.org/10.1053/j.ajkd.2018.05.016>.